Propagation of Harmonics

This is the second chapter, dealing with the first objective. This chapter develops and discusses a propagation model. It begins with a presentation of standing wave definitions. The propagation model is defined and validated, and model elements for components are established. The chapter concludes with a section dedicated to propagation analyses, which contains a guideline for such studies as well as case studies demonstrating voltage propagation in a meshed system and an interpretation of the physical meaning. The Western Danish 400 kV transmission grid is used as a case study.

1 Introduction

In the previous chapters, the harmonic problem of meshed systems was discussed, and the behaviour of both the passive grid and harmonic sources was discussed in relation to the research questions posed. It was concluded that the harmonic analysis techniques widely used today cannot unequivocally explain the physical propagation phenomena in meshed networks. In particular, the introduction of UGC into a transmission network is of great concern. The case with a short UGC section presented in Chapter 1 revealed that the conditions in terms of understanding why a relatively short UGC section can cause such large changes in harmonic distortion in some substations but not in others and what parameters determine it are not fully understood.

Harmonic analysis takes a huge workload and a time-consuming scope, which was also discussed. The complexity of these assessments is increasing, influencing both their speed and the requirement for resources for analysing individual projects regarding power system grid modifications or grid expansions.

Prior to more in-depth analysis, it is useful for a TSO to be able to quantify possible changes in harmonic propagation due to network changes using a quick and simple screening method, such as a feasibility study. The goal of such a screening method is to identify potentially problematic network configurations in the face of harmonic voltage propagation, which can then be rejected in the planning work, or necessary mitigating measures must be incorporated at the overall planning level.

Furthermore, unexpected harmonic amplifications have been measured because of cable laying in the Danish transmission network as discussed in Chapter 1, making understanding the propagation phenomena in a meshed transmission system difficult. Other cases of unexpected increases in harmonic voltages associated with the use of underground cables in power systems have also been documented in the literature [4.1]–[4.3], with examples from Canada, France, and Ireland. They conclude that simulations show that harmonic voltage distortion unexpectedly increases significantly due to underground cables, but the impact due to changed propagation throughout the grid is not discussed, and proper modelling to anticipate this was found to be extremely complicated.

To address these issues, we must understand the impact on harmonic propagation due to increased use of UGC in a meshed system, considering the two topics listed below:

1. How does partially undergrounding transmission lines affect harmonic voltage propagation?
2. What effect does a different degree of meshing have on propagation?

These studies are related to the following four research topics:

(f) Development of a propagation model for the analysis of harmonic standing wave patterns and their propagation in a meshed power system.

(g) Validation of the propagation model, as well as processes for carrying out the validation.

(h) Calculations and analysis for a transmission network with the developed propagation model and for case studies based on known harmonic propagation problems for determining the consequences of increased undergrounding and network topology changes.

(i) Development of a general study procedure for the analysis of harmonic voltage propagation in a cablified meshed grid with the propagation model.

This chapter attempts to map how the use of UGC affects harmonic propagation in a meshed grid, as well as how changes in the passive grid affect harmonic propagation in general, considering the degree of meshing. The use of UGC is emphasised. The following case studies will be used in this study:

- A reference case is required as a baseline against which network changes can be assessed. A model of the Western Danish 400 kV transmission grid in its current state is used in this case, as presented in Section 3 of Chapter 1.

- A variation study in which cable lengths in an existing UGC section of a combined OHL-UGC transmission line are varied to uncover the following conditions:

  - The effect of cable length on harmonic propagation at partial cable laying.
  - Changing a combined line to a pure UGC line.

- A variation study in which a crosswise connection of varying length is established in the meshed transmission network, allowing for the identification of the following conditions:

  - The effect of increased degree of meshing in the transmission network.
  - The effect of the crosswise connection on harmonic propagation.

The actual case studies are defined later in Section 7.4 of this chapter.

The most used harmonic analysis methods in industry are frequency domain methods, primarily frequency scans and harmonic penetration studies, and frequency domain harmonic analysis formulation, which can be very effective and reliable for steady-state solutions [4.4], [4.5]. Because harmonics propagate as standing waves [4.6]–[4.8], the method proposed in this thesis (referred as the *propagation model*) is based on frequency-domain electromagnetic waves that describe standing wave patterns.

The method presented in this chapter differs from traditional harmonic methods of analysis in terms of:

- Assessment of reflection and transmission coefficients as an element to explain the physical circumstances of OHL and UGC that affect the propagation and pinpoint the exact location(s) causing harmonic problems.

- Assessment of propagated waves throughout the network to identify changes in propagation patterns caused by partial cable laying in a transmission line.

- Assessment of changes in propagation patterns in the grid from specific sources.

Because transmission networks will be greatly expanded in the future with underground cables and a greater degree of interconnection, it is critical for a TSO to understand the impact of these changes on harmonic voltage propagation, as voltage is used as a planning threshold limit for TSOs using IEC 61000 for harmonic assessments. The change in standing waves caused by the natural reflections created by the transition between overhead line and cable or between a line and a substation can be quantified with this method.

As a result, evaluating problematic propagation pathways in the grid is possible, and mitigation measures can be budgeted for in projects.

The objective of this chapter is to explain harmonic voltage propagation using the Western Danish transmission network as an example. Tools with a microwave theory background are used because harmonic voltage, when considered as standing waves, can be treated with this theory and, most likely, can provide valuable knowledge on propagation nature that traditional analysis techniques cannot.

In this study, the emphasis is on the use of UGC, so a set of case studies is defined for analysing the effect of UGC on harmonic propagation in a meshed grid and how changes in the passive grid affect harmonic propagation in general. This chapter is outlined as follows:

Section 2 discusses the properties of standing waves, followed by Section 3's justified choice of parameter models. Sections 4 and 5 present the development of S-parameter component models and the propagation model, followed by an outline of case studies and a propagation study model of the Western Danish 400 kV transmission network in Section 6. Section 7 includes a study guide for propagation studies as well as a discussion and analysis of results in relation to case studies. This chapter is summarised in Section 8.

2 The Properties of Standing Waves

The telegraph equations, which are a set of coupled linear equations determining the voltage and current distributions along a linear electrical transmission line, are used to mathematically describe stationary waves or standing waves [4.9]. The equations and solutions are applicable in both time and frequency domains ranging from zero to megahertz and even gigahertz [4.9], [4.10].

Appendix E goes into detail about the telegraph equations, whereas this section only covers the equations required to deal with stationary waves. The properties of incident wave and reflected wave, terminal conditions, and wave extremes are defined after the equations are defined.

This section concludes with an illustrative example that explains what the definitions in relation to a power system transmission line mean.

2.1 The Telegraph Equations

The root-mean-square value (RMS) voltage and current along a transmission line can be calculated using the wave equations (E.7) - (E.8) from Appendix E, which are repeated here:

$$\underline{U}(x) = \frac{1}{2}\left(\underline{U}_S - \underline{Z}_0\underline{I}_S\right)e^{\underline{\gamma}x} + \frac{1}{2}\left(\underline{U}_S + \underline{Z}_0\underline{I}_S\right)e^{-\underline{\gamma}x} \tag{4.10}$$

$$\underline{I}(x) = -\frac{1}{2}\left(\frac{\underline{U}_S}{\underline{Z}_0} - \underline{I}_S\right)e^{\underline{\gamma}x} + \frac{1}{2}\left(\frac{\underline{U}_S}{\underline{Z}_0} + \underline{I}_S\right)e^{-\underline{\gamma}x} \tag{4.11}$$

where $\underline{U}(x)$ is the RMS voltage phasor at a distance x along the line, $\underline{I}(x)$ is the RMS current phasor at a distance x along the line, the sending end RMS voltage and current are denoted by $\underline{U}_S$ and $\underline{I}_S$, respectively, $\underline{Z}_0$ is the characteristic impedance, and $\underline{\gamma}$ is the propagation constant. All parameters are complex numbers. The propagation constant is given in Appendix E (E.9) and repeated here:

$$\underline{\gamma} = \sqrt{\underline{z} \cdot \underline{y}} = \alpha + j\beta \tag{4.12}$$

where $\underline{z} = r + j\omega \cdot l$ and $\underline{y} = g + j\omega \cdot c$ can be interpreted as per-phase frequency dependant values of the series impedance and the shunt admittance. The constant α describes the attenuation (or damping) due to line losses ($\alpha = \sqrt{r \cdot g}$) and is designated as the *attenuation constant (α)* in this thesis. The constant β is the phase constant representing the change in phasor direction per unit length along the path travelled by the wave ($\beta = \omega\sqrt{l \cdot c}$), designated as *phase constant (β)* in this thesis.

The transmission line acts as a constant load with an impedance designated as the complex characteristic impedance $\underline{Z}_0$ (for a lossy line) and this impedance is fixed by the geometry and the insulation material of the conductors and thus appears constant for a specific type of transmission line (at a specific frequency). $\underline{Z}_0$ is given in (E.1) and repeated here:

$$\underline{Z}_0(\omega) = \sqrt{\frac{R(\omega) + j\omega L(\omega)}{G + j\omega C}} \tag{4.13}$$

where $\underline{Z}_0$ is the ratio of the amplitudes of current and voltage propagating along the line. The series impedance is determined by the resistance R and the inductance L. The shunt admittance is determined by the conductance G and the capacitance C of the transmission line, and all four parameters depend on the material used for the line, the line dimensions, the layouts, etc. ω indicates that the parameter value depends on frequency. The parameters R, L, G, and C are designated as *primary line constants* in this thesis.

The parameters $\underline{Z}_0$ and $\underline{\gamma}$ are derived from the primary line constants and in this thesis, we will designate $\underline{Z}_0$ and $\underline{\gamma}$ as *secondary line constants*.

2.2 Incident and Reflected Waves

The voltage and current waveforms along a line are the sum of a forward travelling or *incident wave,* $\underline{U}^+$ and $\underline{I}^+$, and a backward travelling or *reflected wave,* $\underline{U}^-$ and $\underline{I}^-$ [4.9], [4.10]:

$$\underline{U}^+(x) = \frac{1}{2}\left(\underline{U}_S - \underline{Z}_0 \underline{I}_S\right) e^{\underline{\gamma} x} \tag{4.14}$$

$$\underline{U}^-(x) = \frac{1}{2}\left(\underline{U}_S + \underline{Z}_0 \underline{I}_S\right) e^{-\underline{\gamma} x} \tag{4.15}$$

$$\underline{I}^+(x) = -\frac{1}{2}\left(\frac{\underline{U}_S}{\underline{Z}_0} - \underline{I}_S\right) e^{\underline{\gamma} x} \tag{4.16}$$

$$\underline{I}^-(x) = \frac{1}{2}\left(\frac{\underline{U}_S}{\underline{Z}_0} + \underline{I}_S\right) e^{-\underline{\gamma} x}. \tag{4.17}$$

The incident and reflected waves in (4.14) - (4.17) are vectors with a fixed length equal to their RMS value, and as the wave propagates along the line, the vector rotates with $\underline{\gamma} x$.

2.3 Line Termination

When a travelling wave on a transmission line reaches the end of the line, it is either completely absorbed by the load, completely reflected back to the sending end, or partially reflected, with some of the incident wave transmitted into another medium beyond the endpoint junction, that is, a *transmitted wave.* [4.9]. The latter occurs when the transmission line on either side of the junction or the endpoint load has an impedance mismatch.

Endpoint termination can be mathematically described using the *complex reflection coefficient*, which describes how much of a wave is reflected by an impedance discontinuity in the transmission medium. It is defined as the ratio of the amplitude of the reflected wave to the amplitude of the incident wave, both expressed as phasors. The complex reflection coefficient $\underline{\Gamma}$ is defined as [4.9]:

$$\underline{\Gamma} := \frac{\underline{U}^-}{\underline{U}^+} = -\frac{\underline{I}^-}{\underline{I}^+} = \frac{\underline{Z} - \underline{Z}'}{\underline{Z} + \underline{Z}'}. \tag{4.18}$$

When impedances can be defined, $\underline{\Gamma}$ is calculated using characteristic impedances, where $\underline{Z}$ is the characteristic impedance of the transmission line ahead of the discontinuity or the source impedance, and $\underline{Z}'$ is the impedance after the discontinuity or the load impedance as seen from the transmission material between the source and the load.

2.4 Stationary Waves

A standing wave, also known as a stationary wave in physics [4.11], was briefly discussed in Chapter 1. A standing wave is a wave that oscillates in time but does not move in space [4.11]. The peak amplitude of the wave oscillations is constant with respect to time at any point in space, and the oscillations at different points along the wave are in phase [4.11]. Figure 4.1 depicts the properties of a standing wave as a sinusoidal signal formed by an incident and reflected wave. This signal oscillates between a maximum

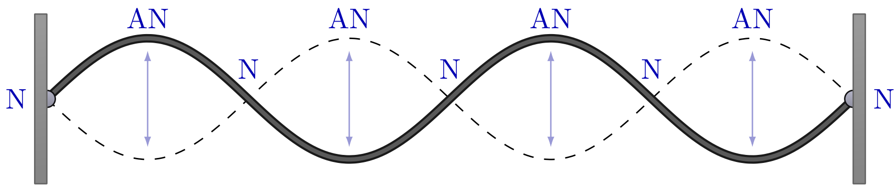


Fig. 4.1: A standing wave that exists between two fixed terminations. The separation between terminations is equal to two wavelengths λ. The wave peaks are referred to as *antinodes (AN)* whereas the zero values are refereed to as *nodes (N)*.

and a minimum value, as indicated by the arrows between the solid line and the dashed line. It oscillates in magnitude but does not propagate down the length of the line like incident or reflected waveforms. Along the line, the amplitude of the standing wave is zero at specific points, indicating that the incident and reflected waves cancel each other out. A "zero point" is known as a *node (N)*, and their positions along the line do not change. An *antinode (AN)* is the point at which the oscillating signal reaches its maximum (or minimum) amplitude and their positions along the line does not change either.

The circumstances for which standing waves form depend on whether the energy fed into a system has an appropriate frequency or not. When the driving frequency applied to a system equals the natural frequency of the system, resonances occur. This means that standing waves are always associated with resonance. It applies to any system in which standing waves can form, that there are numerous natural frequencies. Therefore, the set of all possible standing waves are known as the harmonics of a system [4.11].

In the case of a power system transmission line in normal operation, with a definite length of, for instance, one wavelength and a sinusoidal signal applied to it, the line can be considered a bounded transmission medium, which means that both ends are fixed. In this case, there will be one node at each end of the line and one antinode in the middle. This is also half of the line's wavelength. The next possible standing wave is one full wavelength, which is achieved by adding one node to the centre, doubling the frequency of the signal. If we keep adding nodes, dividing the line into fourths, fifths, sixths, and so on, we get more wavelengths. Figure 4.1 depicts a total of five nodes, which equals two wavelengths. It can be demonstrated that the frequencies of resonance for a line with two fixed ends are multiples of the propagation speed v divided by the wavelength λ [4.10], [4.11], that is:

$$f_{res} = \frac{v}{\lambda} = n\frac{v}{2x}\bigg|_{n=1,2,3,...} \tag{4.19}$$

where x is the length of the line. The propagation speed v given by (E.4) in Appendix E is:

$$v = \frac{1}{\sqrt{L \cdot C}} \tag{4.20}$$

and the propagation speed for a specific transmission line type is equal for all frequencies, assuming that L is constant. Because L in this case is frequency dependant, (4.20) can be written as:

$$v(\omega) = \frac{1}{\sqrt{L(\omega) \cdot C}}. \tag{4.21}$$

Therefore, the propagation speed for a specific transmission line type is constant at a specific frequency, so the node frequency f_N can be calculated by inserting (4.21) into (4.19) as follows:

$$f_N = n\frac{1}{2x}\frac{1}{\sqrt{L(\omega)C}}\bigg|_{n=0,1,2...} \tag{4.22}$$

If, on the other hand, one end is open, i.e. the impedance is reaching infinity, the frequencies of resonance are [4.10], [4.11]:

$$f_{AN} = n\frac{1}{4x}\frac{1}{\sqrt{L(\omega)C}}\bigg|_{n=1,3,5,...} \tag{4.23}$$

For an antinode, n is an odd multiple of a quarter wavelength.

2.5 Illustrative Example

An illustrative example based on numerous examples from literature [4.9], [4.10], [4.12]–[4.14] is given to illustrate the meaning of the definitions presented in this section. Figure 4.2 shows a lossless transmission line perfectly surrounded by air that connects a generator with a voltage $U_G = 2$ per-unit injected at 1 kHz to a load Z_L. The transmission line's characteristic impedance $Z_0 = 300\,\Omega$. The generator impedance Z_G is also $300\,\Omega$.

Assuming the line length equals one wavelength λ, this length can be determined by (4.19) as:

$$\lambda = \frac{v}{f} = \frac{300 \cdot 10^6}{1 \cdot 10^3} = 300\,\text{km} \tag{4.24}$$

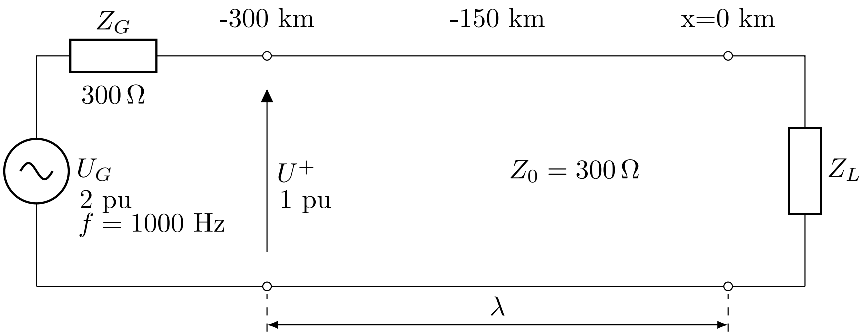


Fig. 4.2: Transmission line with connected generator and load.

that is, a line length of 300 km. Because the line is λ long and due to matching impedances, then the input impedance as seen from the generator is $Z_{in} = Z_L$ and $U(-\lambda) = U(0)$. We define the point $x = 0$ is located at Z_L, where x is the distance along the line. The reflection coefficient Γ_L at the load is given by (4.18) and for various load conditions, Γ_L are given in Table 4.1.

Table 4.1: Calculated values of Γ_L for the example.

Z_L [Ω]	300	0	∞	600	150	$600\angle 70°$
Γ_L	0	-1	+1	$\frac{1}{3}$	$-\frac{1}{3}$	$\frac{1}{3}\angle 70°$

In this case, three extreme values of $\Gamma_L = \{0, -1, +1\}$ are for a matched load, short circuit and open-end terminations, respectively. The latter three values of $\Gamma_L = \{\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}\angle 70°\}$ are for a load impedance twice the size of the characteristic impedance of the line, half the size and a load impedance with an arbitrary direction different from zero degrees, respectively. The incident wave can be determined as:

$$U^+ = \frac{Z_{in}}{Z_{in} + Z_G} \cdot U_S = \frac{Z_0 \frac{1+\Gamma_L(-\lambda)}{1-\Gamma_L(-\lambda)}}{Z_0 \frac{1+\Gamma_L(-\lambda)}{1-\Gamma_L(-\lambda)} + Z_G} \cdot U_S. \tag{4.25}$$

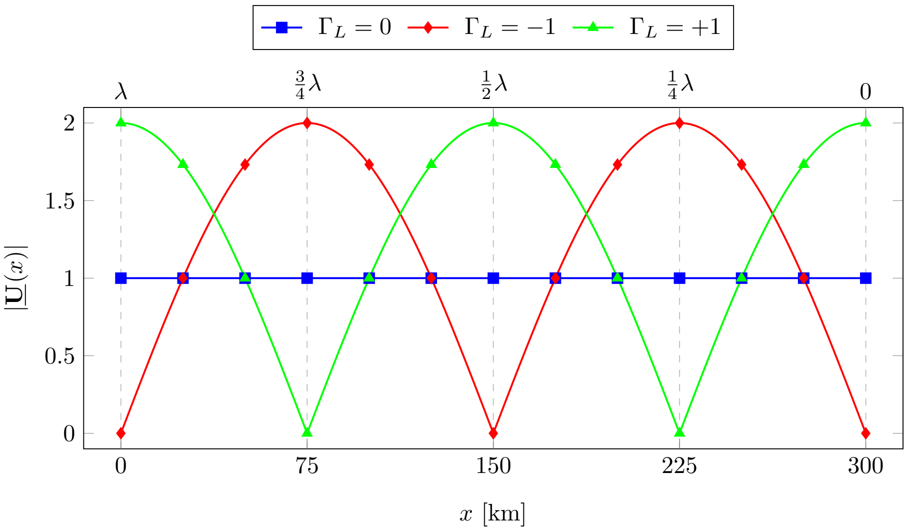
In this case, the incident wave U^+ is half the voltage magnitude of the source, that is, 1 per-unit, because $Z_G = Z_L$ and no reflections occur at the generator terminals. The standing voltage wave along the line can be constructed as:

$$|U(x)| = |U^+| \cdot |1 + \Gamma_L(x)| \tag{4.26}$$

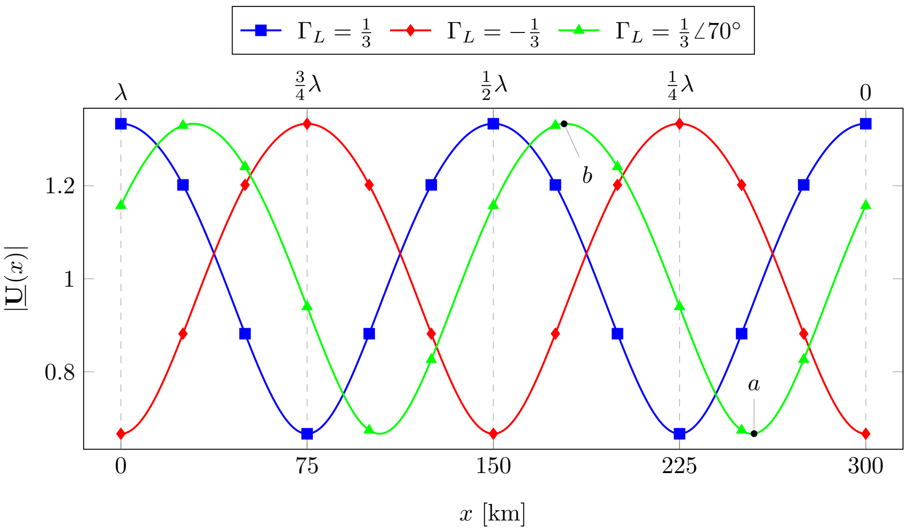
where the value of Γ_L depends on the position along the line, that is [4.10]:

$$\Gamma_L(x) = \Gamma_L \cdot e^{j2 \cdot \frac{2\pi}{\lambda} \cdot (-x)} \tag{4.27}$$

The calculated voltage along the line used in this example is shown in Figure 4.3. The line length x in kilometres is shown on the x-axis, and the absolute value of voltage determined by (4.26) is shown on the y-axis.



(a) $\Gamma_L = \{0, -1, +1\}$



(b) $\Gamma_L = \{\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}\angle 70°\}$

Fig. 4.3: Voltage along the transmission line for load reflections according to Table 4.1. (a) - Voltage waves for extreme cases. (b) - Voltage waves for defined load conditions.

Figure 4.3a depicts the voltage standing wave for the three extreme values of

Γ_L. If there is no reflection at the load ($\Gamma_L = 0$), the voltage along the line equals the input voltage. This is the case for the lossless line used in this example.

If $\Gamma_L = -1$ or $\Gamma_L = +1$, the load termination is either a short circuit or an open-end termination. These terminations causes the highest voltage magnitudes along the line, and the voltage range is between zero and two per-unit, or twice the input voltage U^+. The position of minimum and maximum voltage along the line differs depending on the reflection coefficient.

For open-end termination ($\Gamma_L = +1$), the maximum voltage occurs at the load end and every $\frac{1}{2}\lambda$ along the line. At these positions, antinodes are present. This observation is consistent with (4.23). For short-circuit termination ($\Gamma_L = -1$), the maximum voltage (antinodes) occurs at $\frac{1}{4}\lambda$ from the load end and subsequent every $\frac{1}{2}\lambda$ further along the line. This observation is consistent with (4.22).

Voltage minima occurs at nodes, which are always $\frac{1}{4}\lambda$ away from an antinode. This means that if the distance to a voltage maximum or a voltage minimum is known, the distance to the next extreme value can be calculated by simply adding $\frac{1}{4}\lambda$ to the known distance.

Figure 4.3b depicts the voltage for more realistic reflection coefficients because transmission lines in a power system are most likely never operated with an open-end termination for a longer period of time and almost certainly never with a short circuit termination. In case of reflection coefficients with zero degree angle, the extremes are positioned at multiples of $\frac{1}{4}\lambda$ as for the cases shown in Figure 4.3a, wheres for the case with $\Gamma_L = \frac{1}{3}\angle 70°$, the extremes are shifted, but the distance between nodes and antinodes still equals (4.22) - (4.23).

The voltage minimum and maximum values differs from the extreme cases in that they have values greater than zero and less than two per-unit. The extreme values of $|\underline{U}(x)|$ are found when Γ_L has the largest or the smallest value and are calculated as [4.10]:

$$|\underline{U}(x)|_{max} = U^+(1 + |\Gamma_L|) \tag{4.28}$$

$$|\underline{U}(x)|_{min} = U^+(1 - |\Gamma_L|) \tag{4.29}$$

where the definition of Γ_L is given in (4.27). The maximum and minimum voltages for the cases shown in Figure 4.3b are calculated to be 1.33 per-unit and 0.67 per-unit, respectively. The distance to minima at $x = a$ for the case with $\Gamma_L = \frac{1}{3}\angle 70°$ is calculated as:

$$x(a) \approx -\frac{70° - 180°}{2 \cdot \frac{360°}{\lambda}} = \frac{110}{720}\lambda = 0.153\lambda \qquad \left(\pm n \cdot \frac{\lambda}{2}\bigg|_{n=1,2,\ldots} \right) \tag{4.30}$$

where n is a positive integer. Similarily, the voltage maximum at $x = b$ can be calculated as:

$$x(b) \approx 0.250\lambda + x(a) = (0.250 + 0.153)\lambda = 0.403\lambda \qquad \Big(\pm n \cdot \frac{\lambda}{2}\Big|_{n=1,2,\ldots} \Big). \tag{4.31}$$

As $\lambda = 300\,\mathrm{km}$, points a and b are approximately 46 km and 121 km away from the load terminal.

2.6 Summary on Standing Waves

This section defined standing waves and important parameters in this context and these parameters have been explained using an illustrative example. The definitions provided will be applied in this thesis.

The maxima and minima of both the standing voltage wave and the standing current wave can be determined using this example if the reflection coefficient and incident wave are known

The absolute magnitude of the standing wave is only doubled when compared to the incident wave in extreme cases, such as open-end termination or short circuit. In all other cases, the maximum voltage is less than two times the voltage incident at the transmission line. The voltage along the transmission line equals the incident voltage if the load and transmission line are perfectly matched.

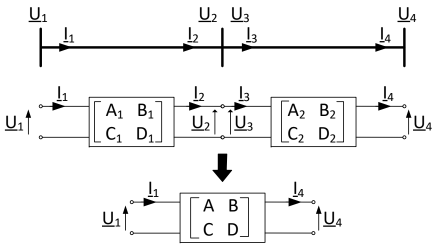
From a practical perspective, the extreme cases and perfectly matched load conditions are unlikely to occur in relation to a power system. As a result, it is reasonable to assume that the harmonic voltage contribution from one line end to the other, terminated in a passive load, will never be greater than twice the incident voltage. This assumption is at least valid for radial or ring connections in this case.

Another viewpoint is required for a meshed grid because the harmonic voltage at the terminals of a transmission line may be superimposed with contributions from adjacent lines connected at a substation. The practical conditions in relation to a meshed system will be discussed further in Section 7 in this chapter.

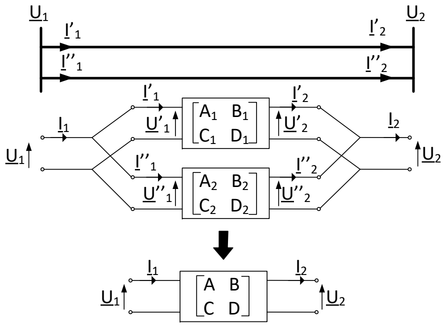
3 Model Parameter Selection

Transmission lines in the Danish transmission grid presented in the introduction in Chapter 1 are not always homogeneous between two substations but consists of several line sections of varying types. This could be an OHL section cascaded with another OHL section with different geometry, or it could be a UGC section. Tables 1.3 and 1.4 shows the various line type configurations that are currently in use in the Western Danish transmission system.

Because of the impact on stationary waves discussed in the previous Section 2 of this chapter, models for terminal conditions at substations or between line sections are of interest in this study, so cascading of transmission line sections is required. The rationale for using Scattering parameters (S-parameters) for modelling is discussed in this section. In general, the combination of transmission line sections can be either parallel lines or cascaded connections of line sections, as shown in Figure 4.4.



(a) Cascaded lines.



(b) Parallel lines.

Fig. 4.4: Single line diagram of combinations of transmission lines. Corresponding two-port models with ABCD-parameters are shown. (a) - Cascade connection of two line sections. (b) - Parallel connection of two lines.

Figure 4.4a depicts a single line diagram of cascade connected line sections, while Figure 4.4b depicts parallel lines. The corresponding modelling with two-port equivalents is also shown and in this case, transmission parameter (or ABCD-parameter) models are depicted. The number of lines presented in Section 3 of Chapter 1 showed that 19 lines and 92 line sections needs to be modelled for this study. Due to this large number of different lines and line sections, an appropriate approach is to use two-port models to describe line sections and the equivalent two-port models are therefore shown in Figure 4.4. Depending on the type of study, every single line section can be modelled by individual two-ports or as cascaded models. The depicted two-port models will be discussed later in this section.

To perform harmonic penetration analysis, a mathematical representation of the power system must be created, and the electrical properties of an interconnected network are found to be difficult to mathematically describe, according to [4.6], [4.15]–[4.17]. This complexity, however, can be overcome by considering individual physical elements with known parameters and connecting them in the manner of a physical network [4.6], [4.15]. For harmonic analyses, the traditional approach is to consider describing the physical elements with impedance or admittance matrices and a per-phase representation as two-port network equivalents or three-phase equivalents using an extended two-port network taking all three phases into consideration [4.6], [4.15], [4.17].

The study objective in this thesis is harmonics propagating in an interconnected network (meshed grid), so this thesis focuses on transmission lines. As a result, the mathematical equivalent using two-port networks of transmission lines is highlighted in the following.

3.1 Choice of Two-port Network Models

Typically, admittance matrices are calculated for each transmission line by means of ABCD-parameters, which are then combined via matrix multiplication to yield the resultant transmission parameters for a complete line [4.15]. These are then transformed into the necessary nodal admittance matrices [4.15]. ABCD-parameters eases the mathematical formulation of equivalent two-port networks due to their ability of cascading lines and line sections to an equivalent model using simple matrix algebra [4.12], [4.13], [4.15], [4.18]. The transmission parameters are easily converted between $\underline{\boldsymbol{Z}}$ and $\underline{\boldsymbol{Y}}$ matrices [4.13], [4.15].

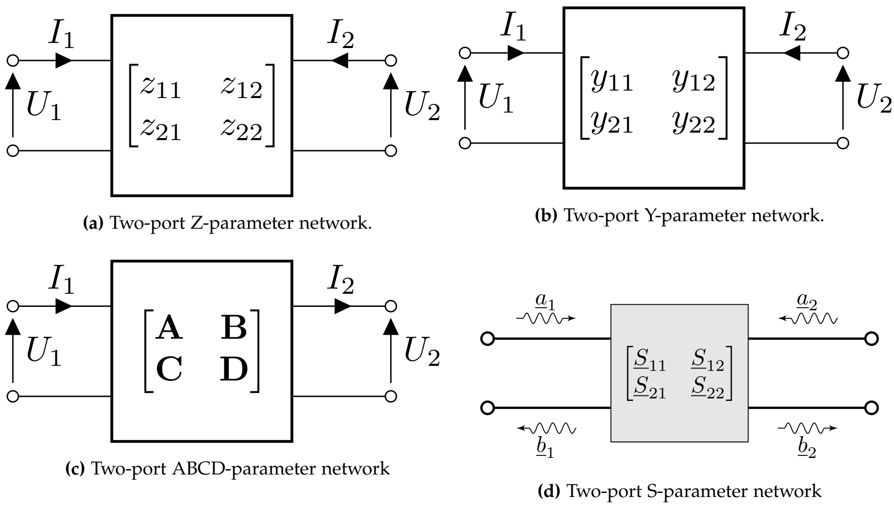
A common concern is the type of parameters to use in the analyses. There are no general guidelines found in the literature, but a practical approach is to first define the purpose of the study and then define the input and output signals of the two-port network. Table 4.2 compares the properties of Z, Y, and ABCD-parameters based on [4.15], [4.17]–[4.19]. The focus is on steady-state signals and linear passive elements. The properties of S-parameters based on [4.12], [4.13] are also shown because these parameters are used for the propagation model, which will be presented later in Section 5. Other parameter types such as for instance hybrid parameters exists [4.19]–[4.21], but these parameter types are not considered in this thesis.

Table 4.2 shows that a common application of Z and Y-parameters is the calculation of port voltages or currents, with the parameters calculated using an open-circuiting and short-circuiting approach, respectively. Input and output are bidirectional, which means that a voltage or current applied at port 1 excites a voltage or current at port 2, and the same magnitude of voltage or current applied at port 2 excites an equal response at port 1. Figure 4.5a depicts the matrix description of the two-port Z-parameters, while Figure 4.5b depicts the matrix description of the two-port Y-parameters. The corresponding current and voltage signals are depicted, with index 1 indicating port 1 and index 2 indicating port 2.

The ABCD-parameters are known as cascade, chain, or transmission parameters because of their ability of mathematical expressing the interconnection of these two-port networks [4.15], [4.16]. The application of ABCD-parameters, according to Table 4.2, is direct calculation of network current and voltage, which can be done for any known impedance and admittance.

Table 4.2: Properties of two-port model parameters. Comparison between ABCD, Z, Y, and S-parameters, based on [4.12], [4.13], [4.15], [4.17]–[4.19].

Dimension	Parameters: Z	Y	ABCD	S
Application	Direct calculation of network voltage	Direct calculation of network current	Direct calculation of network current and voltage	voltage wave characterisation, such as in microwave interconnections
Calculation	Z-parameters are determined by open circuiting port 1 and port 2	Y-parameters are determined by short-circuiting port 1 and port 2	Directly calculable for any known impedance/admittance	Typically calculated in terms of other parameters (Z, Y or ABCD)
Interpretation	Current entering/ exiting a port, voltage measured across a port, wave propagation insensitive	Voltage applied on a port, current exiting a port, wave propagation insensitive	Current entering/ exiting a port, voltage measured across a port, wave propagation insensitive	Reflection carried by a propagating voltage wave
Directionality	Bidirectional with appropriate sign definitions, no reflection considered			Bidirectional at each point, including reflection
Interconnection properties	Series: $\sum Z = Z_1 + Z_2$	Parallel: $\sum Y = Y_1 + Y_2$	Cascaded: $\sum ABCD = ABCD_1 \times ABCD_2$ Parallel: Possible	Possible, but not applicable



(a) Two-port Z-parameter network.

(b) Two-port Y-parameter network.

(c) Two-port ABCD-parameter network

(d) Two-port S-parameter network

Fig. 4.5: Two-port network models. Input and output signals are shown with index 1 indicating port 1 and 2 indicating port 2. (a) - Z-parameters. (b) - Y-parameters. (c) - ABCD-parameters. (d) - S-parameters.

This means that these parameters are determined using Z and Y-parameters. Current and voltage are bidirectional in the same way as for the Z and Y-parameters. The matrix description of the two-port ABCD-parameters is shown in Figure 4.5c. Note the direction of current I_2, which is different compared to Z and Y two-port networks. This direction allows easily interconnection of two-port network models.

According to Table 4.2, the application of S-parameters is the characterisation of wave signals incident and reflected from the ports of a two-port network. Even though their typical association is with microwave frequencies, these parameters are also applicable to power system frequencies and multiples thereof [4.12], [4.13]. Typically, S-parameters are calculated in terms of other parameters such as Z, Y, or ABCD-parameters. S-parameters, on the other hand, have a very different interpretation because they contain information about reflections carried by a propagating wave in an interconnected circuit. S-parameters are also bidirectional, but include information about reflections that Z, Y, and ABCD-parameters do not [4.12], [4.13]. Figure 4.5d depicts the matrix description of a two-port with S-parameters. In this case, the input is an incident wave $\underline{a}_i$, and the output is a reflected wave $\underline{b}_i$.

3.2 Interconnecting Two-port Networks

Returning to the cascade and parallel modelling depicted in Figure 4.4, the interconnection properties of the parameters are as given in Table 4.2. Z-parameter matrices are summed when connected in series, yielding the combined voltage response at ports [4.19]. However, this is not to be interpreted as cascading two-port networks and therefore, for the propagation model cascading, Z-parameters are not applicable. In this case, ABCD-parameters are more appropriate because cascading can be done mathematically as matrix multiplication [4.18]:

$$\begin{bmatrix} \underline{U}_1 \\ \underline{I}_1 \end{bmatrix} = \begin{bmatrix} \underline{A}_1 & \underline{B}_1 \\ \underline{C}_1 & \underline{D}_1 \end{bmatrix} \times \begin{bmatrix} \underline{A}_2 & \underline{B}_2 \\ \underline{C}_2 & \underline{D}_2 \end{bmatrix} \times \begin{bmatrix} \underline{U}_2 \\ \underline{I}_2 \end{bmatrix} \tag{4.32}$$

In this case, $\underline{A}_i$, $\underline{B}_i$, $\underline{C}_i$ and $\underline{D}_i$ are complex numbers, but they could also be matrices. The Y-parameters are the most appropriate parameters for parallel connection of two-ports because the equivalent two-port network is the sum of both Y-matrices. If the model includes ABCD-parameters, a parallel equivalent network can be calculated [4.18]. As an example, (4.33) depicts the equivalent matrix description based on [4.18] using the parameters shown in Figure 4.4b:

$$\begin{bmatrix} \underline{U}_1 \\ \underline{I}_1 \end{bmatrix} = \begin{bmatrix} (\underline{B}_1^{-1} + \underline{B}_2^{-1})^{-1}(\underline{B}_1^{-1}\underline{A}_1 + \underline{B}_2^{-1}\underline{A}_2) & (\underline{B}_1^{-1} + \underline{B}_2^{-1})^{-1} \\ \underline{C}_1 + \underline{C}_2 + (\underline{D}_2 - \underline{D}_1)(\underline{B}_1 + \underline{B}_2)^{-1}(\underline{A}_1 - \underline{A}_2) & \underline{D}_1 + (\underline{D}_2 - \underline{D}_1)(\underline{B}_1 + \underline{B}_2)^{-1}\underline{B}_1 \end{bmatrix} \times \begin{bmatrix} \underline{U}_2 \\ \underline{I}_2 \end{bmatrix} \tag{4.33}$$

In this case, $\underline{B}_1$ and $\underline{B}_2$ are assumed to be nonzero. In case of matrix parameters, $\underline{\boldsymbol{B}}_1$ and $\underline{\boldsymbol{B}}_2$ are assumed to be both nonzero and invertible.

According to Table 4.2, S-parameters are most easily calculated in terms of other parameters; thus, calculating complex models for cascaded or parallel two-port equivalents using S-parameters are not recommended in this thesis. Instead, Y-parameters or ABCD-parameters are used, and the equivalent network is converted. Appendix F contains a conversion table for Z, Y, ABCD, and S-parameters.

3.3 Multi-port Networks

Another important factor to take into consideration is the modelling of multi-port circuits, that is, network components with multiple ports as for instance, a power system substation with N interconnected lines. Then there is a requirement to define the interaction between input and output on all ports, that is, input and output current and voltage on all lines connected to a substation.

Even though ABCD-parameters are a simple set of equations relating current and voltage at the input of an N-port network to the current and voltage at the output of the network, they seem like S-parameters. However, they are conceptually and mathematically different. S-parameters are a type of transfer function with a particular physical meaning [4.12], [4.13], which will be explained later in Section 4 of this chapter. The same can not be said for the application of Z, Y, and ABCD-parameters [4.6], [4.15], [4.17], [4.19], which instead show how voltage and/or current are transformed between each other in a network.

For an N-port network, multiple transfer functions that define signal transfer between each pair of ports is necessary [4.18]. This is not the case for S-parameters as they possesses the characteristics of signal reflection and therefore directly describes the magnitude of transferred and reflected voltage at each port and through the N-port network [4.12], [4.13].

3.4 Summary on Model Parameters

In conclusion, the objective of this thesis is to explain the propagation of harmonics in an interconnected (meshed) grid made up of many line sections of various types (either OHL or UGC). Harmonic voltages are standing wave patterns formed by incident and reflected voltage waves, as well as the reflection caused by line type transition. Therefore, the most appropriate type of parameters for analysing this objective is S-parameters, for the following reasons:

- According to Table 4.2, S-parameters possesses the reflections carried by a propagating voltage wave. The parameters Z, Y and ABCD do not.
- For a N-port network, e.g., a power system substation, S-parameters possesses the magnitude of the transferred voltage carried by an incident voltage wave.
- Multiple transfer functions are required to describe the transferred voltage for Z, Y, and ABCD-parameters through an N-port network, whereas S-parameters do not.

4 S-parameter Model Developments

This chapter's Section 3 presented the characteristics of two-port network models and why S-parameters are used in the modelling approach presented in this thesis. This section presents models for individual network components used in the developed propagation model. The procedure begins with a review of some fundamental microwave theory techniques.

A matrix $\underline{\boldsymbol{S}}$ of S-parameters is used for characterisation of microwave network and transmission line structures, according to [4.12], [4.13], [4.22], [4.23]. S-parameters differ from $\underline{\boldsymbol{Z}}$ and $\underline{\boldsymbol{Y}}$ in that they do not use short-circuit or open-circuit conditions to define a linear electrical network, because such extreme cases of terminal conditions are not always achievable at microwave frequencies. Instead, matched loads are used. S-parameters can be used to analyse microwave circuits that use both lumped and distributed element parameters. At microwave frequencies, the forward and reflected power waves, as well as voltage waves, can be measured, and the S-parameters can be determined.

When applied to steady-state electrical signals, S-parameters describe the electrical behaviour of linear electrical networks. The network is considered a 'black box' in this context, possessing interconnected circuit components or grouped elements such as resistors, capacitors, and inductors. Through ports, these components interact with other circuits. Gain, return losses, voltage-standing wave ratios, reflection coefficients, stability margins, and other network properties are typically addressed in microwave circuit analyses [4.12], [4.13], [4.22].

In the context of a power system transmission grid, we can define *scattering* as the effect (i.e., changes in the waveform) on travelling currents and voltages along a transmission line caused by the insertion of a line section into a transmission line or by transmission line terminal conditions. This is distinct from the use in microwave circuits, and our definition corresponds to a wave along a line impinging an impedance that is different from the line's characteristic impedance. This is a reasonable consideration given that the transmission grid is made up of several interconnected transmission lines and line sections, each with its own characteristic impedance.

S-parameters can describe the reaction on all ports in an N-port network when a given port is excited by a periodic signal and they apply to all frequencies [4.12], [4.13], [4.22]. Even though S-parameters are typically used for high frequency networks, they can also be used for power frequency and low-order multiples of this (in this context, harmonic orders up to the 50th harmonic).

Since it was shown in Chapter 2 (in Section 2) that harmonics in a power system can be considered standing waves, the techniques from microwave theory can be used to quantify harmonic currents and voltages in such a system. The approach we will use here differ from the microwave technology, however, instead of considering a power system as a 'black box' and measuring power waves at the ports, we will model the individual transmission lines and connect them to nodes to determine the propagation of voltages throughout the power system.

The concept of S-parameters may be explained by means of Figure 4.6. According to [4.12], [4.13], the concept depicted in this figure has both network ports 1 and 2 terminated with characteristic impedance $\underline{Z}_{01}$ and $\underline{Z}_{02}$, respectively, and sources $\underline{U}_1$ and $\underline{U}_2$. The two-port network is said to be matched with a reference impedance $\underline{Z}_{0,ref}$ if $\underline{Z}_{01} = \underline{Z}_{02} = \underline{Z}_{0,ref}$, i.e., an identical characteristic impedance at both ports. The incident waves $\underline{a}_1$ and $\underline{a}_2$ enter the ports, while the reflected voltage waves $\underline{b}_1$ and $\underline{b}_2$ exit. The volt-

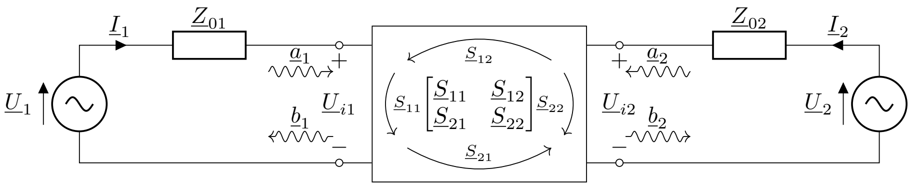


Fig. 4.6: Two-port model of a network with two matched sources described by means of S-parameters (freely after [4.12]).

age at a port, say $\underline{U}_{i1}$ at port 1, can be thought of as the superposition of two opposing waves $\underline{a}_1$ and $\underline{b}_1$. Likewise, the current $\underline{I}_1$ into port 1 can be calculated as the superposition of two opposing waves. In this case, focus will be on the voltage waves.

S-parameters describe the input-output relationships between ports (recall from Section 3) and the first number in the subscript indicates the responding port, while the second indicates the incident port. As a result, $\underline{S}_{21}$ represents the response at port 2 due to a signal at port 1. The arrows within the two-port diagram in Figure 4.6 indicate this. Assuming that each port is terminated in the reference impedance $\underline{Z}_{0,ref}$, the four S-parameters for a two-port are [4.12], [4.13]:

$$\underline{S}_{11} := \frac{\underline{u}_1^-}{\underline{u}_1^+} = \frac{\underline{b}_1}{\underline{a}_1} \tag{4.34}$$

$$\underline{S}_{12} := \frac{\underline{u}_1^-}{\underline{u}_2^+} = \frac{\underline{b}_1}{\underline{a}_2} \tag{4.35}$$

$$\underline{S}_{21} := \frac{\underline{u}_2^-}{\underline{u}_1^+} = \frac{\underline{b}_2}{\underline{a}_1} \tag{4.36}$$

$$\underline{S}_{22} := \frac{\underline{u}_2^-}{\underline{u}_2^+} = \frac{\underline{b}_2}{\underline{a}_2}. \tag{4.37}$$

Here, the designations $\underline{u}_i^+$ and $\underline{u}_i^-$ equals the designations $\underline{a}_i$ and $\underline{b}_i$, respectively, as used in Figure 4.6. We will use the terms $\underline{u}_i^j$ and $\underline{a}_i$ or $\underline{b}_i$ interchangeably, but in all cases we will consider the variables as a voltage. The terms $\underline{a}_i$ and $\underline{b}_i$ are also known as power waves, first defined by [4.23], but in this context we will use the terms for voltage waves and care must be taken not to confuse these with power waves.

An important aspect of S-parameters to understand is that the relative reflection coefficient (please refer to Section 7) at port n is not equal to $\underline{S}_{nn}$ unless all other ports are matched. Likewise, unless all other ports are matched, the transmission coefficient from port m to port n is not equal to $\underline{S}_{nm}$. The S-parameters for a network only characterise network properties (assuming a linear network) and are defined assuming that all ports are matched. Changes in a network's termination or excitation do not affect the scatter parameters of the component itself, but they may affect the relative reflection coefficient seen at a given port. More on this topic follows later in Section 7.

As there is a linear relation between $\underline{b}_i$ and $\underline{a}_i$ we may write the relation in matrix form:

$$\begin{bmatrix} \underline{b}_1 \\ \underline{b}_2 \end{bmatrix} = \begin{bmatrix} \underline{S}_{11} & \underline{S}_{12} \\ \underline{S}_{21} & \underline{S}_{22} \end{bmatrix} \begin{bmatrix} \underline{a}_1 \\ \underline{a}_2 \end{bmatrix} \tag{4.38}$$

According to [4.12], [4.13], the S-parameters of this two-port network may be interpreted as follows:

- $\underline{S}_{11}$ is the input port voltage reflection coefficient
- $\underline{S}_{12}$ is the reverse voltage gain
- $\underline{S}_{21}$ is the forward voltage gain
- $\underline{S}_{22}$ is the output port voltage reflection coefficient

The input reflection coefficient $\underline{S}_{11}$ and output reflection coefficient $\underline{S}_{22}$ describe how much of a wave is reflected by the transmission medium's impedance discontinuity. It equals the ratio of the amplitude of the reflected wave to the incident wave (equations (4.34) and (4.37)). The transmission coefficients, or voltage gain parameters $\underline{S}_{12}$ and $\underline{S}_{21}$, describe the linear ratio of the output reflected wave divided by the input incident wave (equations (4.35) and (4.36)). The scalar value of the parameters is less than one for lossy networks and greater than one for active networks [4.13], [4.22].

The transmission and reflection coefficients for the N-port network depicted in Figure 4.7 can be determined as for the two-port model. The voltage entering the ith port is given in terms of the incident (forward) voltage wave $\underline{a}_i$ and the voltage wave leaving the jth port is given in terms of the reflected (backward) voltage wave $\underline{b}_j$. A part of the voltage entering port i is reflected back at the ith port itself, that is, $\underline{b}_i$ and remaining voltage entering the network exits from the other ports as $\{\underline{b}_1, \ldots, \underline{b}_{i-1}, \underline{b}_{i+1}, \ldots, \underline{b}_N\}$. The outgoing voltage at any port is a linear combination of the transmitted voltage from all other ports. Therefore, the incident and reflected voltages are correlated as follows [4.12]:

$$\begin{bmatrix} \underline{b}_1 \\ \underline{b}_2 \\ \vdots \\ \underline{b}_N \end{bmatrix} = \begin{bmatrix} \underline{S}_{11} & \underline{S}_{12} & \cdots & \underline{S}_{1N} \\ \underline{S}_{21} & \underline{S}_{22} & \cdots & \underline{S}_{2N} \\ \vdots & \vdots & \vdots & \vdots \\ \underline{S}_{N1} & \underline{S}_{N2} & \cdots & \underline{S}_{NN} \end{bmatrix} \begin{bmatrix} \underline{a}_1 \\ \underline{a}_2 \\ \vdots \\ \underline{a}_N \end{bmatrix} \tag{4.39}$$

In the following, network theory is applied assuming a *LTI-system*, which implies [4.12], [4.13]:

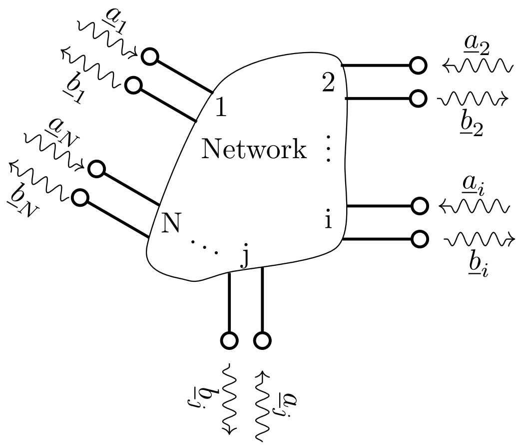


Fig. 4.7: N-port network with incident and reflected waves $\underline{a}_n$ and $\underline{b}_n$.

- **Linear:**
  - If the input increases by x, the output also increases by x
  - Superposition applies: $a \cdot H\{x_1(t)\} + b \cdot H\{x_2(t)\} = H\{a \cdot x_1(t) + b \cdot x_2(t)\}$
- **Causal:**
  - Response in time after excitation only
  - No spontaneous response
- **Time invariant:**
  - The response remains constant regardless of when the system is excited

Because the focus is on propagation patterns of steady-state harmonics due to passive grid changes, it is reasonable to assume the system is an LTI-system.

4.1 Reference Impedance

The characteristic impedance $\underline{Z}_0$ defines the S-parameters. Because $\underline{Z}_0$ depends on the individual line types in a power system, the definition of the specific S-parameters cannot be used directly because the incident and reflected waves are not continuously in the boundary between two lines or line sections unless they possess a common reference impedance.

According to [4.7], [4.8], where a traditional approach uses characteristic impedances of transmission lines and not S-parameters, converting to an impedance at the transition node is required to determine the incident wave and reflected wave in another medium in the following way:

1. Determine the incident wave at the start position of a transmission line segment (using (4.25), for instance, or a 1 per-unit value).
2. Calculate the magnitude and direction of the incident wave at the end of the transmission line before entering the subsequent transmission line segment.
3. Calculate the apparent impedance at the end of the line.
4. Convert the impedance using the impedance base of the subsequent transmission line segment.
5. Calculate a new incident wave magnitude and direction at the beginning of the subsequent transmission line.
6. Calculate the magnitude and direction of the incident wave at the end of the subsequent transmission line.
7. Repeat items 3-6 for each transmission line segment.

Because this is not an appropriate procedure for a large, meshed network, S-parameters are used. A reference impedance $\underline{Z}_{0,ref}$ is chosen and the S-parameters are converted to the same reference. Table 4.3 shows the calculated characteristic impedance at 550 Hz using (4.13) for the line types that were given in Tables 1.3 and 1.4.

Table 4.3: Calculated characteristic impedances at 550 Hz for the line types given in Tables 1.3 and 1.4. The percentage difference in impedances when compared to the line types O1 and U4 is shown. Highlighted OHL type is chosen as reference impedance $Z_{0,ref}$.

ID	$\underline{Z}_0$ [Ω]	$\Delta\underline{Z}_{0,\%}$	ID	$\underline{Z}_0$ [Ω]	$\Delta\underline{Z}_{0,\%}$	ID	$\underline{Z}_0$ [Ω]	$\Delta\underline{Z}_{0,\%}$
				OHL				
O1	295.78	-	O9	341.02	15.3	O17	387.63	31.1
O2	295.96	0.1	O10	355.65	20.2	O18	302.93	2.4
O3	274.48	7.2	O11	271.64	8.2	O19	245.33	17.1
O4	298.11	0.8	O12	271.75	8.1	O20	282.78	4.4
O5	293.05	0.9	O13	271.93	8.1	O21	274.70	7.1
O6	292.31	1.2	O14	404.04	36.6	O22	271.61	8.2
O7	271.78	8.1	O15	299.90	1.4	O23	297.71	0.7
O8	247.90	16.2	O16	271.74	8.1			
				UGC				
U1	59.49	3.7	U4	63.63	-	U7	46.73	18.6
U2	55.72	2.9	U5	57.38	10.9			
U3	63.63	10.9	U6	61.58	7.3			

The percentage difference with regard to the most often used line type is determined to analyse the difference between the values of characteristic impedances for each type. In this case, it is type O1 for OHL (276.8 km) and type U4 for UGC (42.6 km), according to Tables 1.3 and 1.4. The most commonly used types in the grid are chosen because one of these types should also be chosen as the reference impedance for the S-parameters, and thus, the change in wave propagation can be assessed in relation to the most dominant line type. Recommendations for the final choice follow below.

The table shows that the percentage difference in the characteristic impedance of OHL compared to the impedance of the first OHL type O1 ranges from 0.1 % to 36.6 %. The difference is less than 8.2 % for most overhead line types and significantly greater for six of the 23 different types (15.3 % − 36.6 %). For UGC, the difference to the characteristic impedance of the UGC type U4 is between 2.9 % − 10.9 %, but for a single cable type U7, the difference is significantly higher (18.6 %).

For both OHL and UGC, the higher the difference between the characteristic impedance of a line type and the selected reference impedance, the greater the reflection of the incident voltage wave. The significance of this distinction will be discussed in Section 7.7 of this chapter, where the emphasis will be on the UGC type.

For example, if $\underline{Z}_{0,ref}$ is chosen as an OHL's characteristic impedance, the effect of all other line types on wave propagation can be assessed and compared to this type. This thesis includes the following recommendations for selecting $\underline{Z}_{0,ref}$:

- If wave propagation is to be evaluated in a network with both UGC and OHL and where the grid consists mostly of OHL, $\underline{Z}_{0,ref}$ should be chosen as an OHL type of characteristic impedance. The value for the most commonly used type should be chosen; alternatively, a value should be chosen from among the types where the difference in the characteristic impedance is the least.

- If wave propagation is to be evaluated in a network with both UGC and OHL and where the grid consists mostly of UGC, $\underline{Z}_{0,ref}$ should be chosen as a UGC type of characteristic impedance. The value for the most commonly used type should be chosen; alternatively, a value should be chosen from among the types where the difference in the characteristic impedance is the least.

- If wave propagation is to be evaluated in a network that is either a network consisting only of UGC or a network consisting only of OHL, $\underline{Z}_{0,ref}$ should be chosen as the most commonly used type; alternatively, a value should be chosen from the types where the difference in the characteristic impedance is the smallest.

In this study, the meshed grid consists of both OHL and UGC. The most common OHL type is O1, and this type is therefore chosen. The characteristic impedance value is highlighted in Table 4.3.

4.2 Determining S-parameters

The model elements for transmission lines, sources, loads, and junctions in the propagation model are presented in reference [4.24] (paper **J1** of this thesis), but their derivation are not fully explained. These two-port models are discussed in this section using numerical examples to help understand their physical interpretation.

Transmission Line Model

Two-port S-parameters are derived from Z, Y, or ABCD-parameters, as described in Section 3 in this chapter, with the latter being the preferred model due to its cascading ability. A numerical example is given in the following using ABCD-parameters and S-parameters, and the interpretation of the results is discussed.

The line parameters used for the benchmark system were listed in Table B.1, and the same parameters are used in this numerical example, which examines three different line configurations. The ABCD-parameters for 90 km of OHL and 90 km of UGC are calculated using (E.11), as are the parameters of a 90 km combined line consisting of 50 km OHL, followed by 10 km UGC, and lastly followed by additional 30 km OHL. Table 4.4 presents the calculated ABCD-parameters.

Table 4.4: Calculated values of ABCD-parameters for 90 km of OHL, 90 km of UGC and a combined line consisting of 50 km OHL, followed by 10 km UGC and lastly, 30 km OHL.

Line type	Parameters			
	$\underline{A}$	$\underline{B}$	$\underline{C}$	$\underline{D}$
OHL	$-0.35 + 0i$	$0.14 + 279.88i$	$0 + 0i$	$-0.35 + 0i$
UGC	$0.95 + 0.01i$	$0.99 + 0.17i$	$0 + 0.01i$	$0.95 + 0.01i$
Combined	$-2.78 + 0.009i$	$-0.01 - 1.20i$	$-0.005 + 2.11i$	$-1.27 + 0.003i$

For transmission line models, the two-port model presented in Section 3 is used, and the ABCD-parameters presented in Table 4.4 are easily used to determine the S-parameters, as the relationship between ABCD-parameters and S-parameters is given in a conversion table in Appendix F. The equations for parameter conversion are:

$$\underline{S}_{11} = \frac{\underline{A} + \underline{B} \cdot \underline{Y}_0 - \underline{C} \cdot \underline{Z}_0 - \underline{D}}{\underline{A} + \underline{B} \cdot \underline{Y}_0 + \underline{C} \cdot \underline{Z}_0 + \underline{D}} \tag{4.40}$$

$$\underline{S}_{12} = \frac{2(\underline{A}\underline{D} - \underline{B}\underline{C})}{\underline{A} + \underline{B} \cdot \underline{Y}_0 + \underline{C} \cdot \underline{Z}_0 + \underline{D}} \tag{4.41}$$

$$\underline{S}_{21} = \frac{2}{\underline{A} + \underline{B} \cdot \underline{Y}_0 + \underline{C} \cdot \underline{Z}_0 + \underline{D}} \tag{4.42}$$

$$\underline{S}_{22} = \frac{-\underline{A} + \underline{B} \cdot \underline{Y}_0 - \underline{C} \cdot \underline{Z}_0 - \underline{D}}{\underline{A} + \underline{B} \cdot \underline{Y}_0 + \underline{C} \cdot \underline{Z}_0 + \underline{D}} \tag{4.43}$$

The characteristic impedance $\underline{Z}_0$ and characteristic admittance $\underline{Y}_0$ are in this case related to the reference impedance $\underline{Z}_{0,ref}$ as explained in section 4.1, whereby the S-parameters are converted to the same impedance reference plane. The S-parameters for OHL and UGC, as well as the combined line, are calculated using (4.40) - (4.43) and the calculated ABCD-parameter values shown in Table 4.4. The calculated S-parameters are shown in Table 4.5.

Table 4.5: Calculated S-parameters for OHL, UGC and the combined line type in this numerical example.

Line type	Parameters			
	$\underline{S}_{11}$	$\underline{S}_{12}$	$\underline{S}_{21}$	$\underline{S}_{22}$
OHL	$0 + 0i$	$-0.35 + 0.93i$	$-0.35 + 0.93i$	$0 + 0i$
UGC	$-0.45 + 0.44i$	$0.55 + 0.50i$	$0.55 + 0.50i$	$-0.45 + 0.44i$
Combined	$0.18 - 0.85i$	$-0.47 + 0.10i$	$-0.47 + 0.10i$	$-0.53 - 0.70i$

From this table, the following important observations can be made:

- The transmission line will be reciprocal if it is passive and made of isotropic materials, as is the case for power system transmission lines, so that: $\underline{S}_{21} = \underline{S}_{12}$. This means that a wave leaving at the far end of the line propagated from the near end will be equal in magnitude and direction as if the same type of wave propagates from the far end and leaves at the near end.

- Similarly, if the impedance as seen from each end of the line is matched, then $S_{11} = S_{22} = 0$ and no reflection occur. This is seen to be true for the OHL parameters given in Table 4.5, because $\underline{Z}_{0,ref}$ is chosen to be equal to the OHL characteristic impedance. Given the same reference impedance, it is clear that for UGC, the reflection coefficients $S_{11} = S_{22} \neq 0$ and reflections will occur.

- For the combined line consisting of 50 km of OHL followed by 10 km of UGC and again followed by 30 km of OHL it can be seen that $S_{11} \neq S_{22}$.

> This means that the reflection of an incident wave impinging one end of the transmission line is not equal to the reflection of the same type of incident wave impinging the other end of the line. This mismatch is caused by the different transmission media used along the line and the fact that the OHL sections are not of the same length and/or type.

Due to directionality properties (as from Table 4.2), the properties discussed above using this numerical example show that the line types used, as well as the direction of the propagated voltages in the grid, have a significant impact on voltage propagation in a meshed system.

Voltage Source Model

Nonlinear devices connected to the power system are harmonic distortion sources and modelling these sources, according to [4.5], [4.6], [4.25], is complex and involves many trade-offs, as the data basis for modelling may be insufficient. Furthermore, representing all sources in an electricity system is a time-consuming task if all operating scenarios are to be considered [4.26]. Section 3 of Chapter 2 discussed the complexities of the influence of harmonic sources.

The objective of this research is to determine how the insertion of UGC sections affects harmonic voltage propagation in the meshed passive transmission network, as well as to predict how network configuration changes affect the voltage propagation. Because this is not a harmonic penetration analysis, harmonic sources do not need to be highly detailed.

Because the propagation model is based on N-port models with S-parameters, a harmonic voltage source is modelled as a two-port model according to [4.24] as:

$$\begin{bmatrix} \underline{b}_1 \\ \underline{b}_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & \underline{\Gamma}_S \end{bmatrix} \begin{bmatrix} \underline{a}_1 \\ \underline{a}_2 \end{bmatrix} \tag{4.44}$$

where the Thévenin voltage is applied to port 1 (i.e., $\underline{a}_1$) and exits the two-port at $\underline{b}_2$, and the reflection at port 2, determined by the relationship between the network's characteristic impedance and the Thevenin impedance, is known as the complex reflection coefficient $\underline{\Gamma}_S$. Figure 4.8 shows an illustration of this representation.

In this model, $\underline{Z}_{0,th}$ is the Thévenin source's characteristic impedance, $\underline{U}_{th}$ is the Thévenin voltage, and $\underline{I}_{th}$ is current at the source terminal. The grid to which the source is connected is represented by the characteristic impedance $\underline{Z}_0$. The incident wave $\underline{a}_2$ and the reflected wave $\underline{b}_2$, which is the wave that the source transmits into the network, are located at the terminals. The waves $\underline{a}_1$ and $\underline{b}_1$ represent the applied voltage and the reflected voltage generated by the source, respectively.

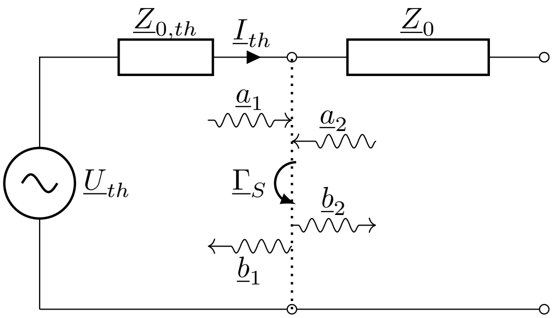


Fig. 4.8: Thévenin equivalent model for propagation model development.

The applied voltage $\underline{a}_1$ is transmitted directly out as the wave $\underline{b}_2$ as shown in (4.44), and the reflection caused by the source itself is zero. The reflection coefficient $\underline{\Gamma}_S$ determines how much of the incident wave $\underline{a}_2$ from the grid is reflected and returned to the grid at the source terminals.

The question for a propagation study is thus what the magnitude of $\underline{a}_1$ should be to represent the voltage contribution from the source and what the magnitude of $\underline{\Gamma}_S$ should be to represent the operating scenario to be analysed. Table 4.6 provides a guideline for parameter value selection.

Table 4.6: Source parameter selection guidelines for propagation models.

Parameter	Comment
$\underline{a}_1 = 1$	Setting the source voltage to 1 per-unit causes the propagation model output to be evaluated against a common source reference voltage, allowing the impact of different network changes in the passive grid to be evaluated against each other while decoupling the impact of the source magnitude on voltage propagation. For the propagation study that follows in Section 7 in this chapter, a source magnitude of 1 per-unit is used.
$\underline{a}_1 = K\angle\theta$	Setting the source voltage to a defined magnitude K and an arbitrary direction θ causes the propagation model output to represent a specific source contribution for a specific operation scenario. In this case, the impact of different network changes in the passive grid can be evaluated for the specified operation scenario. For a meshed system with multiple sources, the propagation of voltages in the grid due to interaction between sources can also be evaluated by changing the magnitude K and/or the direction θ of the sources. In this case, K and θ could be determined by means of measurements from the power system or by simulation. Using harmonic sources with a specific magnitude K and a specific direction θ has been demonstrated in [4.24] as part of the model validation approach, which will be discussed in the following Section 6.

Continued on next page

Table 4.6 – continued from previous page

Parameter	Comment
$\underline{\Gamma}_S = 0$	When the reflection coefficient is zero (matched load termination), no reflection occurs at the source terminal, and the source contribution is $\underline{b}_2 = \underline{a}_1$.
$\underline{\Gamma}_S = 1$	When the reflection coefficient is one (open-end termination), full reflection occurs at the source terminal, and the source contribution is $\underline{b}_2 = \underline{a}_1 + \underline{a}_2$.
$\underline{\Gamma}_S = -1$	When the reflection coefficient is minus one (short circuit termination), negative reflection occurs at the source terminal, and the source contribution is $\underline{b}_2 = \underline{a}_1 - \underline{a}_2$. For the propagation study that follows in Section 7 in this thesis, a magnitude of minus one is used for $\underline{\Gamma}_S$.
$\underline{\Gamma}_S = K\angle\theta$	Setting the reflection coefficient to a fixed magnitude K and a fixed direction θ it is possible to evaluate the impact of the source impedance on the voltage propagation. In this thesis, no harmonic interaction between non-linear devices and the network is assumed, thus providing a relatively simple analytical method because the objective is on passive grid behaviour and not source interactions. When using harmonic penetration solution methods, the same assumptions apply [4.5], [4.25]. Phenomena like cross-coupling between the DC and AC sides of a converter, as well as subsequent frequency coupling introduced by converters, are ignored. Additionally, it implies that the harmonic impedance of the converter and harmonic injection are presumed to be independent of changes that might take place in the operating point of the converter or in the power system itself, such as changes in background distortion level. In some cases, the converter's dependence on the operating point must be considered in the frequency domain for both penetration studies and frequency scans. This can be accomplished using look-up tables or, more precisely, EMT tools. In relation to the method presented in this thesis, a look-up table with specific values of $\underline{\Gamma}_S$ for defined operation scenarios could be implemented, but this is not demonstrated in this research. The source contribution is $\underline{b}_2 = \underline{a}_1 + \underline{a}_2 K\angle\theta$ when specific values for $\underline{\Gamma}_S$ are defined.

The objective behind using scatter parameters is to focus on the propagation of either voltage waves or power waves. The focus in this case has been on voltage propagation, with harmonic voltage being the main concern, partly because the IEC 61000-3-6 standard uses voltage planning thresholds and partly because harmonic voltage is the quantifiable metric utilised by the TSO.

It is feasible to use current source types; however, one will need to convert the current source to a Thévenin voltage source first. Methods for this can be found in the literature [4.19]. Junction models are based on voltage, and the transmission coefficient for currents will be different, which will be seen later in this section.

Load and Transformer Models

Harmonic loads are not considered in this study because the objective is to determine how the insertion of UGC sections affects harmonic voltage propagation in the meshed passive transmission network, as well as to predict how network configuration changes affect voltage propagation.

If harmonic loads were to be modelled, the approach discussed for harmonic sources could be applied. In this case, loads should be modelled as a two-port model according to [4.24]:

$$\begin{bmatrix} \underline{b}_1 \\ \underline{b}_2 \end{bmatrix} = \begin{bmatrix} \underline{\Gamma}_L & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \underline{a}_1 \\ \underline{a}_2 \end{bmatrix} \tag{4.45}$$

where the load reflection coefficient is $\underline{\Gamma}_L$ is the only non-zero parameter. The load model can easily be expanded with transmission coefficients if the wave behaviour behind the load is to be considered, as discussed for the source modelling.

In the case of a transformer, the same modelling method can be used. In this case, transmission coefficients in both directions of the two-port model must be taken into account because of the transfer between the two windings of the transformer. However, this is not dealt with in this thesis.

Table 4.7 provides a guideline for parameter value selection, but no specific modelling details are provided.

Table 4.7: Load parameter selection guideline for propagation models.

Parameter	Comment
$\underline{\Gamma}_L = 0$	When the reflection coefficient is zero, no reflection occurs at the load terminal, and the load contribution is $\underline{b}_1 = 0$.
$\underline{\Gamma}_L = 1$	When the reflection coefficient is one, full reflection occurs at the load terminal, and the load contribution is $\underline{b}_1 = \underline{a}_1$.
$\underline{\Gamma}_L = -1$	When the reflection coefficient is minus one, negative reflection occurs at the load terminal, and the load contribution is $\underline{b}_1 = -\underline{a}_1$.
$\underline{\Gamma}_L = K\angle\theta$	Setting the reflection coefficient to a fixed magnitude K and a fixed direction θ, it is possible to evaluate the impact of the load impedance on the voltage propagation. The contribution is then $\underline{b}_1 = \underline{a}_1 K\angle\theta$ when specific values for $\underline{\Gamma}_L$ are defined.

Junction Models

Where several lines meet, for instance at a power system substation with an incoming feeder and several outgoing feeders, the incident waves will normally be transmitted from the incoming feeder to the outgoing feeders.

First, the conditions for a simple junction with two feeders are explained. Suppose that a uniform loss-free line of characteristic impedance $\underline{Z}_i$ is joined to a similar line of characteristic impedance $\underline{Z}_t$ as shown in Figure 4.9, and that waves of voltage and current $\underline{u}_i$ and $\underline{i}_i$ are incident on the junction from the first line.

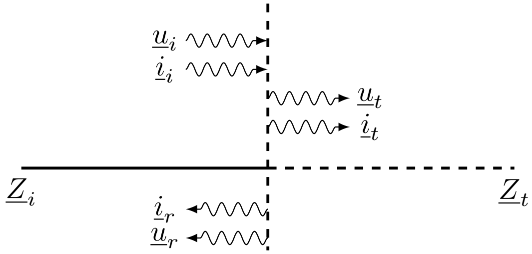


Fig. 4.9: Junction with two feeders. Direction of incident and reflected waves as well as transmitted waves are shown.

The waves $\underline{u}_t$ and $\underline{i}_t$ are transmitted into the second line, and waves $\underline{u}_r$ and $\underline{i}_r$ are reflected back along the first line. The arrows denote the direction of propagation of the waves. Regarding the current flow, the positive direction will be taken as from $\underline{Z}_i$ to $\underline{Z}_t$ for all waves. Similarly, voltages will be regarded as positive if tending to cause current flow in this direction. Using (4.18), the complex reflection coefficient at the junction is:

$$\underline{u}_r = \frac{\underline{Z}_t - \underline{Z}_i}{\underline{Z}_t + \underline{Z}_i}\underline{u}_i = \underline{\Gamma}\underline{u}_i \tag{4.46}$$

$$\underline{i}_r = \frac{\underline{Z}_i - \underline{Z}_t}{\underline{Z}_i + \underline{Z}_t}\underline{i}_i = -\underline{\Gamma}\underline{i}_i. \tag{4.47}$$

In a similar way, the complex transmission coefficient at the junction is:

$$\underline{u}_t = \frac{2\underline{Z}_t}{\underline{Z}_t + \underline{Z}_i}\underline{u}_i = (1+\underline{\Gamma})\underline{u}_i \tag{4.48}$$

$$\underline{i}_t = \frac{2}{\underline{Z}_t + \underline{Z}_i}\underline{i}_i = (1-\underline{\Gamma})\underline{i}_i. \tag{4.49}$$

If $\underline{Z}_i = \underline{Z}_t$, the reflection coefficient $\underline{\Gamma}$ in (4.46) - (4.49) will be zero (no reflection as the transmission media is equal on both sides of the junction), and (4.46) - (4.49) reduces to:

$$\underline{u}_r = 0 \cdot \underline{u}_i, \quad \underline{i}_r = -0 \cdot \underline{i}_i \tag{4.50}$$

$$\underline{u}_t = 1 \cdot \underline{u}_i, \quad \underline{i}_t = 1 \cdot \underline{i}_i. \tag{4.51}$$

Using the relations $\underline{u}_i = \underline{a}_1$ and $\underline{u}_r = \underline{b}_1$ and, as seen from $\underline{Z}_t$ to $\underline{Z}_i$, $\underline{u}_t = \underline{a}_2$ and defining a reflected wave $\underline{u}_{tr} = \underline{b}_2$, the following S-parameter model for a junction with two feeders can be deduced:

$$\begin{bmatrix} \underline{b}_1 \\ \underline{b}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \underline{a}_1 \\ \underline{a}_2 \end{bmatrix}. \tag{4.52}$$

It can be seen that (4.52) leads to full transmission from $\underline{a}$ to $\underline{b}$ and no reflection at either port. The two-feeder model may be expanded to a substation model with N feeders. Now, suppose that a uniform loss-free line of characteristic impedance $\underline{Z}_i$ is joined to N similar lines $\{\underline{Z}_{t1}, \underline{Z}_{t2}, \ldots, \underline{Z}_N\}$ as shown in Figure 4.10, and that waves of voltage and current $\underline{u}_i$ and $\underline{i}_i$ are incident on the junction from the first line. The voltage at the junction must remain equal

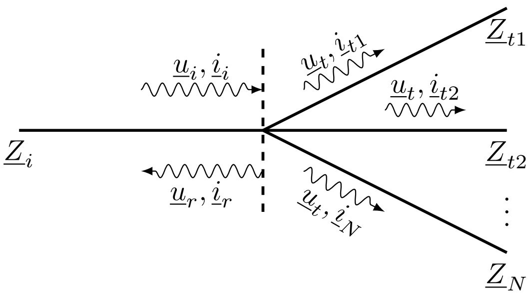


Fig. 4.10: Junction with N feeders. Direction of incident and reflected waves as well as transmitted waves are shown.

for all feeders. Assume waves $\underline{u}_t$ and $\{\underline{i}_{t1}, \underline{i}_{t2}, \ldots, \underline{i}_N\}$ are transmitted into the subsequent lines, and waves $\underline{u}_r$ and $\underline{i}_r$ are reflected back along the first line. The reflection coefficient at the junction is then:

$$\underline{u}_r = \frac{\underline{Z}_i/(n-1) - \underline{Z}_i}{\underline{Z}_i/(n-1) + \underline{Z}_i} \underline{u}_i = -\frac{n-2}{n} \underline{u}_i \tag{4.53}$$

where n is the number of lines. This is, however, only true when $\{\underline{Z}_i = \underline{Z}_{t1} = \underline{Z}_{t2} = \ldots = \underline{Z}_N\}$. In a similar way, the transmission coefficient at the junction is:

$$\underline{u}_t = \frac{2\underline{Z}_i/(n-1)}{\underline{Z}_i/(n-1) + \underline{Z}_i} = \frac{2}{n} \underline{u}_i \tag{4.54}$$

$$\underline{i}_t = \frac{2\underline{Z}_i}{\underline{Z}_i/(n-1) + \underline{Z}_i} \underline{i}_i = \frac{2(n-1)}{n} \underline{i}_i. \tag{4.55}$$

For a junction with one infeed and two outgoing feeders ($n = 3$), (4.53) -

(4.55) can be calculated as:

$$\underline{u}_r = -\frac{3-2}{3}\underline{u}_i = -\frac{1}{3}\underline{u}_i \tag{4.56}$$

$$\underline{u}_t = \frac{2}{3}\underline{u}_i \tag{4.57}$$

$$\underline{i}_t = \frac{2\cdot(3-1)}{3}\underline{i}_i = \frac{4}{3}\underline{i}_i. \tag{4.58}$$

Using the voltage waves, the following S-parameter model for a junction with three feeders can be deduced:

$$\begin{bmatrix}\underline{b}_1\\\underline{b}_2\\\underline{b}_3\end{bmatrix} = \begin{bmatrix}-1/3 & 2/3 & 2/3\\ 2/3 & -1/3 & 2/3\\ 2/3 & 2/3 & -1/3\end{bmatrix}\begin{bmatrix}\underline{a}_1\\\underline{a}_2\\\underline{a}_3\end{bmatrix}. \tag{4.59}$$

It must be emphasized that (4.59) is valid for voltages only, as the coefficient for transmitted voltage and current are not equal, as seen in (4.54) and (4.55), respectively.

The derivation and discussion of the N-port model for a power system substation revealed that propagating voltages are always reflected at the substation busbar and that the transferred voltage from one line to all other lines is always less than the incident wave magnitude. The greater the number of lines connected to a substation, the less voltage is transferred to adjacent lines.

4.3 Summary on S-parameter Models

The characteristics of S-parameter models for use with the propagation model have been discussed in this section. The emphasis has been on models for transmission lines, sources, and junctions, with a brief discussion of load and transformer models. The individual parameters are converted to a common reference impedance, which in this case can be the characteristic impedance of a specific transmission line type, allowing the individual models to be interconnected in a network and thus evaluating the propagated voltage in this network.

For meshed grids, several lines are typically connected to a substation. The greater the number of lines connected to a substation, the less voltage is transferred from one line to adjacent lines, reducing the possible voltage propagation from the substation.

5 The Propagation Model Development

In this section, the development of a propagation model for determining propagation of voltages in a complete network will be set up. The term *"propagation model"* will be used to describe the mathematical model set forth here to determine the propagation of harmonic voltages in a meshed system consisting of several high voltage lines and substations. The goal is to determine the response of any node when a harmonic voltage is applied to any one of the nodes in the grid. In mathematical terms, the goal is as follows:

$$\overrightarrow{\underline{b}} = \underline{\boldsymbol{S}} \cdot \overrightarrow{\underline{a}} \tag{4.60}$$

where $\overrightarrow{\underline{b}}$ is the response when the system $\underline{\boldsymbol{S}}$ is excited with the voltage $\overrightarrow{\underline{a}}$.

The development of this mathematical model will be on the basis of the S-parameter models described in the previous Section 4 of this chapter. The procedure for developing a propagation model is originally given in reference [4.24] (paper **J1** of this thesis) and the modelling approach used in this thesis is depicted in Figure 4.12, based on [4.24].

From this flowchart, it is seen that a given system layout is used to create the propagation model. A network model consisting of transmission line sections, sources and loads, as well as N-port junctions, is built based on the system layout, and ports are labelled in incremental order based on the number of ports of the individual components used in the model. A connection matrix $\boldsymbol{C}$ and a partitioned component matrix, or system matrix, $\underline{\boldsymbol{S}}$ are designed to describe the interconnection of network components. The network matrix $\underline{\boldsymbol{M}}$, incident wave $\overrightarrow{\underline{a}}$, and reflected wave $\overrightarrow{\underline{b}}$ are then computed. The results are saved, and if there are multiple sources, $\underline{\boldsymbol{M}}$ is reset to the next source contribution, and new incident and reflected waves are calculated. Finally, the calculated results are prepared for further analysis.

The approach according to the flow chart in Figure 4.12 is discussed in the following subsections, and explained using a generic model example.

5.1 Network Diagram

Figure 4.11 depicts an example of a system that employs the modelling approach described above. Individual network models must be interconnected

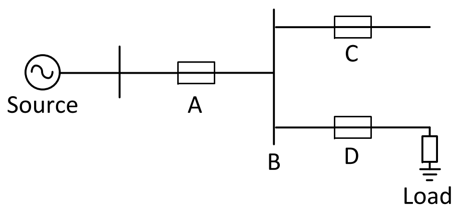


Fig. 4.11: Generic network single line diagram with substation identification and transmission line connections.

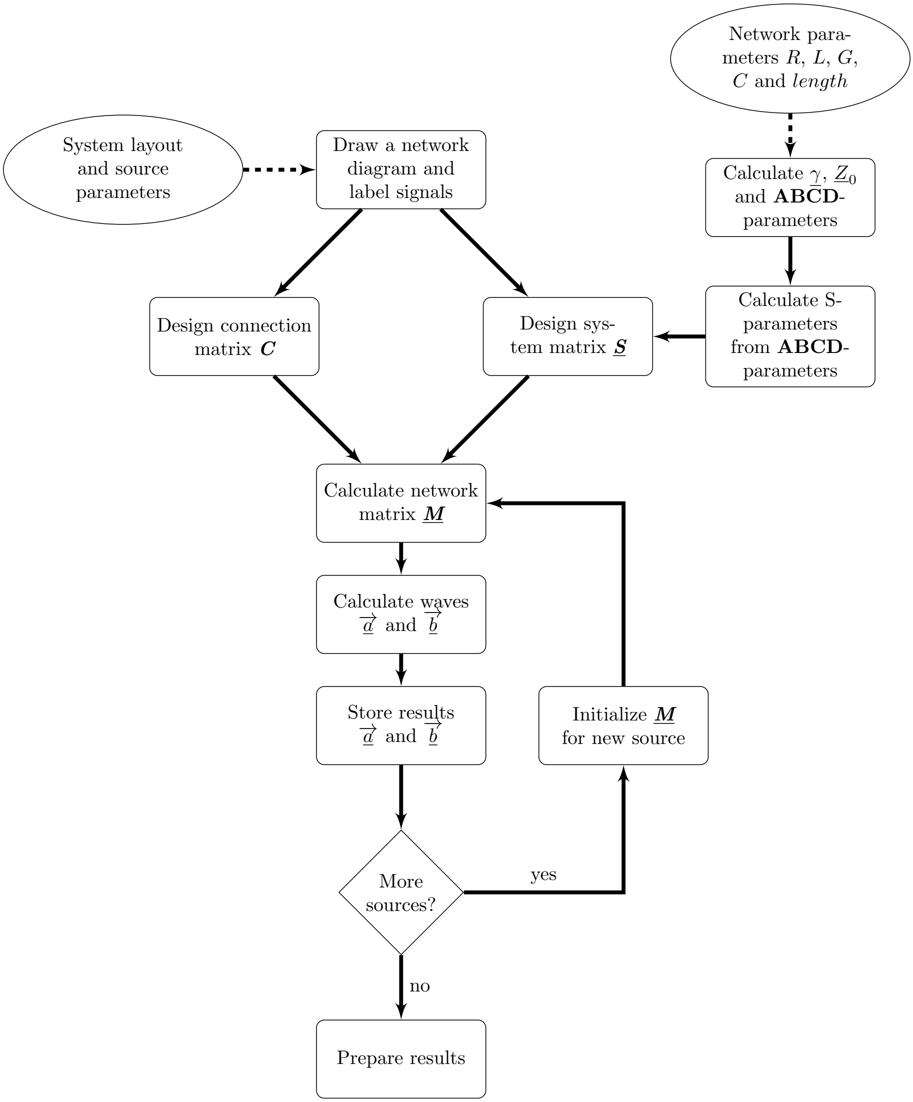


Fig. 4.12: Flow chart for developing the propagation model (freely after [4.24] and adapted to this study).

in the same way as the physical system to transform a system or grid into a propagation model. The individual component models are described in Section 4. The incident and reflected wave at nodes are represented by the N-port model's corresponding node ID.

In Figure 4.11, a source is connected to a bus designated as 'B' through line 'A'. Line 'C' is connected to 'B' and this line has an open-end termination.

Line 'D' is also connected to bus 'B' and this line is terminated in a load.

5.2 Connection Matrix C

Figure 4.13 depicts the corresponding propagation model diagram for the network shown in Figure 4.11. This model consists of 13 nodes, interconnected with five two-port network models and one three-port network model (nodes $= 5 \cdot 2\,\text{ports} + 1 \cdot 3\,\text{ports} = 13$). The source node is in this case labelled as node 2.

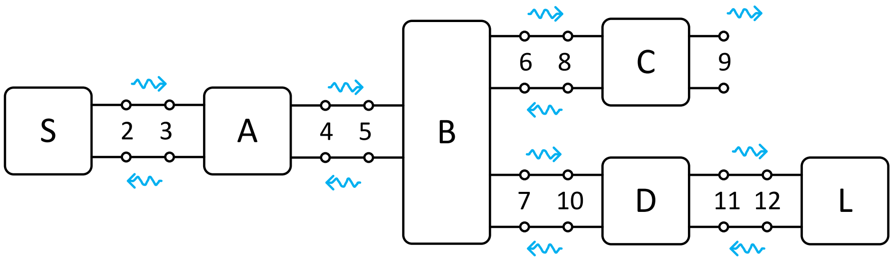


Fig. 4.13: Corresponding propagation model for the generic model depicted in Figure 4.11. Node ID and interconnection of N-port network models are shown (based on [4.24]).

A source S on the far left has the wave $\underline{b}_2$ at the output port. The two-port network 'A' on the left right after the source has input at port 1 (node 3), that is, $\underline{a}_3$ and $\underline{b}_3$, and $\underline{b}_2$ is input to 'A' and the output $\underline{b}_3$ is input to the source on $\underline{a}_2$. Block 'A' has output at port 2 given as $\underline{a}_4$ and $\underline{b}_4$. The response $\underline{b}_4$ from 'A' is input to the three-port network 'B', that is, $\underline{a}_5 = \underline{b}_4$ and the reflected wave $\underline{b}_5$ of block 'B' is input at port 2 of 'A', i.e. $\underline{a}_4 = \underline{b}_5$. The transmitted waves $\underline{b}_6$ and $\underline{b}_7$ at the output ports of 'B' are input to the rightmost networks 'C' and 'D', respectively, that is, $\underline{a}_8 = \underline{b}_6$ and $\underline{a}_{10} = \underline{b}_7$.

The output $\underline{b}_9$, for instance, is the product of the wave $\underline{a}_2$ travelling from the source through the networks 'A', 'B' and 'C' taking the reflections at the ports into consideration. The same type of relationship is true for the output wave $\underline{b}_{11}$ at port 2 of network 'D'. In this case, network 'D' is terminated in a load causing a reflection $\underline{b}_{12}$ which is travelling back to the source through networks 'D', 'B' and 'A'. Some of this wave is also transmitted to network 'C'.

A connection matrix $\boldsymbol{C}$ that contains information how the network blocks are interconnected in the network can be constructed. Mathematically, this can be described as:

$$\overrightarrow{\underline{a}} = \boldsymbol{C}\,\overrightarrow{\underline{b}} \tag{4.61}$$

where $\overrightarrow{\underline{a}}$ is a vector of all incident waves to the blocks and $\overrightarrow{\underline{b}}$ is a vector of all reflected waves. The connection matrix $\boldsymbol{C}$ describes how $\overrightarrow{\underline{a}}$ and $\overrightarrow{\underline{b}}$

are related as shown in Figure 4.13. Because the network has 13 nodes, the connection matrix must be a 13x13 matrix as follows:

$$C = \begin{array}{c}
\begin{array}{ccccccccccccc}
\underline{\mathbf{b}}_1 & \underline{\mathbf{b}}_2 & \underline{\mathbf{b}}_3 & \underline{\mathbf{b}}_4 & \underline{\mathbf{b}}_5 & \underline{\mathbf{b}}_6 & \underline{\mathbf{b}}_7 & \underline{\mathbf{b}}_8 & \underline{\mathbf{b}}_9 & \underline{\mathbf{b}}_{10} & \underline{\mathbf{b}}_{11} & \underline{\mathbf{b}}_{12} & \underline{\mathbf{b}}_{13}
\end{array} \\
\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}
\end{array}
\begin{array}{c}
\\ \underline{\mathbf{a}}_1 \\ \underline{\mathbf{a}}_2 \\ \underline{\mathbf{a}}_3 \\ \underline{\mathbf{a}}_4 \\ \underline{\mathbf{a}}_5 \\ \underline{\mathbf{a}}_6 \\ \underline{\mathbf{a}}_7 \\ \underline{\mathbf{a}}_8 \\ \underline{\mathbf{a}}_9 \\ \underline{\mathbf{a}}_{10} \\ \underline{\mathbf{a}}_{11} \\ \underline{\mathbf{a}}_{12} \\ \underline{\mathbf{a}}_{13}
\end{array} \tag{4.62}$$

The connection between incident and reflected waves is simply denoted by unity. In the second and third columns of $\boldsymbol{C}$, for example, the output $\underline{b}_2$ (of the source 'S') is connected to the input $\underline{a}_3$ (of block 'A') as unity in the third row and the output $\underline{b}_3$ (of block 'A') is connected to the input $\underline{a}_2$ (of the source 'S') as unity in the second row. In (4.62), all connections related to the propagation model diagram in Figure 4.13 are marked with a circle.

5.3 System Matrix S

The individual network models depicted in Figure 4.13 are described by means of the matrix $\underline{S}$ in the following way:

$$\underline{S} = \begin{bmatrix}
\underline{S}_{2x2}^{Source} & 0 & \cdots & \cdots & \cdots & 0 \\
0 & \underline{S}_{2x2}^{A} & \ddots & & & \vdots \\
\vdots & \ddots & \underline{S}_{3x3}^{B} & \ddots & & \vdots \\
\vdots & & \ddots & \underline{S}_{2x2}^{C} & \ddots & \vdots \\
\vdots & & & \ddots & \underline{S}_{2x2}^{D} & 0 \\
0 & \cdots & \cdots & \cdots & 0 & \underline{S}_{2x2}^{Load}
\end{bmatrix}. \tag{4.63}$$

The matrix $\underline{S}$ is also a 13x13 matrix with the individual N-port network S-parameters at the diagonal. It is critical that the individual blocks be arranged in the same order as the node IDs assigned to them by the propagation model. This property is seen in (4.63), where the two-port network model of the source (nodes 1 and 2) is inserted at the first position in the matrix, followed by the transmission line 'A' (nodes 3 and 4), then the three-port network model of 'B' (nodes 5-7), etc., all on the diagonal of the matrix.

Thus, it does not matter whether the model is to be used for a radial or meshed grid, as long as the ports of the individual blocks are numbered consecutively with an ID and placed diagonally in the matrix S. It is the connection matrix C that is used to interconnect the individual ports.

5.4 Network Matrix M

According to [4.24], by inserting (4.60) into (4.61), with $\underline{\boldsymbol{S}}$ as in (4.63), the following expression can be obtained:

$$\overrightarrow{\underline{a}} = \boldsymbol{C}\,\overrightarrow{\underline{b}} = \boldsymbol{C}\underline{\boldsymbol{S}} \cdot \overrightarrow{\underline{a}}. \tag{4.64}$$

It is seen from (4.64) that the input $\overrightarrow{\underline{a}}$ depends on both itself and of its distribution in the network given by $\boldsymbol{C}\underline{\boldsymbol{S}}$. Equation (4.64) can be rewritten to obtain a homogeneous equation:

$$\begin{aligned} \overrightarrow{\underline{a}} &= \boldsymbol{C}\underline{\boldsymbol{S}} \cdot \overrightarrow{\underline{a}} \Rightarrow \\ \overrightarrow{\underline{a}} - \boldsymbol{C}\underline{\boldsymbol{S}} \cdot \overrightarrow{\underline{a}} &= 0 \Rightarrow \\ (\boldsymbol{I} - \boldsymbol{C}\underline{\boldsymbol{S}}) \cdot \overrightarrow{\underline{a}} &= 0 \end{aligned} \tag{4.65}$$

where $\boldsymbol{I}$ is the identity matrix. A network matrix $\underline{\boldsymbol{M}}$ can be defined as:

$$\underline{\boldsymbol{M}} := \boldsymbol{I} - \boldsymbol{C}\underline{\boldsymbol{S}} \tag{4.66}$$

and $\underline{\boldsymbol{M}}$ is the network matrix in the flowchart shown in Figure 4.12.

5.5 Estimating Voltage Propagation

In Figure 4.13, a source is connected to transmission line 'A' and assuming that the incident wave $\underline{a}_1$ is known, the remaining incident waves can be solved after removing the first row in $\underline{\boldsymbol{M}}$, yielding the expression:

$$\underline{\boldsymbol{M}}_{(2:n,2:n)} \cdot \overrightarrow{\underline{a}}_{(2:n)} = -\underline{\boldsymbol{M}}_{(2:n,1)} \cdot \overrightarrow{\underline{a}}_{(1)} \tag{4.67}$$

where the index (*2:n,2:n*) indicates row 2 to row n and column 2 to column n. Index (*2:n,1*) indicates the first column and row 2 to row n. The remaining incident waves can be determined by solving the linear equation (4.67). Lastly, the propagated waves $\overrightarrow{\underline{b}}$ can be obtained from (4.60). In the case of multiple sources, (4.67) is solved for each source separately by removing the row and column where the next source is applied. This process is repeated until all source contributions are included.

5.6 Preparing Results

The flow chart in Figure 4.12 concludes with preparation of results, that is, the incident and propagated voltage at all nodes in the model. The propagation model output has been designed with the variables given in Table 4.8. All output variables are frequency dependant, which means that their values are determined by the harmonic order under consideration.

Table 4.8: The propagation model's output variables.

Variable	Description
ID	Node ID identification according to propagation model diagram.
$\underline{\boldsymbol{A}}$	Matrix with incident voltage waves. For each source contribution k at all nodes j, $\underline{\boldsymbol{A}}$ contains an incident wave vector $\overrightarrow{\underline{a}}^{k}_{j,h}$. Index h represents the harmonic order.
$\underline{\boldsymbol{B}}$	Matrix with propagated voltage waves. For each source contribution k at all nodes j, $\underline{\boldsymbol{B}}$ contains a propagation vector $\overrightarrow{\underline{b}}^{k}_{j,h}$. Index h represents the harmonic order.
$\underline{S}_h$	List with S-parameters for all N-port networks in the model. Index h represents the harmonic order.
$\underline{Z}_{0,h}$	List with characteristic impedance $\underline{Z}_0$ for all line sections in the model. Index h represents the harmonic order.
$\underline{\gamma}_h$	List with propagation constant $\underline{\gamma}$ for all line sections in the model. Index h represents the harmonic order.

There are no examples of results in this section. Later in Section 7, case studies are examined, and the propagation model's results are discussed.

5.7 Propagation Model Development Summary

The steps for developing a propagation model for determining harmonic voltage propagation have been presented in this section. The propagation model is originally described in [4.24] (paper **J1** of this thesis), but this section outlines a process that can be applied to the studies in this thesis. Figure 4.12 shows a flow chart of the process.

The sections that follow define a number of case studies that will help explain harmonic voltage propagation in a meshed grid with a high proportion of UGC. The procedure described in this section is used to create a propagation model for the case studies, and the results will be analysed using the variables listed in Table 4.8.

6 A Propagation Study Model

6.1 Introduction

This section describes the propagation study model of the Western Danish 400 kV transmission system, which has been utilised in this thesis' research. It is based on the developed model described in Section 5. Table 4.9 shows the three case studies used in this study. These cases are defined according to the purpose of study discussed in the introduction of this chapter.

Table 4.9: Case studies to be assessed using the propagation study model. Substation designations according to Figure 4.14.

Case	Case description
3.0	The Western Danish 400 kV grid in its current configuration, with a 7.03 km UGC section in line S5-S6 at Vejle-Aadal (VLA), serving as a reference case study. Sections of the S5-S6 OHL are 79.4 km long.
3.1	Variation of UGC line section length within line S5-S6.
3.2	Substations S6 and S10 are connected across by a UGC. Variations in UGC line lengths in both S6-S10 and S5-S6 lines.

The propagation model developed by [4.24] is used as a reference case in this study and is adapted to the case studies shown in Table 4.9. Case study 3.0 is the reference case as originally documented by [4.24], and case study 3.1 is concerned with the effect on voltage propagation caused by layout variations of the UGC section in line S5-S6. Case study 3.2 is related to a higher degree of meshing in the transmission system due to the implementation of a new crosswise connection S6-S10. Such kinds of UGC variation studies are very important for TSOs in planning studies before deciding on investments. Substation designations are according to Figure 4.14. Several subcases are outlined in cases 3.1 and 3.2, but they will be presented later in Section 7.

Model accuracy is always a point of contention, and the cost of model verification and model validation is usually quite high, especially when extremely high model confidence is required [4.27]. Models are used to predict or compare the future performance of a new system, an existing system under new conditions, or a modified system, according to [4.28]. When models are used for comparison, they are usually compared to a baseline model representing an existing system, someone's idea of how a new or modified system will work (i.e., a baseline design), or current real-world system performance. In any of these scenarios, the model accuracy must be sufficient. Sufficient accuracy means that the model can be used as a substitute for the real system for experimentation and analysis (assuming that the actual system can be experimented with) [4.28], [4.29].

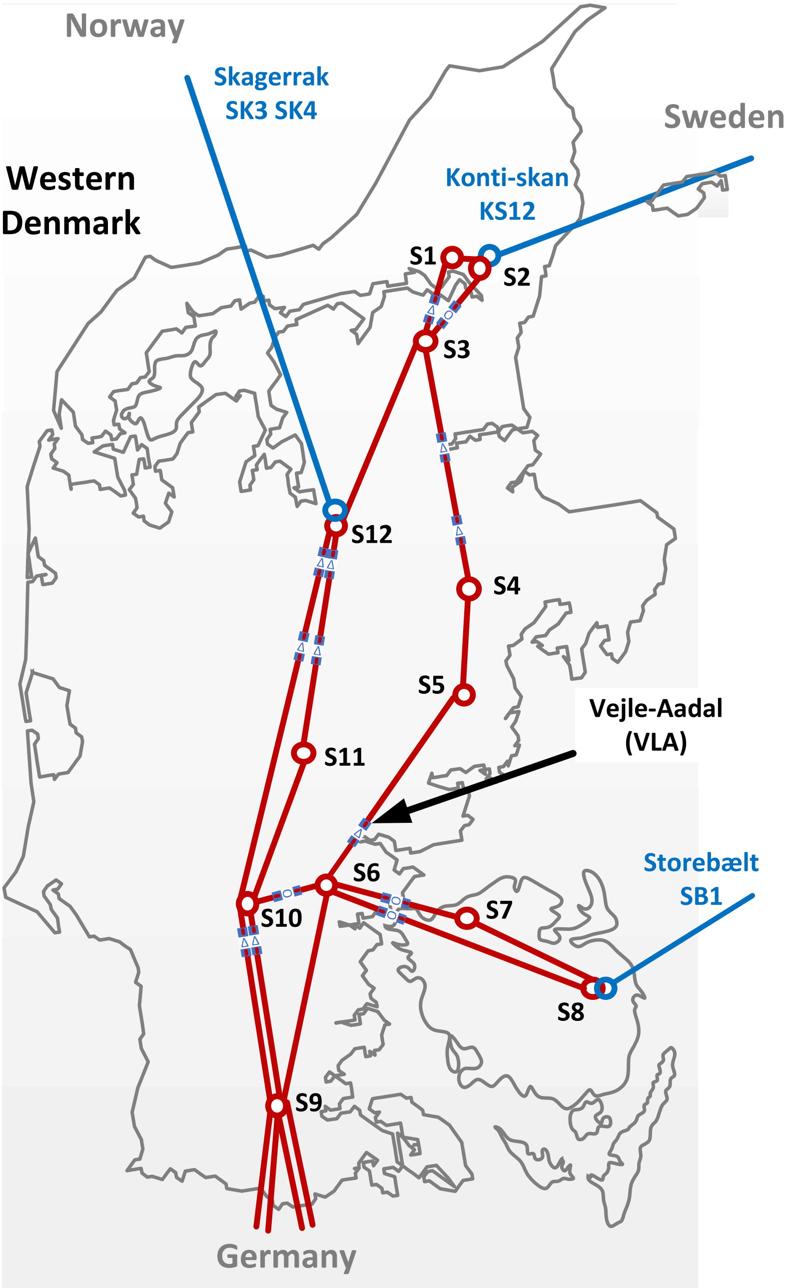


Fig. 4.14: System layout of Western Danish 400 kV transmission grid used for the propagation study model development. The UGC section at Vejle-Aadal (VLA) is highlighted.

A definition of *model verification* and *model validation* in relation to software simulation models is given in [4.27] as:

- "Model verification is often defined as "ensuring that the computer program of the computerized model and its implementation are correct""
- "Model validation is usually defined to mean "substantiation that a computerized model within its domain of applicability possesses a satisfactory range of accuracy consistent with the intended application of the model""

A simple practical consideration could be that if the model possesses a satisfactory range of accuracy, that is, the model has been properly validated, then model verification must have been obtained sufficiently.

The use of measurable parameters from the real system is an important part of model validation [4.27]–[4.29]. Harmonic voltage measurements obtained from the Western Danish 400 kV transmission system are available and documented in both [4.24], [4.26]. In [4.24], a model validation method includes both measurements and simulations, and because the reference case model (case study 3.0) is in fact the developed propagation model from [4.24], the validation procedure is adapted for this study.

Section 6.2 discusses determining system parameters for model development, and Section 6.3 discusses model validation. A summary follows in Section 6.4.

6.2 System Parameters for Model Development

The propagation model development process was described in Section 5, and the flow chart in Figure 4.12 states that a system layout diagram, network parameters R, L, G, C, and line length, as well as harmonic source information, are required. The use of the Western Danish 400 kV transmission grid for these case studies was proposed in the introduction to this chapter.

Figure 4.14 depicts the system layout used for this modelling. Except for the line S6-S10, for which this model will be prepared, this is the same system as in reference [4.24]. The modelling approach is as follows:

1. Energinet provided equivalent network parameters at 550 Hz for each line section in the model for this paper ([4.24]). The same data set is used in this thesis' model, and additionally, Energinet has provided network parameters at 650 Hz for each line section. Tables F.2 and F.3 in Section 2 of Appendix F contains both datasets.

2. The corresponding S-parameters for both 550 Hz and 650 Hz are calculated for all line sections using the method described in Section 4.2, and the results are shown in Tables F.6 and F.7 in Section 3 of Appendix F.

3. Finally, a network diagram is drawn, and the signals are labelled. This diagram is shown in Figure F.2 in Section 4 of Appendix F and represents the reference case model described in Table 4.9 concerning case study 3.0. This model differs slightly from the model presented in [4.24] in that it is prepared for the crosswise connection between substations S6 and S10, as defined in Table 4.9 for case study 3.2. Figure F.3 in Section 4 of Appendix F depicts the augmented network diagram to be used with case study 3.2.

4. The MATLAB software version listed in Section 7 of Chapter 1 is used, and the propagation model is built using the data and the network diagrams from Appendix F and the method described in the flow chart of Figure 4.12.

Because the propagation model is identical to that presented in reference [4.24], the same model validation method is used in this thesis. In [4.24], four harmonic sources are applied to the model, and thus the same source magnitudes and directions are used. Table 4.10 shows the percentage value of harmonic distortion at source buses according to [4.24], and the source magnitudes were measured quantities from the power system, whereas the source voltage directions were determined through simulations by Energinet.

Table 4.10: Percentage value of harmonic distortion at source buses according to [4.24].

Substation	S2	S8	S9	S12
U_{11} [%]	$0.46\angle 0°$	$1.08\angle 135°$	$0.46\angle 0°$	$0.34\angle -135°$

6.3 Propagation Study Model Validation

According to reference [4.24], the propagation model method is validated against a real-life study case based on harmonic voltage level measurements and simulation results provided by Energinet, as originally documented in [4.26].

Figure 4.14 depicts the Western Danish transmission system's 400 kV lines and substations, as well as three HVDC converter stations located at substations S2, S8, and S12, respectively. In addition, there is an interconnection towards Germany at substation S9.

Energinet has used this real-life case to validate a simulation model developed for harmonic assessment studies [4.26]. Both power system measurements and simulation results are available in [4.26], and the developed simulation model documented by Energinet is successfully validated against the measurements. The propagation model developed in this research has used both sets of data for validation [4.24]. Furthermore, the grid model was implemented in a power system simulation software tool to compare the propagation model output to a simulation model. In [4.24], this simulation model is referred to as the *evaluation model*.

Because the developed propagation model is intended for voltage propagation screening rather than detailed harmonic assessment studies, it is less detailed than the comprehensive model developed by Energinet and, for simplicity, does not account for geometric asymmetries, transformers, shunt reactors, or loads. As a result, asymmetry and damping caused by loads and reactive units are not represented, and the results cannot be expected to be completely comparable with measurements. It must be emphasised, that the evaluation model uses distributed parameters.

As a result, the validation includes values from four different data sources: measurements from the power system, simulation results from [4.26], simulation results from the evaluation model and propagation model results from [4.24]. These values are presented in Table 4.11, along with the results from the propagation model case study 3.0 presented in this thesis, utilised with the source values given in Table 4.10. All values are the 11th harmonic distortion in percentage of the fundamental frequency voltage.

Table 4.11: Measured and simulated 11th harmonic distortion in percentage of the fundamental voltage for the validation of models, as presented in [4.24]. The results obtained for case study 3.0 are also shown for comparison.

Substation	Results from [4.26]		Results from [4.24]		Case
	Measured	Simulated	Evaluation	Propagation	3.0
S1	-	-	0.41 %	0.40 %	0.40 %
S3	-	-	0.21 %	0.20 %	0.20 %
S4	1.47 %	1.42 %	1.55 %	1.55 %	1.55 %
S5	-	-	0.69 %	0.69 %	0.69 %
S6	-	-	1.26 %	1.24 %	1.24 %
S7	-	-	0.53 %	0.52 %	0.52 %
S10	-	-	2.14 %	2.09 %	2.09 %
S11	-	-	2.35 %	2.30 %	2.30 %

Only the values for substation S4 are presented by [4.26], but it does not include values for the additional substations S1, S3, S5-S7, and S10-S11. For substation S4, a value of 1.47 is measured and a value of 1.42 is simulated by [4.26]. The results for both the evaluation model and the propagation model by [4.24] are approximately 1.55, which corresponds to a 9.1 % difference between [4.26] and [4.24], as well as a 5.4 % deviation from the actual measured harmonic distortion at substation S4. The table also includes the results obtained with the evaluation model and the propagation model for other substations, excluding those with connected sources.

Reference [4.24] finds that there is good agreement between the obtained results (less than 5.4 % deviation between the evaluation model and the propagation model) and sufficient agreement between the individual results when model differences are considered (that is, the differences between results documented by [4.26] and [4.24]). Because the objective is to evaluate the propagation behaviour caused by the insertion of UGC sections or lines in the grid rather than harmonic assessment studies, the less detailed modelling approach was found to be sufficient.

Table 4.11 also shows the results obtained with the propagation study model developed in this study. Because case study 3.0 is identical to the model presented in [4.24], the obtained results should be equal, as shown in the table when comparing the propagation model results from [4.24] with the case study 3.0 results. As a result, the propagation study model presented in this section has been found to be applicable to the research presented in this thesis.

6.4 Summary on Propagation Study Model

This section presented the necessary steps for developing a propagation study model for case studies to explain the impact of UGCs in interaction with OHLs in the meshed transmission network. The model is prepared according to the flow chart in Figure 4.12, which describes the development of a propagation model.

General methods and the complexity of verification and validation of simulation models are superficially touched upon, and the model is validated both against measurements from the power system and several simulation models implemented in commercial power system simulation software, documented in the literature.

It has been concluded that the model satisfactorily represents the transmission network to be evaluated in this thesis. The deviations between power system measurements and simulations have been found to be less than 5.4 % under the established assumptions.

7 Case Studies and Study Guideline

7.1 Introduction

The objective of this chapter is to explain harmonic voltage propagation in the Western Danish transmission network using tools with a background in microwave theory, because harmonic voltage, when considered as standing waves, can be treated with this theory. Most of this chapter has focused on theories and definitions, as well as the derivation of relevant parameters for use in the propagation model.

Table 4.12: Collected and calculated propagation model variables. All variables are complex values

Variable	Description
$\underline{a}^k_{j,m}$	Incident voltage at node j as consequence of source k, evaluated for case m
$\underline{b}^k_{j,m}$	Propagated (or reflected) voltage at node j as consequence of source k, evaluated for case m
$\sum \underline{a}_{j,m}$	Sum of incident voltages at node j, evaluated for case m: $\sum_{j=1}^{n}(\underline{a}_{1,m} + \underline{a}_{2,m} + \ldots + \underline{a}_{n,m})$
$\sum \underline{b}_{j,m}$	Sum of propagated voltages at node j, evaluated for case m: $\sum_{j=1}^{n}(\underline{b}_{1,m} + \underline{b}_{2,m} + \ldots + \underline{b}_{n,m})$
$\lvert\Delta \sum \underline{a}_{j,1m}\rvert$	Difference in the sum of incident voltages at node j for case m compared to the reference case (case 1): $\lvert \sum \underline{a}_{j,1}\rvert - \lvert \sum \underline{a}_{j,m}\rvert$
$\lvert\Delta \sum \underline{b}_{j,1m}\rvert$	Difference in the sum of propagated voltages at node j for case m compared to the reference case (case 1): $\lvert \sum \underline{b}_{j,1}\rvert - \lvert \sum \underline{b}_{j,m}\rvert$
$\lvert\Delta \underline{b}_{f:g,1m}\rvert$	Difference in the absolute value of the sum of propagated voltage for case m compared to case 1 evaluated at nodes $f : g$: $\lvert\Delta \underline{b}_{f:g,1m}\rvert = \lvert \sum \underline{b}_{f:g,m}\rvert - \lvert \sum \underline{b}_{f:g,1}\rvert$
$\lvert \sum \underline{b}^{s,k}_{f:g,1m}\rvert$	Ratio of the sum of propagated propagated voltage for case m compared to the reference case (case 1), confined of nodes f to g and evaluated for substation s and source k: $\lvert \sum \underline{b}^{s,k}_{f:g,1m}\rvert = \left\lvert \frac{\sum \underline{b}^{s,k}_{f:g,m}}{\sum \underline{b}^{s,k}_{f:g,1}} \right\rvert$
$\Delta\underline{\zeta}_{j,1m}$	Propagation coefficient at node j, evaluated for case m compared to the reference case (case 1).
$\sum \underline{\zeta}_{j,m}$	Sum of propagation coefficients at node j, evaluated for case m
$\chi^s_{f:g,m}$	Propagation ratio in relation to the reference case at substation s, consisting of nodes f to g and evaluated for case m: $$\chi^s_{f:g,m} = \frac{\lvert \sum \underline{b}^s_{f,m} + \sum \underline{b}^s_{f+1,m} + \ldots + \sum \underline{b}^s_{g-1,m} + \sum \underline{b}^s_{g,m}\rvert}{\lvert \sum \underline{b}^s_{f,1} + \sum \underline{b}^s_{f+1,1} + \ldots + \sum \underline{b}^s_{g-1,1} + \sum \underline{b}^s_{g,1}\rvert}$$

The starting point in this context has been the propagation model originally presented in [4.24] (paper **J1** of this thesis), where the process for establishing the actual model, as well as the validation of the model, has been described and elaborated in detail in this chapter compared to the description given in [4.24].

It remains to establish a proper method in the form of a guideline to describe a propagation study and to carry out such a study in which the conditions for the Western Danish transmission grid are analysed. As a result, this section provides a guideline that the TSO can use for such studies.

The objective, as stated in the introduction in Section 1 of this chapter, is to answer the following two questions to explain the phenomenon of propagation:

1. How does partially undergrounding transmission lines affect harmonic voltage propagation?
2. What effect does a different degree of meshing have on propagation?

This section begins with a discussion of the key parameters associated with the model. Subsection 7.2 provides a guideline for propagation studies. Modelling, as well as presenting study cases, validating the model, and calculating a dataset for analysis, are all covered in Subsections 7.3 - 7.6. Subsection 7.7 is devoted to analysis, which includes a presentation and discussion of results, as well as partial conclusions for the individual parts, highlighting the significant and important conclusions. Finally, in Subsection 7.8, the methodology and guidelines' applicability are assessed.

Throughout this section, the variables presented in Table 4.12 are widely used and explained in more detail. Initially, the table show the variables and a short, descriptive explanation. In the sections that follow, more details are presented, where the variables find their use.

7.2 Methodology

A study approach for propagation studies in a power system has been developed and used in this thesis. The study approach is depicted on Figure 4.15. From the formulation of a propagation model to the analysis of the outcomes, this guideline consists of six general steps. The following tasks are included in each step:

Initialisation: Based on a system layout and the corresponding network parameters, a propagation model is first outlined. Section 6 of this chapter contains a description of the process. Both the S-parameters and secondary line constants are collected for analysis.

Case Definitions: The preparation of case studies is the next step. Typically, a TSO defines a common reference case, or "snapshot," in time and evaluates other cases against this reference case. This is not strictly necessary for the use of the propagation model, but this approach is used in this case.

A reference case is drawn up that serves as a baseline model. A set of case studies are defined, and these cases can be compared to the baseline model.

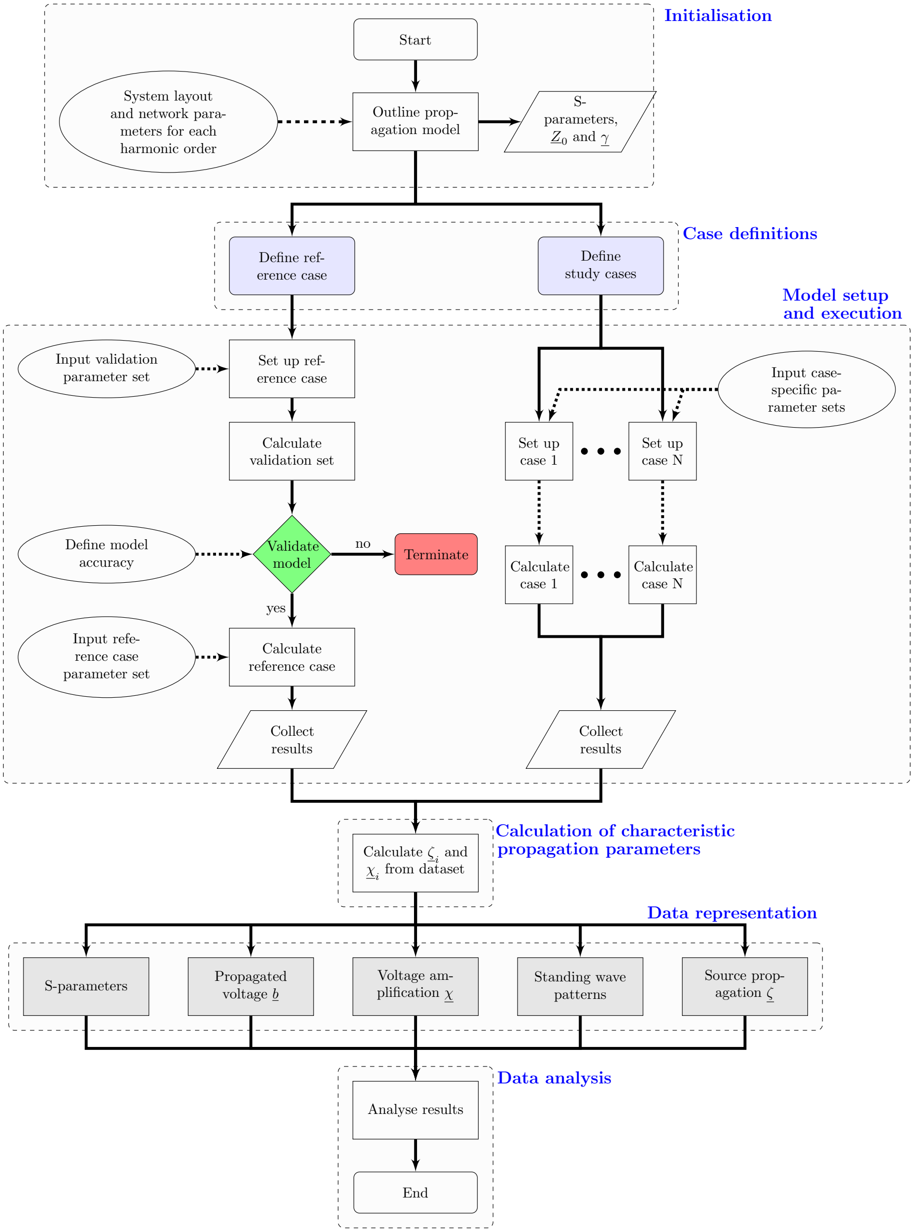


Fig. 4.15: Developed guideline for propagation studies in an arbitrary power system.

Model Setup and Execution: After defining the cases, the model is set with the desired parameters:

1. Parameters for each source are set ($\underline{a}_i$ and $\underline{\Gamma}_S$).
2. Parameters of individual loads are set ($\underline{\Gamma}_L$).
3. Line configurations to be analysed.

To provide a data set for the model's validation, the reference case is first set up with one or more sets of validation parameters. This dataset is compared with predetermined model accuracy requirements. If the model fails to comply to the established requirements, then the execution is terminated and the model, the validation set, or both must be re-evaluated. The reference model is then set up with the parameter set that will reflect the baseline model if the model's accuracy can be approved. The calculations are carried out, and the results are gathered.

If the model passes the validation criteria, case studies $\{1, \ldots, N\}$ are set up with case-specific parameter sets, the calculations are executed, and the corresponding data is collected.

Calculation of Characteristic Propagation Parameters: For each node in the propagation model, the propagation coefficient $\underline{\zeta}$ is calculated and the propagation ratio χ is calculated for each busbar. These parameters will be detailed later in Subsection 7.6.

Data Representation: The propagation model layout specified in the initialising state is used to group the S-parameters, the collected dataset, and the characteristic propagation parameters for visualisation. The following results are visualised:

- Line section S-parameters.
- Propagated voltage patterns.
- Voltage amplification at nodes.
- Standing wave patterns.
- Source propagation characteristics.

Data Analysis: The final part of this study approach is data analysis and discussion of results.

In the following, each step is elaborated using the case studies outlined for this research.

7.3 Initialisation

The first step of the study approach in Figure 4.15 is definition of a propagation model. In this case, two case studies from the Western Danish 400 kV grid are analysed according to Table 4.9. Figure 4.16 depicts the system layout and the two areas for the case studies are highlighted. The grid is also divided into three regions: I, II, and III, with regions I containing substations S6-S9, II containing substations S10-S12, and III containing substations S1-S5.

Recall from Section 6.1, that the first case 3.1 is related to the combined OHL and UGC line between substations S5 and S6. The second case 3.2 is a new 400 kV UGC line between substations S6 and S10.

The propagation model formulated in Section 6 will be applied to these two scenarios since it accurately represents the Western Danish 400 kV transmission grid in its current configuration, including the UGC section of 7 km at Vejle-Aadal. As a result, this section will not go into detail about how to build a propagation model. Both the 11th and 13th harmonics will be assessed to show the propagation model's general application.

Section 4 of Appendix F contains the specific network diagrams for cases 3.1 and 3.2, and Section 2 of Appendix F contains the equivalent line parameters at 550 and 650 Hz. These models and the corresponding data were retrieved in Section 6 of this chapter.

7.4 Case Definitions

To evaluate the source propagation on an equal basis, a source voltage of 1.0 per-unit is applied to all four sources in the model, located at buses S2, S8, S9 and S12. With this source voltage, each source contribution at a specific node or bus can be evaluated quantitatively. As stated in Section 7.1, the purpose is to evaluate the harmonic voltage propagation throughout the grid and not to conduct a harmonic assessment and therefore, a 1.0 per-unit source voltage is applicable, but not to be compared to a real-life harmonic assessment. The case studies and the set of subcases are given in Table 4.13.

Five subcases are outlined in study case 3.1. The UGC section in this case study is increased in steps from 7.03 km to the full UGC length between substations S5 and S6. The UGC section is extended to 20 km, 30 km, 40 km, 50 km, and the full distance of the line of 79.43 km. For all subcases, two parallel cables are used. To keep the total line length at 79.43 km for all subcases, the OHL section from the UGC section to S5 is similarly reduced for each subcase. The harmonic source voltages are 1.0 per-unit for all subcases.

Study case 3.2 outlines three subcases. Between substations S6 and S10 in the first subcase A, a 20 km single-circuit UGC line of the same type as the UGC section used at Vejle-Aadal is established. Line S5-S6's UGC section is restored to its original length of 7.03 km. In subcase B, the UGC length

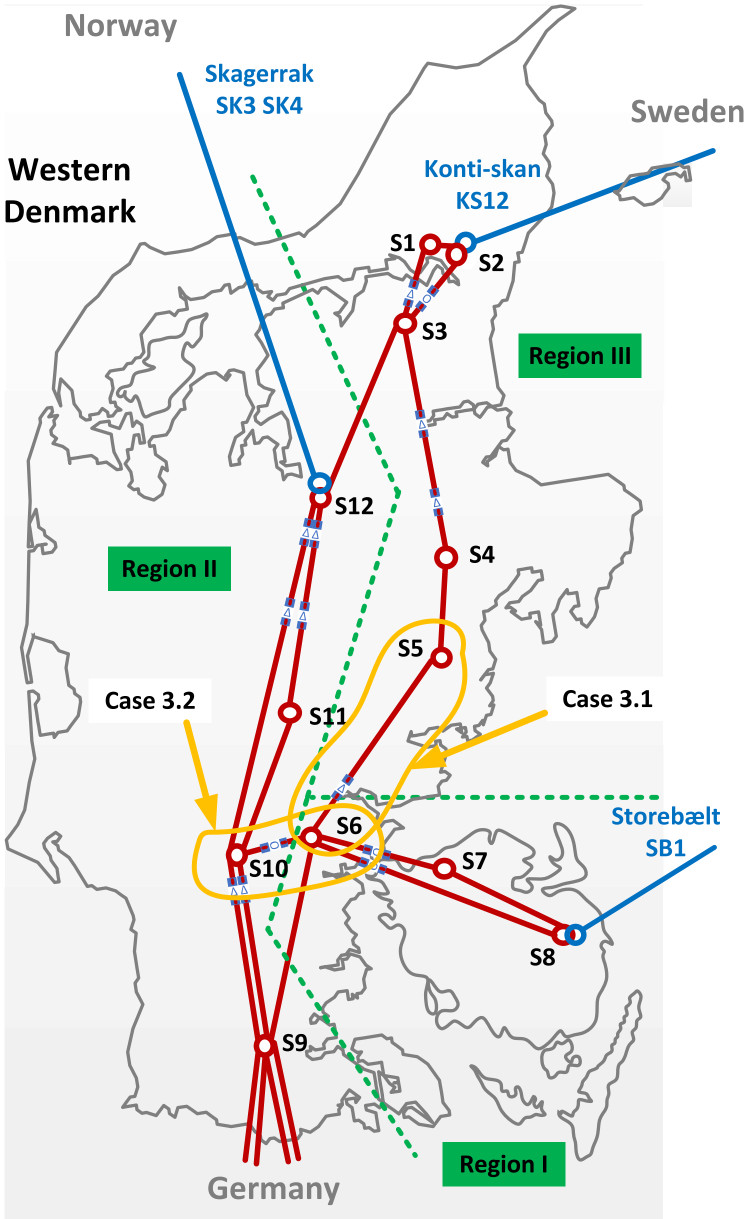


Fig. 4.16: System layout of Western Danish 400 kV transmission grid used for the propagation study model development. Areas concerning case studies 3.1 and 3.2 are highlighted. The grid is divided into three regions: I, II, and III, with regions I containing substations S6-S9, II containing substations S10-S12, and III containing substations S1-S5.

Table 4.13: Case studies 3.0 (reference case), 3.1 with combined OHL and UGC line and 3.2 with pure UGC line.

Case	ID	Case description
3.0		Reference case. The Western Danish 400 kV grid in its current state with a 7.03 km UGC section in line S5-S6 at Vejle-Ådal. The OHL section length between substation S6 and the UGC section is 12.8 km and the OHL section between the UGC section and substation S5 is 59.6 km.
3.1	A	UGC line section in line S5-S6 is 20 km. OHL section length from the UGC to substation S5 is 46.6 km.
	B	UGC line section in line S5-S6 is 30 km. OHL section length from the UGC to substation S5 is 36.6 km.
	C	UGC line section in line S5-S6 is 40 km. OHL section length from the UGC to substation S5 is 26.6 km.
	D	UGC line section in line S5-S6 is 50 km. OHL section length from the UGC to substation S5 is 16.6 km.
	E	UGC line section in line S5-S6 is 79.43 km (full line length).
3.2	A	UGC line S6-S10 is 20 km. UGC line section in line S5-S6 is 7.03 km.
	B	UGC line S6-S10 is 25 km. UGC line section in line S5-S6 is 7.03 km.
	C	UGC line S6-S10 is 25 km. UGC line section in line S5-S6 is 20 km and OHL section length from the UGC to substation S5 is reduced to 46.6 km.

between S6 and S10 is increased to 25 km, and in subcase C, the UGC line between S6 and S10 is also 25 km, but the UGC line section in line S5-S6 is increased to 20 km and the OHL section is decreased similarly, as in study case 3.1A. The harmonic source voltages are 1.0 per-unit for all subcases.

The 11th and 13th harmonic impedance and admittance for the changed line sections in each study case is given in Tables F.4 and F.5, respectively, in Section 2 of Appendix F.

7.5 Model Setup and Execution

The reference case is first utilised to validate the model in this step. The validation process was addressed in Section 6.3 in this case, so it is not included in this section.

After that, the reference case is utilised with the defined parameter set for the study under consideration. In this case, the focus is on propagation behaviour in the meshed grid and the source parameters are set as follows:

$$\underline{a}_1 = 1.0\angle 0^\circ \text{ per-unit}$$
$$\underline{\Gamma}_S = -1$$

that is, the source model acts as an ideal harmonic source with no reflection at the source terminal.

The propagation model is configured with the study cases shown in Table 4.13, and each study case is initialised and run. For each study case, the result variables given in Table 4.14 are collected.

Table 4.14: Collected propagation model variables. All variables are complex values

Variable	Description
$\underline{a}^k_{j,h}$	The calculated incident voltage at node j is collected for each node designated in the propagation model schematic depicted in Section 4 of Appendix F. The incident voltage contribution for each individual source k is collected. Index h represents the harmonic order, e.g., 11 or 13.
$\underline{b}^k_{j,h}$	The calculated propagated voltage at node j is collected for each node designated in the propagation model schematic depicted in Section 4 of Appendix F. The incident voltage contribution for each individual source k is collected. Index h represents the harmonic order.
$\underline{S}_{XY,h}$	Calculated S-parameters for each line section in the model. Index XY is the port designations, which were explained in Section 4. Index h represents the harmonic order.
$\underline{Z}_{0,h}$	The calculated characteristic impedance of each line section in the model. Index h represents the harmonic order.
$\underline{\gamma}_h$	The calculated propagation constant for each line section in the model. Index h represents the harmonic order.

The parameters shown in the table are used for the analyses presented later in Section 7.7.

7.6 Calculation of Characteristic Propagation Parameters

According to Section 7.1, both the propagation coefficient and the propagation ratio given in Table 4.12 are calculated using the gathered variables according to Table 4.14. In this study, the voltage propagation is analysed at substations S1, S3, S4, S5, S6, S7, S10, and S11. These substations were chosen because they do not have sources connected.

Each substation is modelled with a set of nodes ranging from ports f to g, depending on the number of ports, where indexes f and g are as shown in Table 4.15. The node IDs can be seen from the network diagrams in Section 4 of Appendix F.

Table 4.15: Propagation ratio node index for evaluated substations according to network diagram in Section 4 of Appendix F.

Substation	Index		Substation	Index		Substation	Index	
	f	g		f	g		f	g
S1	131	132	S5	152	153	S10	49	53
S3	114	117	S6	6	10	S11	81	82
S4	154	155	S7	12	13			

7.7 Data Representation and Discussion

The next step in the process is data representation. The five different categories of results will be discussed in the subsections that follow, along with the data analysis for each category, so that the final two steps in the guideline can be evaluated jointly. The definitions given in Table 4.12 are further discussed in this subsection.

Chapter 3 focused on measurable indicators from the Western Danish transmission network. In terms of harmonic distortion, the correlation and regression analyses revealed that the meshed grid appeared divided into three regions, I, II, and III, where variations in harmonic voltage in each region are not very strongly correlated, but for regions I and III, a negative correlation between substations in the regions has been demonstrated. Figure 4.16 depicts the breakdown into these three regions. As the study is centred on these specific areas, the three regions of the meshed grid come into play in the following discussion. The following are the five categories of results:

1. A variation study of S-parameters for UGC types are evaluated due to parameter changes caused by changed line lengths, as shown in Table 4.13.

2. Propagated voltage $\underline{b}$ at buses as consequence of propagated harmonic voltages from surrounding substations.

3. Voltage amplification estimated at buses shown in Table 4.15 due to network changes. To quantify the impact of case studies on harmonic voltage, the propagation ratio χ is used.

4. Evaluate standing wave patterns along line S4-S5 and up to the UGC section in line S5-S6 for each study case to estimate expected voltage range at a bus or node, as well as extreme minimum and maximum voltage values and their distance from a bus/node.

5. Evaluate the propagation of source-specific voltage across the grid. The propagation coefficient $\underline{\zeta}$ is used to quantify the impact of case studies of source-specific propagation. Substations S3 and S6 are evaluated.

Evaluation of S-parameters

A sensitivity analysis of S-parameters in relation to UGC length is carried out to determine the reflection and transmission parameters $\underline{S}_{11}$ and $\underline{S}_{21}$. As discussed in Section 4.2, $\underline{S}_{11} = \underline{S}_{22}$ and $\underline{S}_{12} = \underline{S}_{21}$ due to reciprocity. According to Table 4.3, the Western Danish 400 kV grid contains seven different UGC types, which are displayed with their maximum length and formation in Table 4.16. The maximum UGC system length is 8.4 kilometres. The UGC

Table 4.16: UGC types in the Western Danish 400 kV grid.

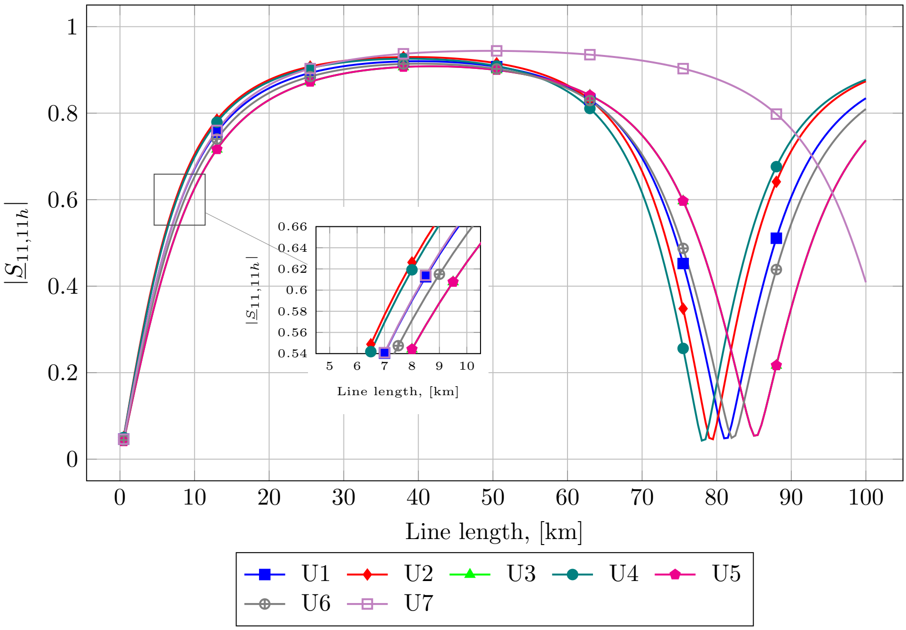
Type	Length [km]	Formation
U1	3.9	Flat
U2	1.6	Flat
U3	4.5	Flat
U4	8.4	Flat
U5	7.4	Flat
U6	4.4	Flat
U7	7.3	trefoil

length in the case studies will be varied up to approximately 80 kilometres according to Table 4.13. As a result, the sensitivity analysis is performed for UGC lengths ranging from 0.5 to 100 kilometres in 0.5-kilometer increments.

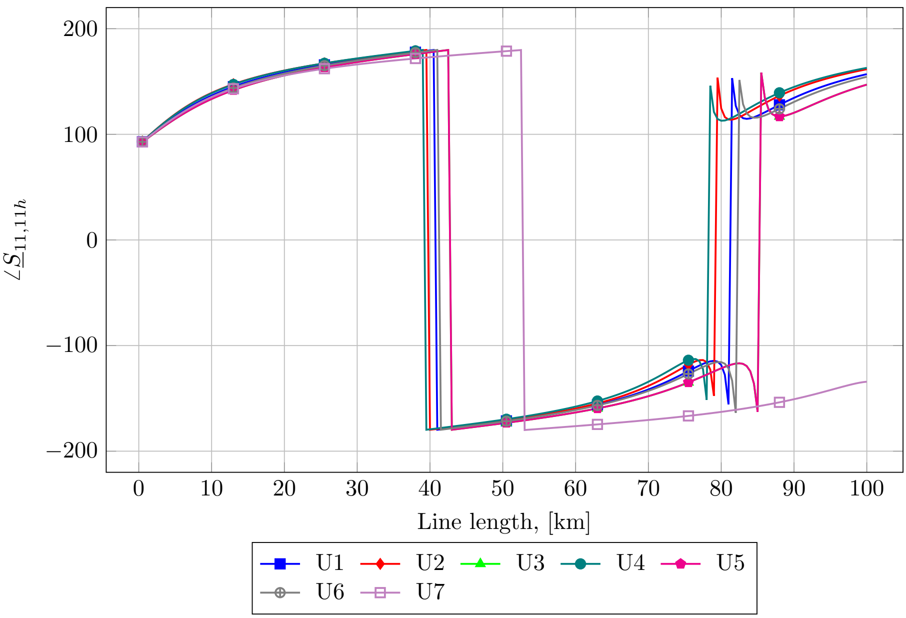
Figure 4.17 depicts the magnitude and direction of $\underline{S}_{11}$ for the 11th harmonic, while Figure 4.18 shows the same for $\underline{S}_{21}$. The x-axis shows the UGC length, and the y-axis shows the magnitude of the reflection coefficient and the direction in degrees, respectively.

Figure 4.17a depicts the reflection coefficient $\underline{S}_{11}$ as function of length for the seven UGC types. For short UGC lengths (< 2 kilometres), the difference in reflection between the UGC types is minor. The difference increases with length, and the zoom plot at 8 kilometres shows that the span between the UGC type with the largest reflection coefficient (type 2) and the UGC type with the smallest reflection coefficient (type 5) is approximately 0.09, or about a 9% difference. The maximum reflection for UGC types 1 – 6 is seen at approximately 38-44 kilometres and the exact maximum is in fact at $\frac{1}{4}\lambda$ for each UGC type. For UGC type 7, the maximum reflection is seen at a slightly larger distance (approximately 53 kilometres).

The direction depicted in Figure 4.17b shows increasing angle with UGC length and because of the positive angle reflection, the direction of the reflected wave as seen from $x = 0$ for each UGC type moves in positive direction. The direction becomes negative at $\frac{1}{4}\lambda$ (approximately 38-44 km for UGC types 1-6 and approximately 53 km for UGC type 7), that is, the imaginary part of $\underline{S}_{11}$ becomes negative, and thus the direction of the reflected wave moves in the negative direction. At $\frac{1}{2}\lambda$ (approximately 78–85 km for

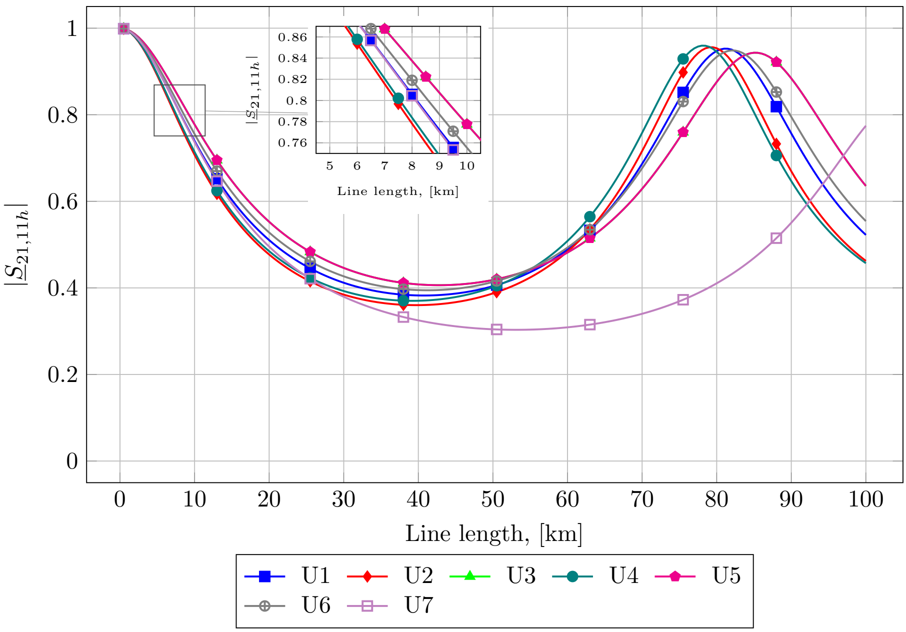


(a) $|\underline{S}_{11,11h}|$

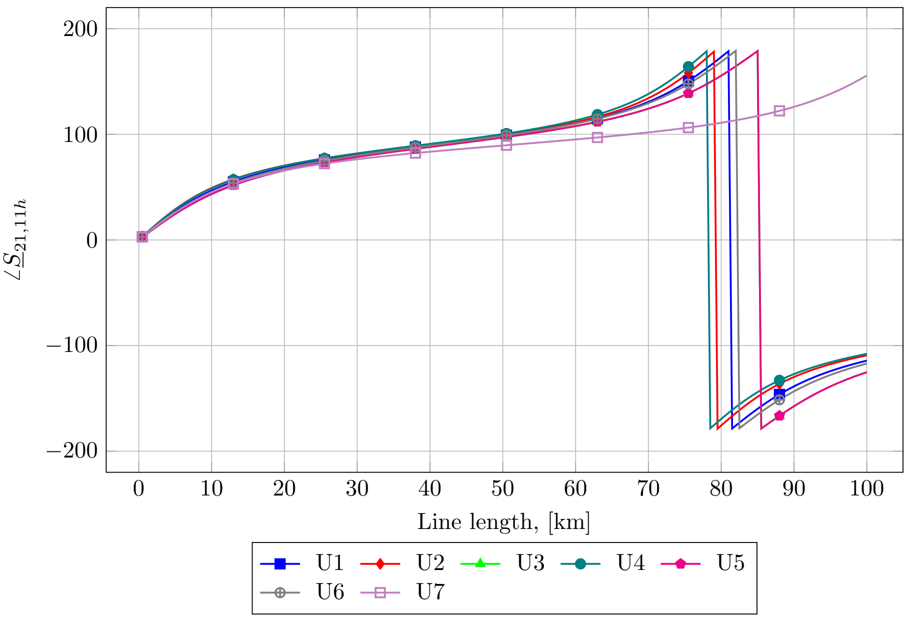


(b) $\angle\underline{S}_{11,11h}$

Fig. 4.17: Magnitude and direction of $\underline{S}_{11}$ for the 11th harmonic.



(a) $|\underline{S}_{21,11h}|$



(b) $\angle\underline{S}_{21,11h}$

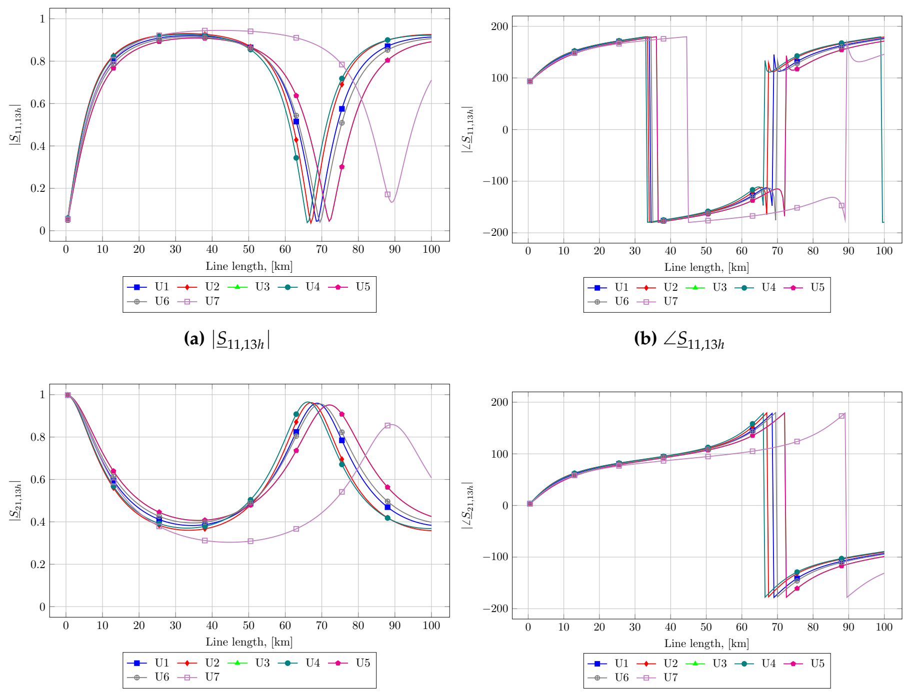
Fig. 4.18: Magnitude and direction of $\underline{S}_{21}$ for the 11th harmonic.

UGC types 1-6 and more than 100 km for UGC type 7), the imaginary part of $\underline{S}_{11}$ becomes positive, and thus the direction of the reflected wave moves in the positive direction. Because the S-parameters are hyperbolic functions, the direction oscillates around $\frac{1}{2}\lambda$. The UGC length increments in steps of 0.5 kilometres on these plots.

An example will be presented later in this section to illustrate the interpretation in relation to a power system.

$\underline{S}_{11}$ returns to a minimum at a UGC length of $\frac{1}{2}\lambda$, and for longer distances, $\underline{S}_{11}$ increases with a larger difference between the individual UGC types due to their different wavelengths and attenuation. This means that the magnitude of the reflection coefficient is highly dependent on the UGC type for UGC lengths above $\frac{1}{2}\lambda$.

The transmission coefficient $\underline{S}_{21}$ as a function of length is shown in Figure 4.18a for the seven UGC types. The difference in transmission between the UGC types is minor for short UGC lengths (<2 kilometres), and as the length increases, the transmitted part from one end to the other decreases to a minimum value at $\frac{1}{4}\lambda$.



(a) $|\underline{S}_{11,13h}|$

(b) $\angle\underline{S}_{11,13h}$

(c) $|\underline{S}_{21,13h}|$

(d) $\angle\underline{S}_{21,13h}$

Fig. 4.19: Magnitude and direction of $\underline{S}_{11}$ and $\underline{S}_{21}$ for the 13th harmonic.

This is due to the increase in the reflected part of the incident voltage, as was shown in Figure 4.17a. The zoom plot at 8 kilometres shows that the span between the UGC type with the largest transmission coefficient (type 5) and the UGC type with the smallest reflection coefficient (type 2) is approximately 0.07, or about a 7% difference.

Figure 4.18b depicts the direction of $\underline{S}_{21}$, and the characteristic is the same as discussed above for $\underline{S}_{11}$, besides that the imaginary part of $\underline{S}_{21}$ becomes negative at $\frac{1}{2}\lambda$.

Figure 4.19 depicts $\underline{S}_{11}$ and $\underline{S}_{21}$ for the 13th harmonic for the sake of completeness in this discussion.

The magnitude and direction of the 13th harmonic $\underline{S}_{11}$ are depicted in Figure 4.19a and Figure 4.19b. The magnitude and direction of the 13th harmonic $\underline{S}_{21}$ are depicted in Figure 4.19c and Figure 4.19d. The 13th harmonic characteristics are similar to the 11th harmonic characteristics. The difference is due to the shorter wavelength for the 13th harmonic compared to the 11th harmonic, because the UGC wavelength decreases by approximately 10-15 kilometres when the frequency increases by 100 Hz (two harmonic orders).

As regards the transmission coefficient $\underline{S}_{21}$, the characteristics are also like the 11th harmonic $\underline{S}_{21}$, taking the decrease in wavelength into account.

An Example of Interpretation in Relation to Power Systems: As previously stated, $\underline{S}_{11}$ and $\underline{S}_{21}$ both influence the harmonic voltage at the UGC section's terminals. Equation (4.26) gives the standing wave pattern $|\underline{a}| \cdot |1 + \underline{S}_{11}|$, and Figure 4.20 shows the 11th and 13th harmonics at the terminals of a type 1 UGC as a function of UGC length when an incident voltage wave of 1 per-unit is applied and the UGC line has an open-end termination. The x-axis represents UGC length, while the y-axis represents the absolute value of harmonic voltage $|U_h|$ in per-unit at terminals.

The voltage at the terminal of the incident wave decreases as the UGC length increases, until a UGC length of $\frac{1}{4}\lambda$ is reached. As previously discussed, as the UGC length increases, the imaginary part of S11 becomes negative, resulting in an increase in voltage. The magnitude difference between the 11th and 13th harmonic voltages at a given distance x demonstrates the difficulty in predicting the outcome of a specific UGC project, as a "decent" 11th harmonic voltage at distance x may not be as "decent" for the 13th harmonic voltage. The term "decent" could, for example, refer to a harmonic distortion level that is below or above the assessment threshold, but for the purposes of this interpretation, we will leave this unanswered.

Figure 4.20 also show that the transmitted wave opposite from the terminal of the incident wave decreases as the UGC length increases. $\underline{S}_{21}$ reaches a minimum at $\frac{1}{4}\lambda$ and then supposedly must increase due to increasing size of $\underline{S}_{21}$, but because the direction of $\underline{S}_{21}$ first becomes negative as the UGC length increases beyond $\frac{1}{2}\lambda$, the magnitude of the transmitted wave keeps decreas-

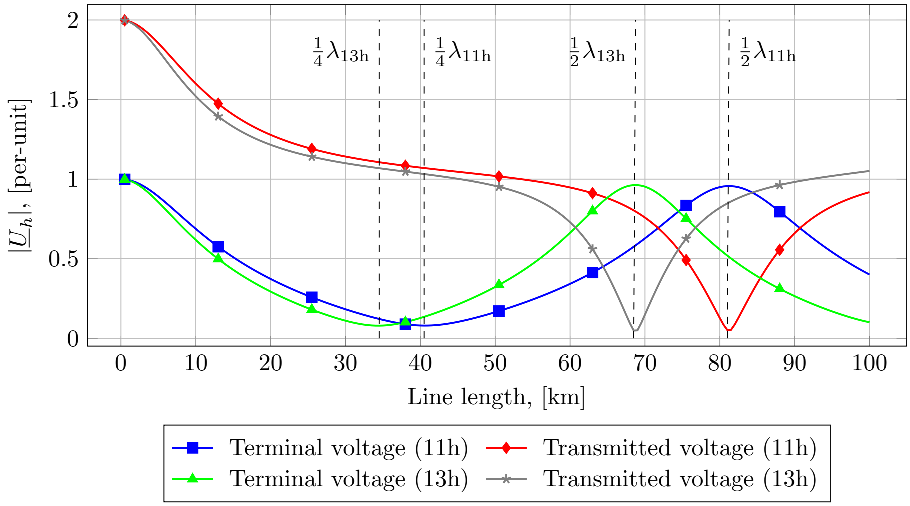


Fig. 4.20: 11th and 13th harmonic voltage at the terminals of a type 1 UGC as function of UGC length. An incident wave of 1 per-unit is applied. The "Terminal voltage" is at the incident terminal and the "Transmitted voltage" at the opposite terminal beyond the UGC section. The voltage is the sum of incident and reflected voltage waves $|\underline{a} + \underline{b}|$ as a function of length.

ing until $\frac{1}{2}\lambda$. At an UGC length of $\frac{1}{2}\lambda \pm$ approximately 10 kilometres, the voltage drops significantly towards minimum for a length of $\frac{1}{2}\lambda - 10$ kilometres to $\frac{1}{2}\lambda$ and rises significantly for a length of $\frac{1}{2}\lambda$ to $\frac{1}{2}\lambda + 10$ kilometres.

From a power system perspective, these properties indicate that a UGC line section (or line) of a certain length reduces the voltage transfer from one terminal to the other. This is an important observation, as the UGC line section then acts as a limiting component regarding the propagation throughout a complete power system. Similarly, this effect may lead to increased harmonic voltages at the terminal of the incident voltage, due to major reflections.

To summarise, the following important properties of UGC response to applied harmonic distortions can be understood in this context:

Summary

- The Western Danish 400 kV transmission grid is composed of several combined OHL and UGC sections, and the dataset identifies seven different UGC types. The typical UGC length ranges between 1.6 and 8.4 kilometres.
- The type of UGC is more or less irrelevant for short lengths in terms of wavelength ($< \frac{1}{4}\lambda$) because the difference in both reflection and transmission between cable types is less than 7-

10%, but for long cables ($> \frac{1}{4}\lambda$), the cable type is important for both reflection and transmission. The difference is greatest between cables in flat formation and cables in trefoil formation.

- Sensitivity analyses have revealed a difference in the magnitude of the reflection coefficient $\underline{S}_{11}$ of up to 9% for UGC lengths up to approximately 40 kilometres ($\frac{1}{4}\lambda$), with the most severe difference in the range of 10 - 20 kilometres.
- At UGC lengths equal to $\frac{1}{4}\lambda$, the direction of $\underline{S}_{11}$ changes from positive to negative.
- The voltage transmitted from the incident terminal to the terminal opposite the incident voltage decreases with UGC length, reaching a minimum at $\frac{1}{2}\lambda$. The most significant variation in transmitted voltage is observed for line lengths near $\frac{1}{2}\lambda$.
- Certain UGC lengths create voltage transfer barriers, resulting in increased voltage on the terminal of the incident wave and lower after the barrier.

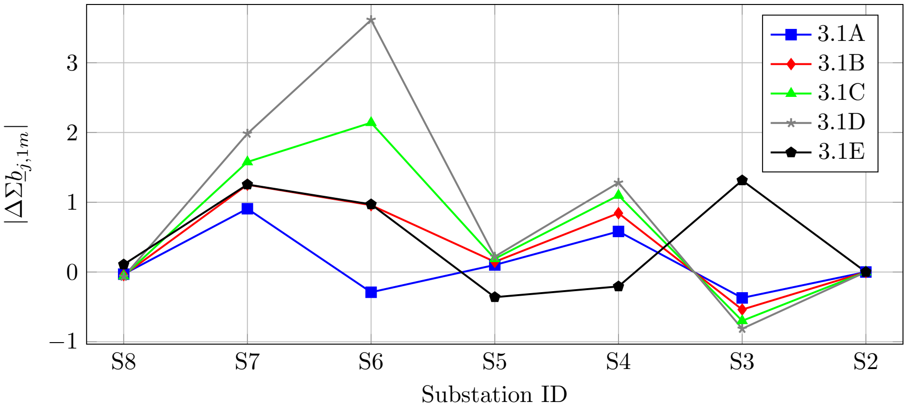
Voltage Propagation in The Grid

Figure 4.21 depicts the difference in the sum of propagated voltage $\Delta\Sigma\underline{b}_{j,1m}$ at buses S2 - S8 because of 11th harmonic propagation from surrounding substations. It is the difference in propagated voltage as compared to the reference case, therefore, it is denoted by Δ (according to the definitions in Table 4.12). The outcomes for all study cases are displayed. Figure 4.21a shows the propagated voltage in regions I + III for cases 3.1A - 3.1E, while Figure 4.21b shows the propagated voltage in regions I + III for study cases 3.2A - 3.2C. Substation ID is shown on the x-axis and the absolute value of $|\Delta\Sigma\underline{b}_{j,1m}|$ is shown on the y-axis.

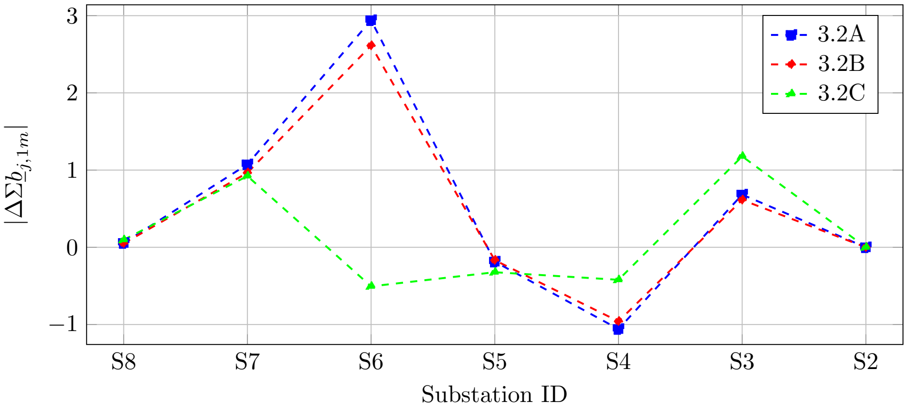
Figure 4.21a shows that as the UGC section length in line S5-S6 increases, so does the difference in propagated voltage compared to the reference case along lines S2, S3, S4, S5, S6, S7, and S8 for study cases 3.1A - 3.1E. Positive values of $|\Delta\Sigma\underline{b}_{j,1m}|$ indicate increased voltage $\underline{b}$ and negative values indicate reduced voltage $\underline{b}$.

Figure 4.22 depicts the sum of propagated voltage $|\Delta\Sigma\underline{b}_{j,1m}|$ at buses S9 - S2 along buses S10 - S12 because of 11th harmonic propagation from surrounding substations. The outcomes for all study cases are displayed.

Figure 4.22a shows the propagated voltage in region II for cases 3.1A - 3.1E, while Figure 4.22b shows the propagated voltage in region II for study cases 3.2A - 3.2C. Substation ID is shown on the x-axis and the value of



(a) Cases 3.1A - 3.1E.



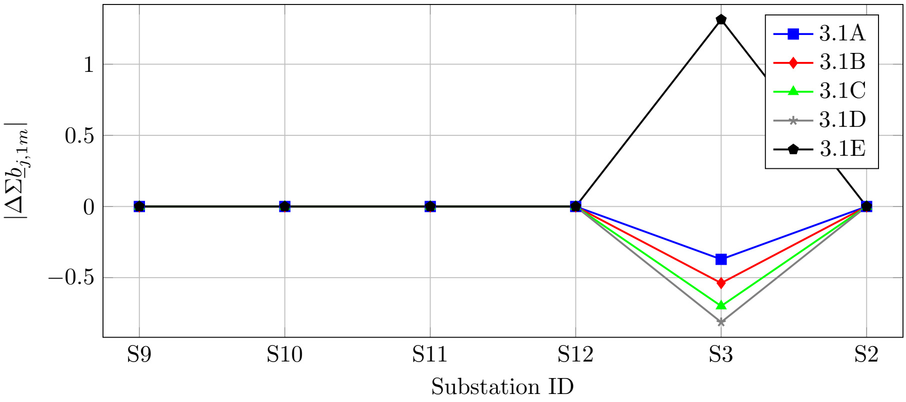
(b) Cases 3.2A - 3.2C.

Fig. 4.21: 11th harmonic voltage propagation in regions I + III for all study cases. (a) - Propagated voltage for study cases 3.1A - 3.1E. (b) - Propagated voltage for study cases 3.2A - 3.2C.

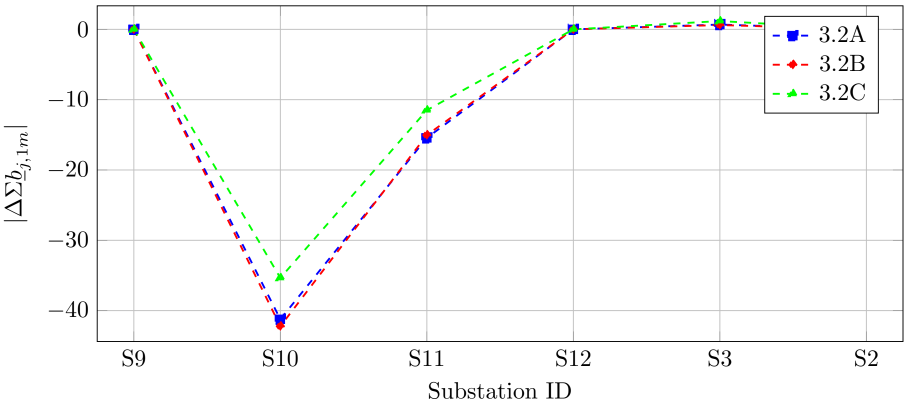
$|\Delta\Sigma\underline{b}_{j,1m}|$ is shown on the y-axis.

Figure 4.22a shows that as the UGC section length in line S5-S6 increases for study cases 3.1A - 3.1E, it has no effect on the propagation at substations S9 - S12. For substation S3, the voltage propagation experiences a change because this substation connects regions II and III. For study cases 3.2A - 3.2C, the propagated voltage is affected due to the crosswise connection between regions II and III at S10.

Evaluation of Study Cases 3.1A – 3.1E: For study cases 3.1A - 3.1D, where the UGC section length is increased from approximately 7 kilometres to 20 – 50 kilometres in incremental steps of 10 kilometres, the change in propagated



(a) Cases 3.1A - 3.1E.



(b) Cases 3.2A - 3.2C.

Fig. 4.22: 11th harmonic voltage propagation in region II for all study cases. (a) - Propagated voltage for study cases 3.1A - 3.1E. (b) - Propagated voltage for study cases 3.2A - 3.2C.

voltage is caused by an increase in the reflection coefficient ($\underline{S}_{11}$) from 0.86 to 0.92 at its maximum. Similarly, the voltage transfer ($\underline{S}_{21}$) decreases from 0.51 to 0.38. Note that these S-parameters can be seen in Table F.8 in Section 3 of Appendix F. Because of the increased $\underline{S}_{11}$, more voltage propagates from bus S6 towards bus S7 and similarly, the transferred voltage from bus S6 to bus S5 decreases due to decreasing $\underline{S}_{21}$. As $\underline{S}_{22}$ also increases, more voltage incident from bus S4 also propagates from bus S5 towards bus S2.

In study case 3.1E, line S5-S6 is a pure UGC line consisting of two parallel UGG of approximately 80 km length. When compared to the reference case, the change in propagated voltage is like study case 3.1B at buses S6 and S7, but it is reduced at buses S5 and S4 and increased at bus S3. Parameter $\underline{S}_{11}$ is low (0.16) in comparison to study cases 3.1A-3.1D, whereas the transmission coefficient $\underline{S}_{21}$ is high (0.94). Because of the high transmission coefficient $\underline{S}_{21}$ (or $\underline{S}_{12}$), more voltage is transferred from bus S6 to bus S5 (or vice versa). The direction of $\underline{S}_{21}$, however, is shifted significantly (from approximately -36 degrees to approximately -170 degrees, according to Table F.8, and thus the sum of propagated voltages at buses S5 - S2 for this study case is opposite to study cases 3.1A - 3.1D, because the voltage is a superposition of all propagated voltages at buses. The influence of the line length on the transmission coefficient was discussed in the previous subsection about S-parameter evaluation.

From Figure 4.22a, there is no change in voltage propagation between substations S9 - S12. This means that region II appears to be distinct from regions I and III in terms of harmonic voltage propagation, implying that no changed terminal conditions exist. According to Table F.6, the S-parameters for line S3-S12 are $|\underline{S}_{11}| \approx 0.05$ and $|\underline{S}_{21}| \approx 1.0$, resulting in almost no reflection on this line and thus no change in propagation. The same is true for the S6-S9 line.

Evaluation of Study Cases 3.2A – 3.2C: Figure 4.21b shows the difference in propagated voltage for study cases 3.2A – 3.2C, where a crosswise UGC line is introduced between buses S6 and S10. In this case, the difference in propagated voltage increases at bus S6 in study cases 3.2A - 3.2B but decreases for study case 3.2C. The new connection between bus S6 and bus S10 is a UGC line of 20 or 25 kilometres in length.

From the sensitivity analysis as part of the S-parameter evaluation discussed in the previous subsection, a UGC of this length has a high reflection coefficient $\underline{S}_{11}$ (approximately 0.86 – 0.89 according to Table F.8) and a relatively modest transfer coefficient $\underline{S}_{21}$ (approximately 0.45 – 0.51 according to Table F.8). Therefore, the propagated voltage at bus S6 is primarily reflected and hence, not transferred to bus S10. Therefore, the difference in propagated voltage at bus S6 is high as compared to the reference case.

The UGC section in line S5-S6 is increased to 20 kilometres in study case 3.2C, as in study case 3.1A. $\underline{S}_{11}$ for the UGC section is increased significantly (from approximately 0.55 to 0.86) in this case, and more voltage is reflected at bus S6. However, because the direction of $\underline{S}_{11}$ has also been shifted, the resulting voltage is lower than in the reference case. This means that the propagation is influenced by the individual UGC lengths connected to a substation. The magnitude of the propagated voltage increases when all cable lengths are less than $\frac{1}{4}\lambda$.

Figure 4.22b shows the difference in propagated voltage along S9 - S12, with a significant change at bus S10 when the crosswise connection is introduced. This is because when this new connection is inserted, a significant difference compared to the reference case is obvious because the reference case value is zero. However, despite the change in UGC length between S6-S10, the propagated voltage is nearly identical for the three subcases, owing to the minor changes in $\underline{S}_{11}$ and $\underline{S}_{21}$ for this cable length, as discussed in the previous subsection.

Summary

The method described here can be used to evaluate harmonic propagation in the passive grid using the following principles:

- Variations in propagated voltage $\underline{b}$ in the system can be evaluated against a reference case or between individual cases.
- Evaluation can be performed as a detailed study for many study cases to evaluate minor changes such as line length or line type variations, or as a network screening study with many contemporary changes. The impact of network changes on voltage propagation between points in the grid or between sources and points in the grid can be understood.
- The S-parameters describe the specific network response to the network changes.
- The Western Danish transmission grid is divided into three regions in terms of harmonic propagation, which means that changes in regions I and III have no or little effect on propagation in region II.
- The magnitude of the propagated voltage is influenced by the individual UGC lengths connected to a substation. The consequence of cable lengths on propagated voltage can be understood.

Voltage Amplification at Buses

Changes in bus voltage as a result of network changes presented in the study cases in Table 4.13 can be evaluated as part of this propagation study. The propagation model is an important property for a screening study at start of a harmonic assessment study. The effect of each individual network change on the harmonic bus voltage can thus be evaluated.

The voltage is the sum of the incident and reflected voltages at a node in the system, so $\underline{u} = \underline{a} + \underline{b}$. Because $\underline{a}$ and $\underline{b}$ have constant amplitudes when travelling along a line, the magnitude of the propagation coefficient is fixed and only the direction varies as the waves travel along a line, according to the definitions presented in Section 2.2 of this chapter.

According to the modeling approach described in Section 5 of this chapter, the response $\underline{b}$ of a network component model is the incident wave $\underline{a}$ on the subsequent network component model. As a result, at each node, the ratio of $\underline{a}$ to $\underline{b}$ is fixed, and the ratio of change of $\underline{b}$ compared to the reference case is the same as the change in voltage $\underline{u}$ at the node under evaluation. These conditions hold true for steady-state analysis.

The *propagation ratio* χ is defined in this thesis (Table 4.12) as a measure of the relative voltage change, or voltage amplification, according to the definition given in Section 4 of Chapter 2, where χ is the propagation ratio in relation to the reference case at substation s consisting of nodes f to g, evaluated for case m. χ is repeated here:

$$\chi^s_{f:g,m} = \frac{|\sum \underline{b}^s_{f:g,m}|}{|\sum \underline{b}^s_{f:g,1}|} = \frac{|\sum \underline{b}^s_{f,m} + \sum \underline{b}^s_{f+1,m} + \ldots + \sum \underline{b}^s_{g-1,m} + \sum \underline{b}^s_{g,m}|}{|\sum \underline{b}^s_{f,1} + \sum \underline{b}^s_{f+1,1} + \ldots + \sum \underline{b}^s_{g-1,1} + \sum \underline{b}^s_{g,1}|} \tag{4.68}$$

where $\sum \underline{b}_{f:g,m}$ is the sum of propagated voltages from all sources at nodes $f : g$, evaluated for study case m.

In this case study, the 11th harmonic propagation ratio for the study cases presented in Table 4.13 is calculated using (4.68), and the results are shown in Table 4.17. For comparison, the reference case propagation voltage $|\sum \underline{b}_{3.0}|$ and harmonic voltage $|\sum \underline{u}_{3.0}|$ are given (as study case 3.0 in the table).

Table 4.17: Calculated 11th harmonic propagation ratio χ compared to the reference case at substation S1, S3, S4, S5, S6, S7, S10 and S11 for study cases in Table 4.13. The reference case (3.0) propagated voltage $|\sum \underline{b}_{3.0}|$ and terminal voltage $|\sum \underline{u}_{3.0}|$ are given.

ID / Case	S1	S3	S4	S5	S6	S7	S10	S11
$\lvert\sum \underline{b}_{3.0}\rvert$	1.05	2.32	1.13	1.35	1.88	0.03	15.00	15.89
$\lvert\sum \underline{u}_{3.0}\rvert$	1.05	1.16	1.13	1.35	0.63	0.03	14.38	15.89
3.1A	0.97	0.84	1.51	1.07	0.85	35.81	1.00	1.00
3.1B	0.96	0.77	1.74	1.11	1.51	49.02	1.00	1.00
3.1C	0.95	0.70	1.97	1.14	2.14	61.53	1.00	1.00
3.1D	0.94	0.65	2.13	1.16	2.92	77.12	1.00	1.00
3.1E	1.10	1.57	0.82	0.73	1.52	49.13	1.00	1.00
3.2A	1.05	1.29	0.07	0.86	3.08	42.20	0.05	0.03
3.2B	1.05	1.26	0.15	0.88	2.87	38.01	0.03	0.05
3.2C	1.09	1.51	0.63	0.76	0.88	36.39	0.22	0.28

In this evaluation, only the results related to substation S6 and S10 are discussed (highlighted in the table), because these results can easily be interpreted when combined with the discussion on voltage propagation above. However, a note to the very high values calculated for substation S7 is that the reference case voltage at this bus is extremely low (0.03 per unit) as seen from Table 4.17.

Figure 4.21 showed that the propagated voltage at substation S6 was less than the reference case voltage for study cases 3.1A and 3.2C (negative values). As a result of the decreased voltage propagation, the expected harmonic voltage for these two grid configurations can be expected to be less than the reference voltage, i.e., approximately 85% and 88% of the reference case voltage, as shown in Table 4.17. The opposite is the case for the other study cases. Here, the harmonic voltage is expected to increase 151% - 308%, most severe for study case 3.2A. This is also expected compared to the voltage propagation shown in Figure 4.21b, where the voltage difference for these cases has the highest value of the three cases 3.2A – 3.2C (approximately 3.0 per unit).

Table 4.17 shows that when the crosswise connection S6-S10 is established, there is a significant reduction in voltage at substation S10. Due to the superposition of propagated voltage from S6 to S10, as well as contributions from substations S9, S11, and S12, the voltage is reduced to values ranging from approximately 3% to 22% in study cases 3.2A - 3.2C. Similarly, as previously discussed, the voltage at S6 increases 151% - 308% due to the new contribution from S10.

To summarise, the expected change in harmonic voltage at substations (or other nodes under evaluation) as a result of specific network changes can be calculated using the propagated voltages $\underline{b}$ across the grid and the propagation ratio χ. The sum of propagated voltages at a substation was evaluated, but the same method applies to each individual source contribution. Instead of (4.68), the propagation ratio in this case is:

$$\chi^{s,k}_{f:g,m} = \frac{|\sum \underline{b}^{s,k}_{f:g,m}|}{|\sum \underline{b}^{s,k}_{f:g,1}|} = \frac{|\underline{b}^{s,k}_{f,m} + \underline{b}^{s,k}_{f+1,m} + \ldots + \underline{b}^{s,k}_{g-1,m} + \underline{b}^{s,k}_{g,m}|}{|\underline{b}^{s,k}_{f,1} + \underline{b}^{s,k}_{f+1,1} + \ldots + \underline{b}^{s,k}_{g-1,1}\underline{b}^{s,k}_{g,1}|} \tag{4.69}$$

where k is the source index under consideration.

It should be noted that the magnitude of all sources in this evaluation is set to 1 per unit and the direction is set to 0 degrees. The angular rotation of sources influences the harmonic level at a node or substation, as discussed in Chapter 2, which is an important consideration for harmonic assessment studies. The focus in this case is on harmonic propagation in the passive network and the possible contribution of individual sources to propagation at a given point in the network, which is why the method is applicable to this type of study.

Summary

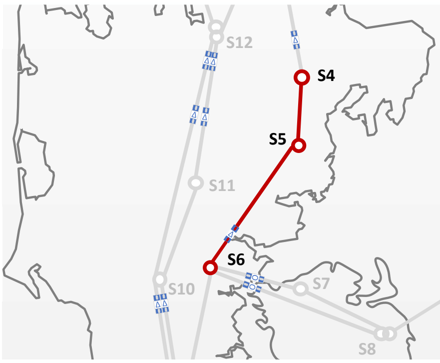
The method described here can be used to evaluate the propagation ratio in the passive grid using the following principles:

- The ratio of change χ of voltage at specific nodes or substations can be evaluated against a reference case or between individual cases.

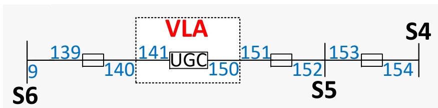
- Evaluation can be performed as a detailed study for many study cases to evaluate minor changes such as line length or line type variations, or as a network screening study with many contemporary changes.
- To compare across study cases, the evaluation is performed using sources with fixed magnitude and direction.
- The propagation ratio χ describes the amplification at a node as defined in Section 4.2 of Chapter 2.

Evaluating Voltage Extremes

Section 2 of this chapter discussed the behaviour of standing waves and provided an illustrative example to demonstrate the importance of the reflection coefficient $\underline{\Gamma}$ for the voltage wave shape and how minimum and maximum voltage values along a line are identified. These parameters are quantified for a section of the transmission grid used in this study. The minimum and max-



(a) Segment of transmission grid.



Case	OHL [km]	UGC [km]	OHL [km]	OHL [km]
3.1A	12.8	7	59.6	30.3
3.1B	12.8	20	46.6	30.3
3.1C	12.8	30	36.6	30.3
3.1D	12.8	40	26.6	30.3
3.2A	12.8	7	59.6	30.3
3.2B	12.8	7	59.6	30.3
3.2C	12.8	20	46.6	30.3

(b) Corresponding network diagram.

Fig. 4.23: Part of the transmission grid used for evaluating expected minimum and maximum voltage values along transmission lines. (a) – Segment of the Western Danish transmission grid (highlighted). (b) - Network diagram with node ID and line length based on the study cases being evaluated. Line section length in kilometres given in table.

imum voltages, as well as the distance between a substation and its respective location along line S4-S5, and a section of line S5-S6 from S5 to the UGC section at node 150 (according to the network diagram provided in Figure F.2), are calculated.

Figure 4.23 depicts the transmission grid segment as well as a single line diagram with the corresponding node ID and line parameters from the propagation model. The evaluation in this study is limited to the 11th harmonic voltage. The study cases 3.1A–3.1D and 3.2A–3.2C, as well as the reference case, are used for this evaluation.

For each study case, the incident and propagated voltage $\underline{a}$ and $\underline{b}$ determined with the propagation model are used as reference voltages at the node under evaluation. These values are given in Table 4.18 for nodes 152 and 154 for each study case.

Table 4.18: Complex values of incident and propagated voltage $\underline{a}$ and $\underline{b}$ at nodes 152 and 154 obtained for each study case. Values are used for evaluation of standing wave patterns.

Case	$\underline{a}_{152}$	$\underline{b}_{152}$	$\underline{a}_{154}$	$\underline{b}_{154}$
3.0	-0.5575 - 0.4622i	-0.5757 + 0.4619i	-0.6684 - 0.2216i	-0.6857 + 0.2171i
3.1A	-0.8818 + 0.2261i	-0.8326 - 0.2647i	-0.7580 + 0.5434i	-0.6969 - 0.5659i
3.1B	-1.0320 + 0.5342i	-0.9439 - 0.5935i	-0.8030 + 0.8875i	-0.6973 - 0.9186i
3.1C	-1.1926 + 0.8266i	-1.0365 - 0.9183i	-0.8630 + 1.2198i	-0.6809 - 1.2615i
3.1D	-1.3519 + 1.0160i	-1.0570 - 1.1684i	-0.9548 + 1.4531i	-0.6194 - 1.5088i
3.2A	-0.0679 - 1.6180i	-0.1047 + 1.6429i	-0.5706 - 1.4886i	-0.6160 + 1.5070i
3.2B	0.2913 - 2.7182i	0.4204 + 2.6908i	-0.5801 - 2.6600i	-0.4509 + 2.6865i
3.2C	0.3466 - 2.9945i	0.5774 + 2.9291i	-0.6160 - 2.9431i	-0.3771 + 2.9670i

To begin, standing voltage waves along 600 kilometres of OHL are presented for completeness in order to demonstrate the difference between study cases from a large-scale perspective and illustrate the general behaviour of the standing waves for a full wavelength line. In this example, the reflection $\underline{S}_{11}$ at the UGC section at node 150 for each study case are used to construct the voltage waves.

The standing voltage wave are constructed using (4.26) and Figure 4.24 depicts the waves for each study case. The voltage minimums all occur at different distances along the line, and the distance between the minimum points for reference case 3.0 and case 3.1A, for instance, is approximately 70 kilometres. Because this is a significant span, the voltage at a specific node at a specific distance x along the line also varies within a certain range. This will be discussed later in this section.

It can also be seen that the voltage magnitude varies along the line. The length between minimum and maximum voltage magnitudes is $\frac{1}{2}\lambda$, as was discussed for the numerical example given in Section 2.5 of this chapter.

The distance to the maximum and minimum voltages as seen from $x = 0$ can be calculated using (4.28) and (4.29).

Evaluation of Standing Voltage Wave Along Unchanged Line: The parameters of the line S4-S5 do not change in any of the study cases, but the voltage along the line is affected by changes in other parts of the grid. The standing

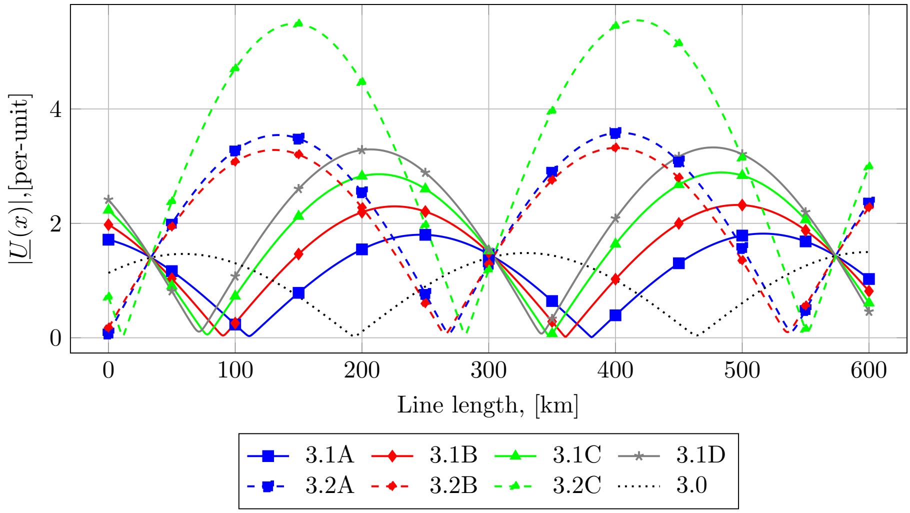


Fig. 4.24: Standing voltage waves along 600 km of OHL for study cases 3.1A – 3.1D, 3.2A – 3.2C and the reference case 3.0 according to Table 4.18.

voltage wave along this line is depicted in Figure 4.25 for the study cases. The line is about 30 kilometres long, and the reference incident voltages listed in Table 4.18 are used. Equation (4.26) is used to compute the standing wave pattern. The standing wave pattern along line S4-S5 for study cases 3.0, 3.1A-

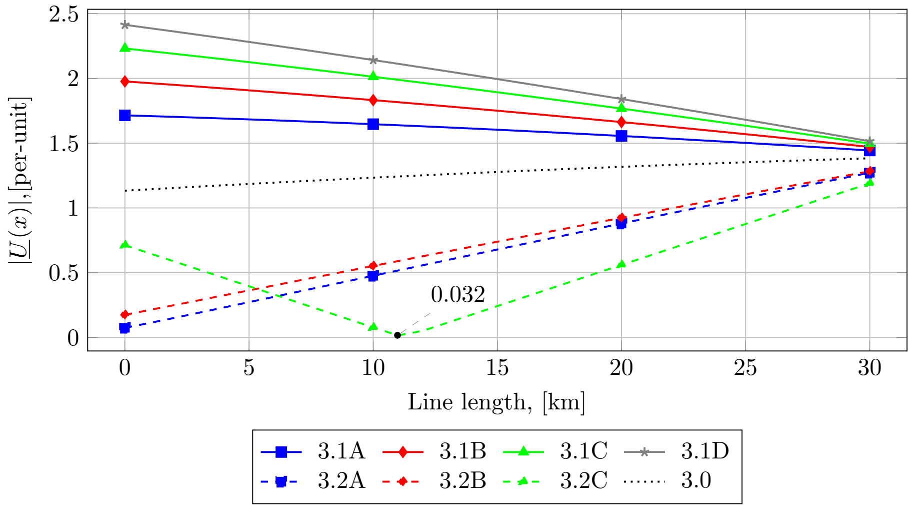


Fig. 4.25: Standing wave pattern along line S4-S5 for study cases 3.0, 3.1A-3.1D and 3.2A-3.2C. Substation S4 is located at zero kilometres and substation S5 is located at 30 kilometres. Minimum values for cases 3.2B and 3.2C are given.

3.1D and 3.2A-3.2C are shown. Substation S4 is located at zero kilometres and substation S5 is located at 30 kilometres. A single minimum point is seen at approximately 11 from substation S4 and it is related to study case 3.2C.

The voltage range from an expected minimum voltage to an expected maximum voltage can be read to be approximately 0.1 per unit (case 3.2A) to 2.4 per unit (case 3.1D) at a distance of zero kilometres, which is substation S4. This isn't particularly interesting in and of itself, because these values have already been determined by the propagation ratio χ discussed in the previous subsection.

More interestingly, it is seen that the range of expected voltage at the end of line S4-S5, that is, at substation S5, lies within the range of approximately 1.2 per unit (case 3.2C) and 1.5 per unit (case 3.1D) for all study cases under evaluation, and most likely, for this set of study cases, the applied network changes will not cause severe voltage variations at this bus. Although the term "severe" is debatable, if a specific voltage range is defined, this range could be used for evaluation. This study, however, does not specify such a range.

Evaluation of Standing Voltage Wave Under Changed Line Conditions: The length of the OHL section of line S5-S6 changes for study cases 3.1A – 3.1D, and 3.2C as compared to the reference case 3.0. For all study cases, the terminal conditions changes, either at the UGC section or at substation S6. These changes affect the standing voltage wave along both lines S4-S5 and S5-S6. Figure 4.26 depicts the standing voltage wave along S5 – node 150 for the study cases under evaluation, where S5 is located at $x = 0$. The voltage magnitude at the end of the line section (at node 150) is shown for each case.

From the reference case, the line is about 60 kilometres long, and the reference voltages listed in Table 4.18 are used. Equation (4.26) is used to compute the standing wave pattern.

The voltage for study cases 3.1A-3.1D decreases towards a minimum voltage value beyond a line length of 40 kilometres from S5, as shown in this figure. However, because the OHL section length decreases with increasing UGC section length (as shown in the table in Figure 4.23b), the resulting voltage at the end of the OHL section should be read at the corresponding line section distance from S5. In this case, the voltage never reaches a possible extreme minimum value, and the expected variation is between 0.63 and 0.99 per-unit, which is less than the reference case voltage value of 1.32 per-unit.

In comparison to study case 3.1, the voltage increases in study cases 3.2A-3.2C. This is due to the significant changes at substation S6, which was previously discussed when evaluating S-parameters. When comparing subcases

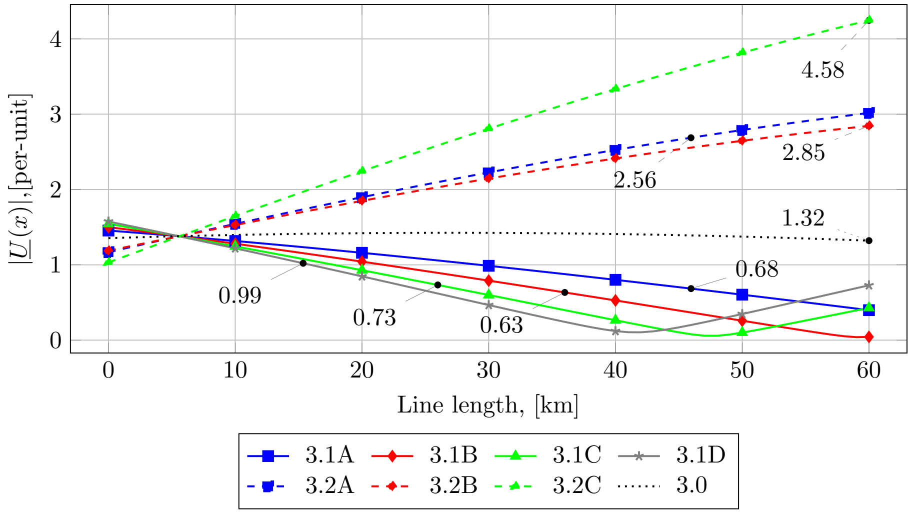


Fig. 4.26: Standing wave pattern along lines S5–S6 up to node 150 for study cases 3.0, 3.1A–3.1E, and 3.2A–3.2C. Substation S5 is located at zero kilometres. The voltage at the end of the OHL section (at node 150) for each study case is shown in the figure.

in 3.2 to subcases in 3.1, the slope of these three voltage curves rises rapidly, resulting in an expected voltage range of 2.56 - 4.58 per-unit, which is significantly higher than for study cases 3.1A - 3.1D.

To put this analysis into context, suppose the TSO needed to evaluate the harmonic distortion level at this location (at the end of the OHL section, that is, at node 150) for analysis of the impact of distortion level on component ageing and was required to know how these grid configurations influenced the expected harmonic level at this location. Figure 4.26 depicts a specific voltage range that is most likely to affect this location. Another important observation is that harmonic resonance is present at extremes, such as a possible minimum or maximum point, and the line distance to resonances can be calculated. This is, however, not a part of the evaluation in this thesis. The results also indicate whether certain grid configurations or line lengths, i.e., study cases, possibly should be avoided.

Summary

Evaluating the standing voltage waves along the lines for a set of study cases allows for the quantification of expected voltage ranges in the passive grid due to specific network configurations using the following principles:

- The expected minimum and maximum voltages and the distance from the point of evaluation can be determined using (4.28) - (4.29).

- The expected rate of change in voltage at specific distances along a line can be quantified for each study case and compared to a reference case or between individual cases.
- Specific voltage ranges or possible extreme ranges can be identified, allowing for the identification and rejection of potentially problematic grid configurations due to problematic harmonic propagation phenomena prior to detail-specific analyses, reducing the number of studies required.
- The presence of resonances along the lines can be assessed because they are linked to the location of voltage extremes.
- To compare across study cases, the evaluation is performed using sources with fixed magnitude and direction.

Evaluating Source-Specific Voltage Propagation

The evaluation of source-specific voltage propagation in this case concerns study cases 3.1A, 3.1E, and 3.2A. The propagation coefficient is used to assess the propagation of voltages across the grid. This evaluation can be source-specific or based on the total of contributions. The evaluation is related to specific grid nodes or busbars. The propagation coefficient was initially defined in [4.24] as:

$$\underline{\zeta}_j = \frac{\sum \underline{u}_j^-}{\sum \underline{u}_j^+}. \tag{4.70}$$

At node j, the propagation coefficient $\underline{\zeta}_j$ is defined as the ratio of the sum of reflected voltage waves $\sum \underline{u}_j^-$ to the sum of incident voltage waves $\sum \underline{u}_j^+$ [4.24]. According to Section 4 in this chapter, the designation $\underline{a}_j$ is used for incident waves and $\underline{b}_j$ is used for reflected or propagated waves, so (4.70) is defined as follows:

$$\underline{\zeta}_j := \frac{\sum \underline{b}_j}{\sum \underline{a}_j} \tag{4.71}$$

For source-specific evaluation, the change in $\underline{\zeta}_j$ for a specific study case compared to the reference case is used and this difference is defined according to [4.24] as:

$$\Delta\underline{\zeta}_{j,1m} := \underline{\zeta}_{j,1} - \underline{\zeta}_{j,m} \tag{4.72}$$

where $\Delta\underline{\zeta}_{j,1m}$ represents the propagation coefficient difference evaluated at node j for case m, $\underline{\zeta}_{j,1}$ represents the propagation coefficient computed by (4.71) at node j for study case 1 (reference case), and $\underline{\zeta}_{j,m}$ represents the propagation coefficient computed by (4.71) at node j for study case m. This means that $\Delta\underline{\zeta}_{j,1m}$ is the sum of the difference in propagated voltages determined at node j in the network. $|\Delta\underline{\zeta}_{j,1m}|$ can be evaluated as follows:

- If $|\Delta\underline{\zeta}_{j,1m}| < 1$ then, according to the definition in (4.71), $|\sum \underline{a}_i| > |\sum \underline{b}_i|$, which means that more voltage is transmitted into the subsequent network element than reflected from it, and thus more voltage propagates through the node for the study case under evaluation than the reference case.
- If $|\Delta\underline{\zeta}_{j,1m}| > 1$, then $|\sum \underline{b}_i| > |\sum \underline{a}_i|$, implying that more voltage is reflected from this node and thus less voltage propagates through the node for the study case under consideration when compared to the reference case.
- If $|\Delta\underline{\zeta}_{j,1m}| = 0$, then $|\sum \underline{b}_1 / \sum \underline{a}_1|$ equals $|\sum \underline{b}_m / \sum \underline{a}_m|$, indicating that no change in propagated voltage occurs at the evaluated node. This means that the ratio of the voltage propagating through the node for the study case under consideration is the same as the ratio of the voltage propagating through the node for the reference case.

The propagation coefficient difference given by (4.72) can be augmented to evaluate the propagation behaviour from each individual source, say, N sources in the grid, and for a set $f : g$ of nodes as:

$$\Delta\underline{\zeta}^{s,k}_{f:g,1m} = \begin{bmatrix} \underline{\zeta}^{s,k}_{f,1} - \underline{\zeta}^{s,k}_{f,m} \\ \underline{\zeta}^{s,k}_{f+1,1} - \underline{\zeta}^{s,k}_{f+1,m} \\ \vdots \\ \underline{\zeta}^{s,k}_{g-1,1} - \underline{\zeta}^{s,k}_{g-1,m} \\ \underline{\zeta}^{s,k}_{g,1} - \underline{\zeta}^{s,k}_{g,m} \end{bmatrix} \tag{4.73}$$

Where $f : g$ represents the range of evaluated nodes, m represents the study case under consideration, s represents the substation ID, and k represents the source ID. Defining $\underline{h}^{s,k}_{1(f:g),m} = \underline{\zeta}^{s,k}_{f:g,1} - \underline{\zeta}^{s,k}_{f:g,m}$ (4.73) can be written as follows for N sources:

$$\Delta\underline{\zeta}^{s,k}_{f:g,1m} = \left.\begin{bmatrix} \underline{h}^{s,1}_{1(f),m} & \underline{h}^{s,2}_{1(f),m} & \cdots & \underline{h}^{s,N-1}_{1(f),m} & \underline{h}^{s,N}_{1(f),m} \\ \underline{h}^{s,1}_{1(f+1),m} & \underline{h}^{s,2}_{1(f+1),m} & \cdots & \underline{h}^{s,N-1}_{1(f+1),m} & \underline{h}^{s,N}_{1(f+1),m} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \underline{h}^{s,1}_{1(g-1),m} & \underline{h}^{s,2}_{1(g-1),m} & \cdots & \underline{h}^{s,N-1}_{1(g-1),m} & \underline{h}^{s,N}_{1(g-1),m} \\ \underline{h}^{s,1}_{1(g),m} & \underline{h}^{s,2}_{1(g),m} & \cdots & \underline{h}^{s,N-1}_{1(g),m} & \underline{h}^{s,N}_{1(g),m} \end{bmatrix}\right|_{k=1,2,\ldots,N} . \tag{4.74}$$

In this thesis, (4.74) is used to evaluate the propagation coefficient at substations S3 and S6 for the four sources and nine study cases. Using the node ID's of Table 4.15, (4.74) applied to substation S3 is:

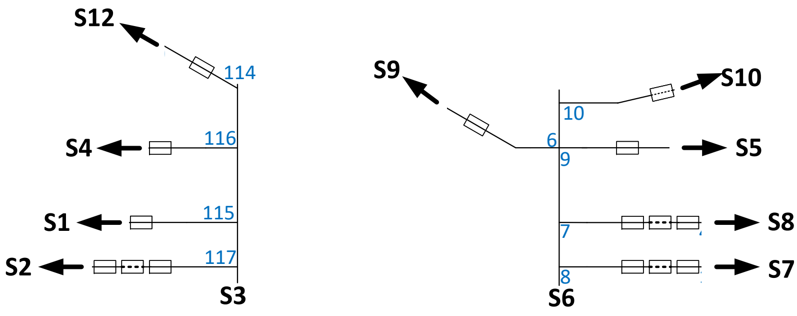
$$\Delta\underline{\zeta}^{S3,k}_{114:117,1m} = \begin{bmatrix} \underline{h}^{S3,1}_{1(114),m} & \underline{h}^{S3,2}_{1(114),m} & \underline{h}^{S3,3}_{1(114),m} & \underline{h}^{S3,4}_{1(114),m} \\ \underline{h}^{S3,1}_{1(115),m} & \underline{h}^{S3,2}_{1(115),m} & \underline{h}^{S3,3}_{1(115),m} & \underline{h}^{S3,4}_{1(115),m} \\ \underline{h}^{S3,1}_{1(116),m} & \underline{h}^{S3,2}_{1(116),m} & \underline{h}^{S3,3}_{1(116),m} & \underline{h}^{S3,4}_{1(116),m} \\ \underline{h}^{S3,1}_{1(117),m} & \underline{h}^{S3,2}_{1(117),m} & \underline{h}^{S3,3}_{1(117),m} & \underline{h}^{S3,4}_{1(117),m} \end{bmatrix} \tag{4.75}$$

and, for substation S6:

$$\Delta\underline{\zeta}^{S6,k}_{6:10,1m} = \begin{bmatrix} \underline{h}^{S6,1}_{1(6),m} & \underline{h}^{S6,2}_{1(6),m} & \underline{h}^{S6,3}_{1(6),m} & \underline{h}^{S6,4}_{1(6),m} \\ \underline{h}^{S6,1}_{1(7),m} & \underline{h}^{S6,2}_{1(7),m} & \underline{h}^{S6,3}_{1(7),m} & \underline{h}^{S6,4}_{1(7),m} \\ \underline{h}^{S6,1}_{1(8),m} & \underline{h}^{S6,2}_{1(8),m} & \underline{h}^{S6,3}_{1(8),m} & \underline{h}^{S6,4}_{1(8),m} \\ \underline{h}^{S6,1}_{1(9),m} & \underline{h}^{S6,2}_{1(9),m} & \underline{h}^{S6,3}_{1(9),m} & \underline{h}^{S6,4}_{1(9),m} \\ \underline{h}^{S6,1}_{1(10),m} & \underline{h}^{S6,2}_{1(10),m} & \underline{h}^{S6,3}_{1(10),m} & \underline{h}^{S6,4}_{1(10),m} \end{bmatrix} \tag{4.76}$$

Evaluating Propagation Patterns at Substations S3 and S6: Figure 4.27 depicts the network diagram for S3 and S6, respectively. The network diagram for substation S3 is shown in Figure 4.27a, with nodes 114-117 designated as its ports. Figure 4.27b depicts the network diagram for substation S6, with nodes 6-10 designated as its ports. Lines to neighbouring substations are also depicted in each figure.

The propagation coefficient is calculated for each source contribution and each port at substations S3 and S6 using (4.75) and (4.76), respectively. Remember from earlier that if propagated voltage ratio does not change, $\underline{\zeta}$ is zero. As a result, only non-zero values of $\underline{\zeta}$ must be evaluated.



(a) Substation S3.

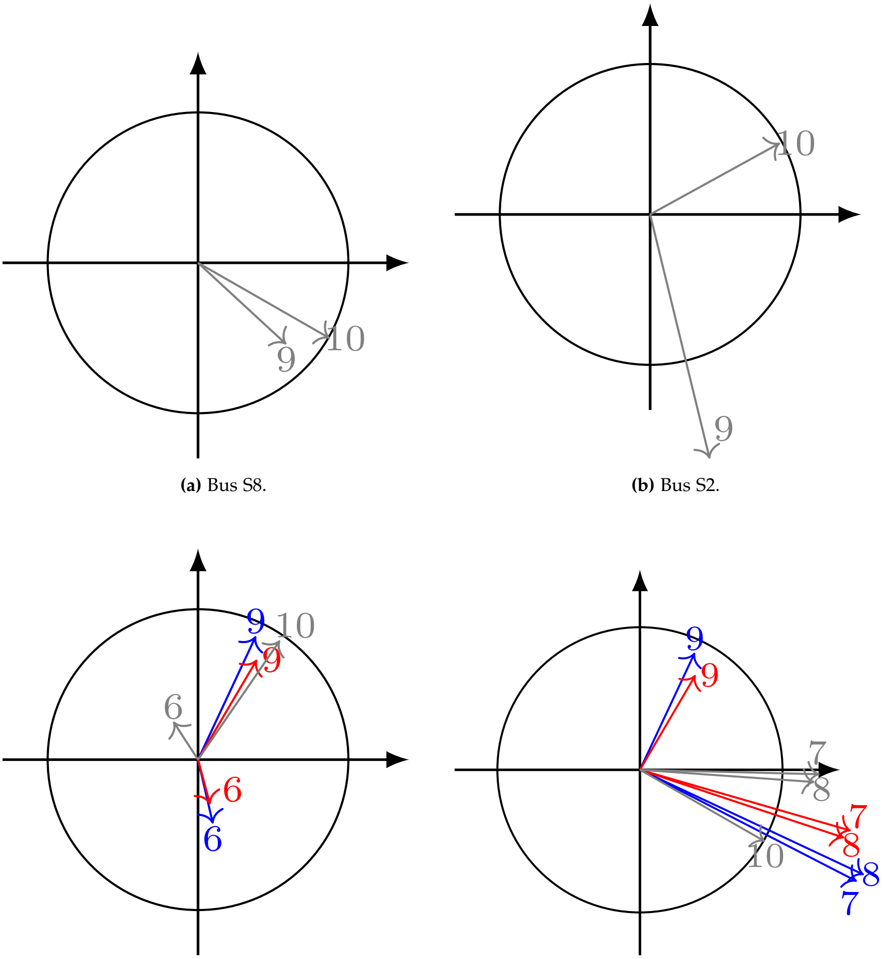
(b) Substation S6.

Fig. 4.27: Network diagram for substations S3 and S6. Node ID is shown. (a) - Substation S3 with node ID 114-117. (b) - Substation S6 with node ID 6-10.

Propagation Pattern at Substation S6: The magnitude and direction of the propagation coefficient due to voltage propagation from each of the four sources are depicted in Figure 4.28. The source at bus S8 is depicted in Figure 4.28a, the source at bus S2 is depicted in Figure 4.28b, the source at bus S12 is depicted in Figure 4.28c, and the source at bus S9 is depicted in Figure 4.28d. The blue phasor represents study case 3.1A, the red phasor represents study case 3.1E, and the grey phasor represents study case 3.2A. The circle in the diagrams corresponds to $\underline{\zeta} = 1$, that is, a circle with radius one.

The propagation coefficient related to the sources located at bus S9 and bus S12 is affected in study cases 3.1A, 3.1E, and 3.2A. When compared to the reference case, the source contribution from S12 at ports 6 and 9 is less than 1, indicating that less voltage is reflected and thus less voltage propagates back along the lines connected to ports 6 and 9. The same property is observed at port 6 in case 3.2A (grey), whereas at port 10, the propagation coefficient is close to one. This is because, in this case, a new line is connected at this port towards substation S10. The magnitude of the phasors represents the ratio of change, or how large the changed propagation pattern for the specific source at the ports is because of the specific grid change. Another observation is that only the propagation along lines S6-S10 and S5-S6 is affected by this source, as well as line S6-S10 for study case 3.2A, which is a new line connection between these two substations.

For the source located in S9, propagation is affected at ports 7, 8, and 9 for study cases 3.1A (blue) and 3.1E (red), and ports 7, 8, and 10 for study case 3.2A. Because the phasors at ports 7 and 8 exceed one, more voltage is reflected and propagates back on the lines connected to these ports. This means that when longer UGC are introduced beyond substation S6 (towards S5), more voltage is reflected and, as a result, the voltage at the source bus will most likely increase in this case. Table 4.17 shows that this is the case, as



(a) Bus S8. (b) Bus S2. (c) Bus S12. (d) Bus S9.

Fig. 4.28: Magnitude and direction of $\underline{\zeta}$ at substation S6 as a consequence of voltage propagation from each of the four sources. Blue phasor represents study case 3.1A, red phasor represents study case 3.1E, and grey phasor represents study case 3.2A. The circle corresponds to $\underline{\zeta} = 1$. (a) - Source located at S8. (b) - Source located at S2. (c) - Source located at S12. (d) - Source located at S9.

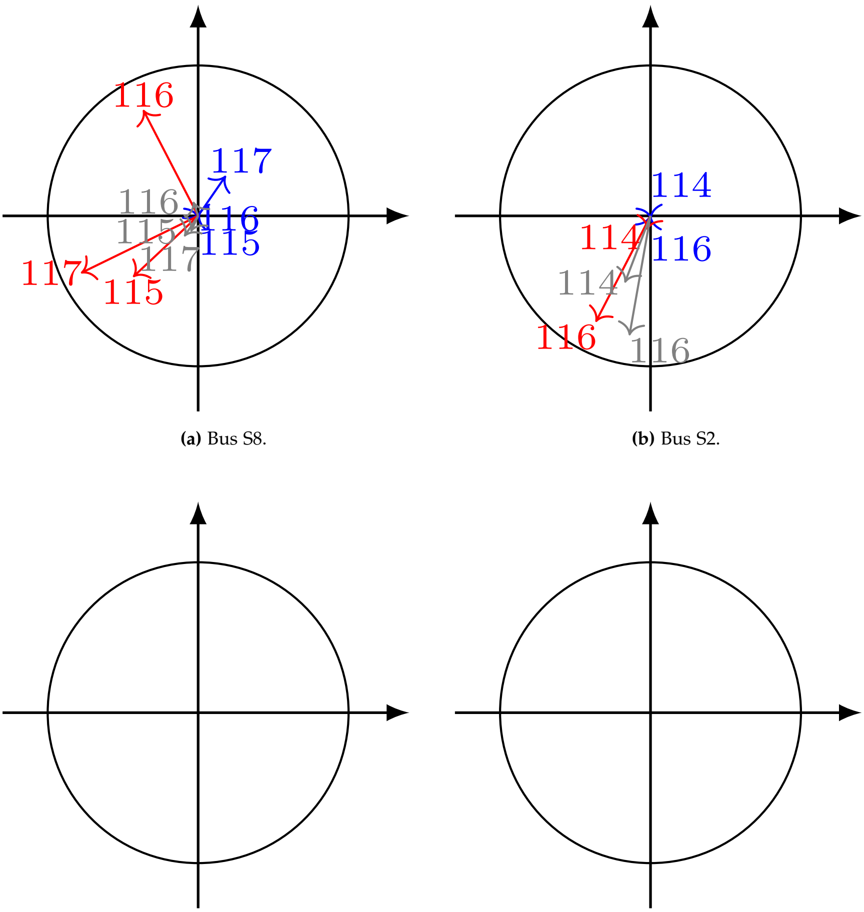
the propagation ratio increases at bus S7, which is in the grid between bus S6 and the source bus S8. When the grid changes in study cases 3.1 and 3.2 are implemented, the result is less voltage propagating along the line connected at port 9.

In terms of the sources at buses S2 and S8, the propagation pattern at bus S6 appears to be affected only for study case 3.2A. The propagation along the line connected at port 9 is affected because of the new line connected at port 10. The source contribution from S2 causes more reflection at port 9, resulting in more voltage propagating back on this line to substation S9.

Propagation Pattern at Substation S3: Figure 4.29 depicts the propagation coefficients calculated for each source contribution at the ports of substation S3. In this case, no phasors are visible in Figure 4.29c for the source located on bus S12, nor in Figure 4.29d for the source located on bus S9. This means that, despite grid changes, the propagation from these two sources remains unchanged at substation S3. This is consistent with the findings in [4.8], in which the standing voltage wave patterns were only affected in the changed portion of the grid. This was also covered in Section 2.2 of Chapter 2. The propagation model presented in [4.24] (paper **J1** of this thesis) has also shown that for the reference case, there is only a minor change in propagated voltage from substation S12 towards substation S4, confirming that the transfer gain factor between S12 and S4 is not accurate since there is no significant change in voltage propagation between these stations.

However, the propagation from the sources located in both S2 and S8 is affected at substation S3. The propagation coefficient is changed in all three study cases. The propagation of voltage by the source in S2 is affected along lines towards S4 (port 116) and S12 (port 114), indicating that less voltage is reflected; however, the impact is minor for study case 3.1A (blue) than for study case 3.1E (red), and the impact is greatest towards bus S4 at port 116 for 3.1E. The propagation path towards the line insertion point S6-S10 is affected on both paths in study case 3.2A, as the propagation coefficient at ports 114 and 116 is changed.

Because of the increased UGC length along the path towards S3 as seen from S8, propagation is mostly affected for study case 3.1E (red) for the source located at bus S8. Less voltage reflection is observed as a result.



(a) Bus S8.

(b) Bus S2.

(c) Bus S12.

(d) Bus S9.

Fig. 4.29: Magnitude and direction of $\underline{\zeta}$ at substation S3 as a consequence of voltage propagation from each of the four sources. Blue phasor represents study case 3.1A, red phasor represents study case 3.1E, and grey phasor represents study case 3.2A. The circle corresponds to $\underline{\zeta} = 1$. (a) - Source located at S8. (b) - Source located at S2. (c) - Source located at S12 (empty due to unchanged propagation patterns). (d) - Source located at S9 (empty due to unchanged propagation patterns).

Summary

Evaluating the propagation coefficient at busbars for a set of study cases allows for the quantification of expected voltage propagation changes from each source due to specific network configurations. The following principles apply:

- The propagation coefficient at a node or port can be calculated using (4.72).
- If the propagation coefficient is zero, no changes in propagation ratio is observed for the evaluated case compared to the reference case.
- If the propagation coefficient is less than one, more voltage is transmitted through the substation for the evaluated case compared to the reference case.
- If the propagation coefficient is more than one, less voltage is transmitted through the substation and more voltage is reflected for the evaluated case compared to the reference case.
- Using the propagation coefficient, one can compare several cases to a reference case and determine case-specific changes to the propagated voltage at all nodes. This shows how much the harmonic voltage changes at nodes and from which source the voltage is changed at a node.

7.8 Summary on Case Studies

This section has presented methods for evaluating harmonic propagation in a network using the propagation model. Figure 4.15 depicts a flowchart that describes the process from data collection to model development to analysis and finally to results evaluation.

The results have helped to clarify the propagation of harmonics in the meshed grid and the impact of UGC on propagation. In general, the procedure outlined in this section can serve as a guideline for harmonic analyses using the propagation model.

The guideline in Figure 4.15 shows the individual steps in setting up a propagation model, defining case studies, setting up the model and performing calculations, presenting data, and data analysis. With regard to the presentation of data, five categories of results are indicated. The five categories discussed in detail in Section 7.7 are presented in Table 4.19. These are the evaluation of S-parameters, voltage propagation in the grid, voltage amplification at buses, evaluating voltage extremes, and evaluating source-specific voltage propagation.

Table 4.19: Selection of results for evaluation based on field of study.

Results to be evaluated		Study types				
		Propagation study	Harmonic assessment	Screening study	Source impact study	Mitigation study
1.	Evaluation of S-parameters	X		X		
2.	Voltage propagation in the grid	X				X
3.	Voltage amplification at buses		X	X	X	X
4.	Evaluating voltage extremes		X			
5.	Evaluating source-specific voltage propagation	X			X	X

Five study types are also shown in the table, where the propagation model can advantageously be used for analyses and harmonic studies. These are:

- **Propagation studies**, where the propagation of harmonic voltage waves in a network is quantitatively determined and evaluated and problematic grid configurations in relation to propagation patterns are identified, as well as the evaluation of changing propagation conditions from harmonic sources.

- **Harmonic assessment**, which is largely comparable to harmonic penetration analyses, with the additional result that voltage extremes along individual lines can be determined and evaluated.

- **Screening study**, with the purpose of evaluating problematic grid configurations in relation to voltage amplification at buses in the grid, as well as quantitative evaluation of S-parameters in order to identify the impact of specific components on voltage propagation in the grid.

- **Source impact study**, to evaluate the impact of individual sources on harmonic voltage propagation in the grid, preferred propagation routes, and contribution to voltage amplification in nodes.

- **Mitigation studies**, in which suitable locations in the grid for the installation of harmonic filters are evaluated quantitatively based on expected amplification, the primary paths of harmonic voltage propagation, and the magnitude of the contribution of the individual sources to harmonic distortion in nodes in the network

The last two studies, the source impact study and the mitigation study, are related to the second objective of this thesis and will therefore be discussed in Chapter 5. However, the techniques for providing data for these studies are described in Section 7.7 of this chapter.

8 Chapter Summary

This chapter started by introducing the scenarios examined in the study as well as the methodology used. This introduction was followed by an analytical review of the properties of standing waves, on which the chosen method is based. It was followed by an analysis of the properties of Z, Y, ABCD and S-parameters, in which the choice of S-parameters was justified. The procedure for establishing a propagation model with S-parameters is described in [4.24] (paper **J1** of this thesis) and in this chapter the modelling has been elaborated in detail. Finally, the results for the case studies are presented and discussed, where a dedicated section is structured as a guideline that generally presents the procedure and interpretation of results for a propagation study.

The purpose of these scenarios has been to determine the significance of the interaction between UGC and OHL in relation to the propagation of voltage in a meshed transmission network, and here both cable lengths and the position of the cables in a network have been analysed. Location refers in this case partly as a section of cable on a line consisting primarily of overhead line, as a replacement for an overhead line section, or as a new crosswise connection that makes the network more meshed. This is based on the 400 kV transmission network in Western Denmark. Because there are known sources of harmonic voltages in the existing grid, the focus has been on the 11th and 13th harmonic voltages.

The model has been used to analyse the following five aspects:

1. Effect of UGC in combination with OHL on propagation, in the form of voltage reflections, as well as the influence of cable lengths on propagation.

2. Determination of voltage propagation because of changes in grid configurations.

3. Determination of voltage amplification because of changed grid configurations.

4. Evaluation of the standing voltage waves along lines to determine expected extremes (possible minimum and maximum voltages) as well as distances from substations to their location in the network.

5. Determination of the propagation of harmonic voltage from individual sources in the network and their impact at a point because of changed network configurations.

In its current state, the UGC sections used in the Western Danish transmission grid have line lengths ranging from 1.6 kilometres to 8.4 kilometres. Even though these line lengths appear to be short in comparison to the total length of lines in the grid, their impact on harmonic propagation is significant. A sensitivity analysis of S-parameters for various UGC types revealed (in this context, 11th and 13th harmonic wavelengths) that even short UGC lengths (less than approximately 15 kilometres) cause a change in reflection at its terminals ranging from 0% to approximately 80%, regardless of UGC type. For cable lengths ranging from 15 to 60 kilometres, the change in reflection is in the range of 80% to 90%, indicating that the voltage reflection in this range of line lengths may be considered high but almost constant, or at least with less variation. Because of the significant reflection, even for short UGC lengths, the voltage at the cable section's terminals and in subsequent substations may rise due to the increased reflected voltage. This effect is quite comparable to the voltage amplification phenomena described in harmonic analyses.

The voltage propagating through the UGC section demonstrates the opposite behaviour. The transmitted voltage is reduced by 45-50% for cable lengths less than 15 kilometres, and another 10-15% is observed for lengths greater than 15 kilometres. A voltage sensitivity analysis revealed that the transmitted voltage decreases until a cable length (in terms of the frequency-dependent wavelength) of about $\frac{1}{2}\lambda$, causing the UGC section to act as a limiting component in the grid in terms of harmonic voltage propagation. The evaluation of harmonic propagation revealed that the UGC length has a significant influence on the change in voltage propagation within the respective region experiencing grid changes, and that the impact is global.

This study also revealed that the Western Danish transmission grid is divided into three areas in terms of harmonic propagation, which means that changes in regions I and III have no effect on propagation in region II and vice versa, due to the values of transmission and the reflection coefficient direction on the lines connecting these two areas and their directionality properties. This was also demonstrated by observing the impact of grid changes on propagated voltage in one region and connecting the two areas with a crosswise connection.

As previously discussed, the effect of reflection described by S-parameters is comparable to the voltage amplification phenomena described in harmonic analyses. In Chapter 2, the term *"amplification"* was defined, and a propagation ratio χ is defined as a quantitative measure of this amplification. χ is calculated at substations for the case studies and indicates whether the propagated voltage rises or falls because of grid configuration changes.

The standing voltage waves along two lines in the grid has also been evaluated to demonstrate assessment of possible voltage ranges a node or substation can be expected to be exposed for. In this case, the presence of resonances close to a node or substation can be evaluated and problematic grid configurations can possibly be avoided.

To evaluate the change in propagation caused by grid changes, a propagation coefficient ζ is used, and the change in propagation from each source can be evaluated individually. In this case, ζ is calculated at substations S3 and S6, and it turns out that voltage propagation at S6 is most affected by sources at buses S9 and S12, whereas voltage propagation at S3 is most affected by sources at S2 and S8. In fact, when evaluated at substation S3, the grid changes appear to have no effect on the propagation of voltage from buses S9 and S12. This confirms the findings by Energinet from the assessment of gain factors described in Section 1.4 of Chapter 2 and the statistical analyses in both [4.30] (paper **J2** of this thesis) and by the PCA analysis in Chapter 3 of this thesis. It has also been found that increasing the length of the UGC results in more voltage passing through the evaluated substations due to increased reflection at the terminals of the UGC section, which is consistent with the evaluation of S-parameters.

The investigated case studies do not address all aspects of UGC in the Western Danish 400 kV transmission network. Instead, this chapter should be used as a guideline for conducting harmonic voltage propagation studies. However, by conducting in-depth analyses for these case studies, it was possible to gain a better understanding of propagation behaviour in a meshed power system, particularly the impact of UGC on propagation phenomena.

The results obtained with the propagation model described in this chapter demonstrate its usage. The propagation model varies from traditional techniques in the following ways:

- The meshed grid can be assessed by finding the optimal paths for harmonic voltage propagation or by assessing the propagation coefficient $\underline{\zeta}$ of individual sources.
- S-parameters, in the form of reflection and transmission coefficients, provide the physical conditions that influence both the magnitude and propagation of the harmonic voltage wave through the grid. These coefficients can be used to quantify the effect of different grid topologies on propagating voltage. This has to do with how much voltage is transferred along a line segment as well as how much voltage a particular line blocks or reflects back down the line.
- Across the grid, incident and reflected waves, as well as reflection coefficients, are determined, and standing wave profiles along lines can be

constructed. Extremes along the lines are easily identifiable.

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